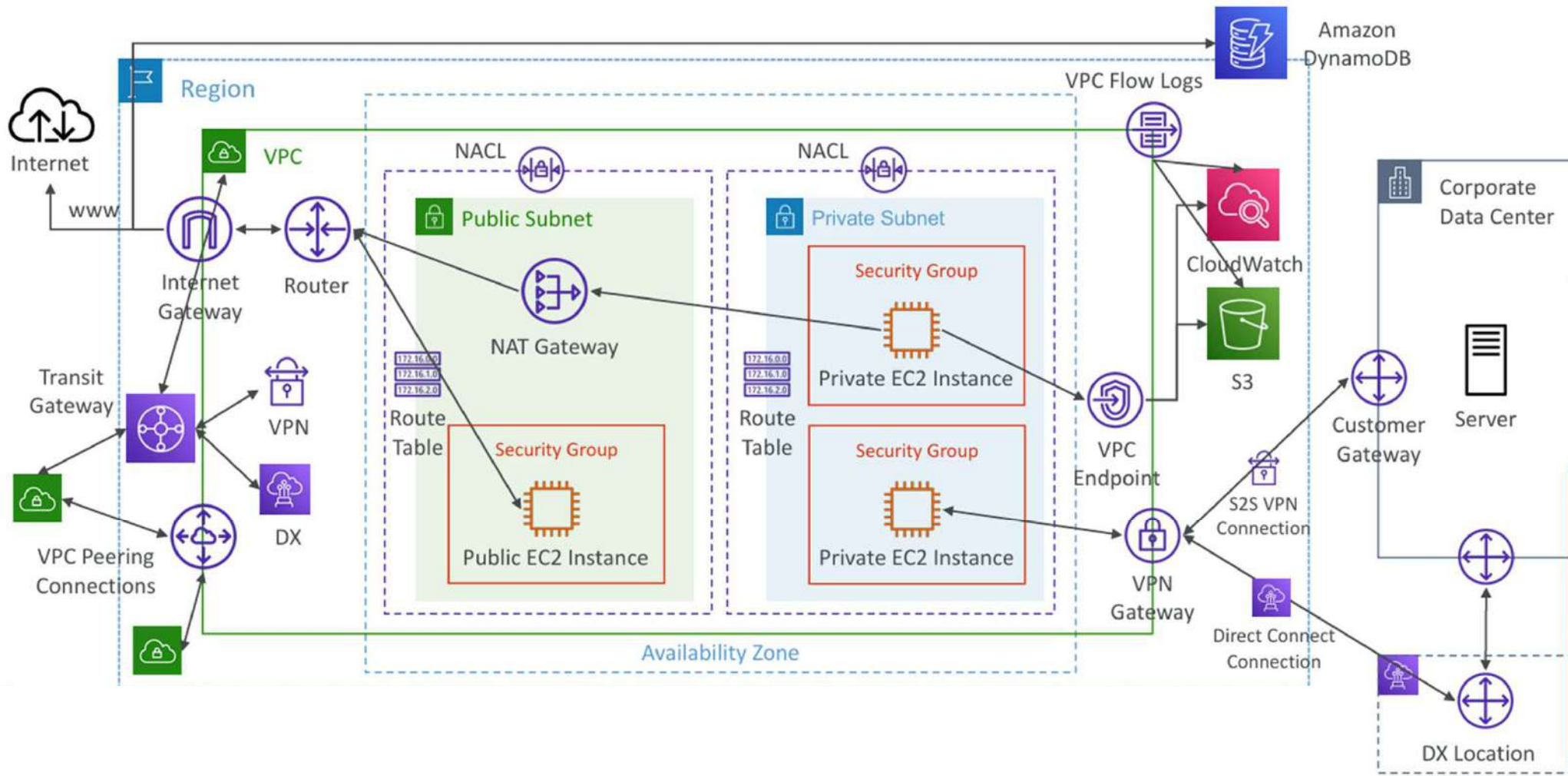


Networking – VPC

**AWS Certified Solutions Architect Associate
Certification SAA-C03**

<https://www.youtube.com/@techwithtwinkle>

VPC Components Diagram



<https://www.youtube.com/@techwithtwinkle>

Understanding CIDR – IPv4

- ▶ **Classless Inter-Domain Routing** – a method for allocating IP addresses
- ▶ Used in Security Groups rules and AWS networking in general

IP version	Type	Protocol	Port range	Source	Description
IPv4	SSH	TCP	22	122.149.196.85/32	–
IPv4	HTTP	TCP	80	0.0.0.0/0	–

- ▶ They help to define an IP address range:
 - We've seen WW.XX.YY.ZZ/32
 - We've seen 0.0.0.0/0 => all IPs => one IP
 - But we can define: 192.168.0.0/26 => 192.168.0.0 – 192.168.0.63 (64 IP addresses)

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Understanding CIDR – IPv4

- ▶ A CIDR consists of two components
- ▶ **Base IP**
 - Represents an IP contained in the range (XX.XX.XX.XX)
 - Example: 10.0.0.0, 192.168.0.0, ...
- ▶ **Subnet Mask**
 - Defines how many bits can change in the IP
 - Example: /0, /24, /32
 - Can take two forms:
 - /8 ó 255.0.0.0
 - /16 ó 255.255.0.0
 - /24 ó 255.255.255.0
 - /32 ó 255.255.255.255

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Understanding CIDR – Subnet Mask

- ▶ The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP

192	168	0	0	/32 => allows for 1 IP (2^0)	→ 192.168.0.0
192	168	0	0	/31 => allows for 2 IP (2^1)	→ 192.168.0.0 -> 192.168.0.1
192	168	0	0	/30 => allows for 4 IP (2^2)	→ 192.168.0.0 -> 192.168.0.3
192	168	0	0	/29 => allows for 8 IP (2^3)	→ 192.168.0.0 -> 192.168.0.7
192	168	0	0	/28 => allows for 16 IP (2^4)	→ 192.168.0.0 -> 192.168.0.15
192	168	0	0	/27 => allows for 32 IP (2^5)	→ 192.168.0.0 -> 192.168.0.31
192	168	0	0	/26 => allows for 64 IP (2^6)	→ 192.168.0.0 -> 192.168.0.63
192	168	0	0	/25 => allows for 128 IP (2^7)	→ 192.168.0.0 -> 192.168.0.127
192	168	0	0	/24 => allows for 256 IP (2^8)	→ 192.168.0.0 -> 192.168.0.255
...					
192	168	0	0	/16 => allows for 65,536 IP (2^{16})	→ 192.168.0.0 -> 192.168.255.255
...					
192	168	0	0	/0 => allows for All IPs	→ 0.0.0.0 -> 255.255.255.255



Quick Memo

Octets

1st . 2nd . 3rd . 4th

- /32 – no octet can change
- /24 – last octet can change
- /16 – last 2 octets can change
- /8 – last 3 octets can change
- /0 – all octets can change

<https://www.youtube.com/@techwithtwinkle>

Understanding CIDR – Little Exercise

- ▶ **192.168.0.0/24 = ... ?**
- 192.168.0.0 – 192.168.0.255 (256 IPs)
- ▶ **192.168.0.0/16 = ... ?**
- 192.168.0.0 – 192.168.255.255 (65,536 IPs)
- ▶ **134.56.78.123/32 = ... ?**
- Just 134.56.78.123
- ▶ **0.0.0.0/0**
- All IPs!
- ▶ When in doubt, use this website
<https://www.ipaddressguide.com/cidr>

Public vs. Private IP (IPv4)

- ▶ The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses
- ▶ Private IP can only allow certain values:
- ▶ 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) ← in big networks
- ▶ 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) ← AWS default VPC in that range
- ▶ 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) ← e.g., home networks
- ▶ All the rest of the IP addresses on the Internet are Public

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Default VPC Walkthrough

- ▶ All new AWS accounts have a default VPC
- ▶ New EC2 instances are launched into the default VPC if no subnet is specified
- ▶ Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
- ▶ We also get a public and a private IPv4 DNS names

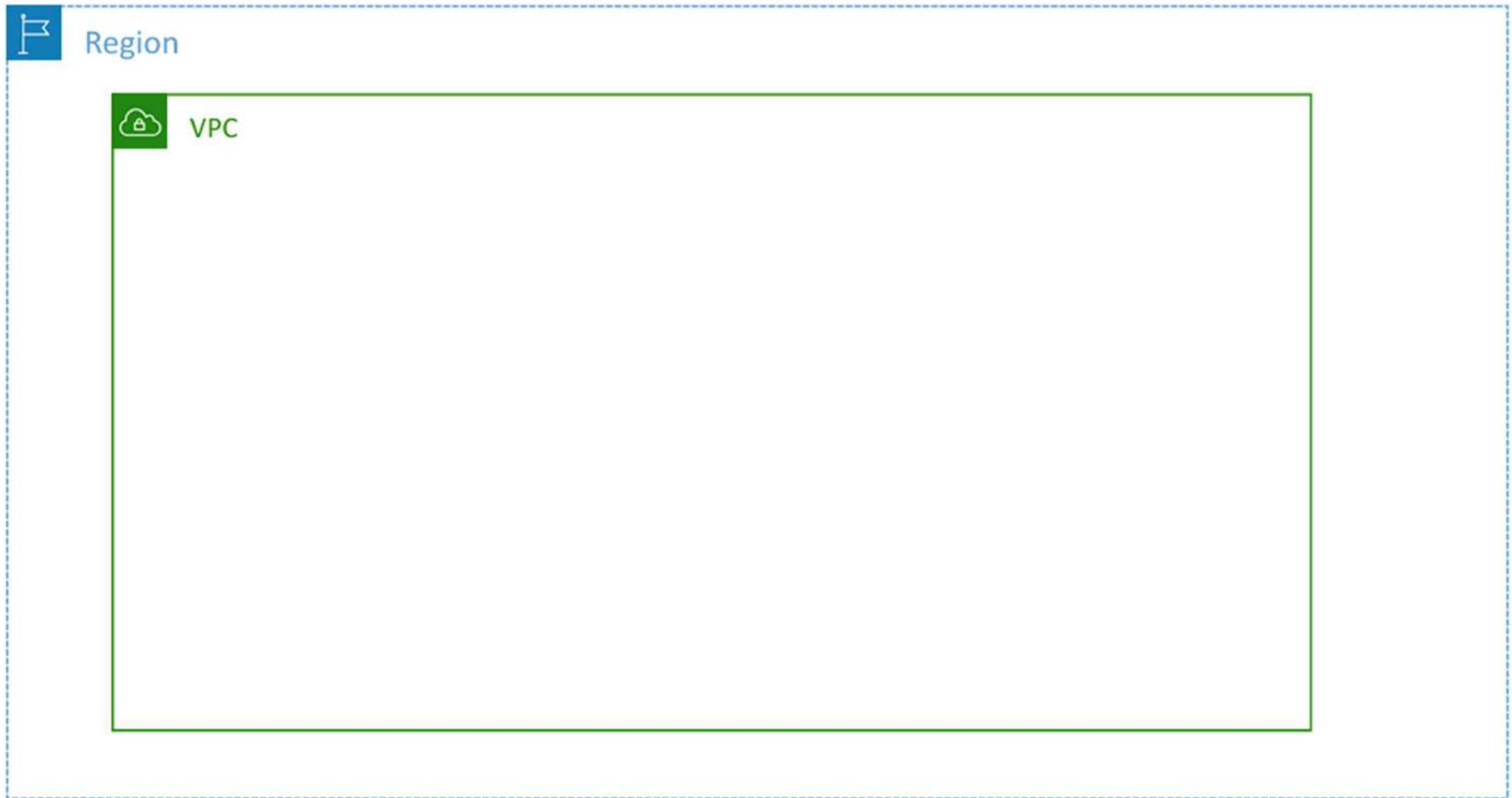
<https://www.youtube.com/@techwithtwinkle>

VPC in AWS – IPv4



- ▶ VPC = Virtual Private Cloud
- ▶ You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- ▶ Max. CIDR per VPC is 5, for each CIDR:
 - **Min. size is /28 (16 IP addresses)**
 - **Max. size is /16 (65536 IP addresses)**
- ▶ Because VPC is private, only the Private IPv4 ranges are allowed:
 - ▶ 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
 - ▶ 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
 - ▶ 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)
- ▶ **Your VPC CIDR should NOT overlap with your other networks (e.g., corporate)**

State of Hands-on



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VPC – Hands On

Search VPC

The screenshot shows the AWS Management Console interface. At the top, the search bar contains the text 'vpc'. Below the search bar, a dropdown menu displays search results. The 'Services' section is highlighted, showing three results: 'VPC' (Isolated Cloud Resources), 'AWS Global View', and 'AWS Firewall Manager'. An orange arrow points to the 'VPC' service result, with the text 'Click Here' next to it. The 'Features' section below shows a 'Dashboard' with a 'VPC feature' listed. At the bottom left, there is a feedback section asking 'Were these results helpful?' with 'Yes' and 'No' buttons. The background shows the AWS console sidebar with 'Console' and 'Recent' sections, and the top navigation bar with the AWS logo, search bar, and account information.

Services

- VPC** Isolated Cloud Resources
- AWS Global View** AWS Global View provides a global dashboard and search functionality that lets you ...
- AWS Firewall Manager** Central management of firewall rules

Features

- Dashboard**
 - VPC feature

Were these results helpful?

Yes No

<https://www.youtube.com/@techwithtwinkle>

 **VPC dashboard** < AWS Global View  Filter by VPC ▾

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Create VPC  **Launch EC2 Instances**

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region Refresh Resources

You are using the following Amazon VPC resources

Resource	Count
VPCs ▶ See all regions	Mumbai 1
NAT Gateways ▶ See all regions	Mumbai 0
Subnets ▶ See all regions	Mumbai 3
VPC Peering Connections ▶ See all regions	Mumbai 0
Route Tables ▶ See all regions	Mumbai 1
Network ACLs ▶ See all regions	Mumbai 1

Service Health
[View complete service health details](#) 

Settings
[Block Public Access](#)
[Zones](#) 
[Console Experiments](#)

Additional Information 
[VPC Documentation](#)

 [VPC](#) > [Your VPCs](#) > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only  ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

DemoVPC  **Enter name**

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16  **Enter IP Address**

<https://www.youtube.com/@techwithtwinkle>

- ☐ No IPv6 CIDR block
- ☐ IPAM-allocated IPv6 CIDR block
- ☐ Amazon-provided IPv6 CIDR block
- ☐ IPv6 CIDR owned by me

Tenancy

Default

Select this



Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q DemoVPC X

Remove tag

Add tag

You can add 49 more tags

Click on create VPC



Create VPC

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

You successfully created vpc-06622afdfcf55aef7 / DemoVPC

Your VPCs (2)

Last updated
2 minutes ago

Actions

Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR
☐	-	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16
☐	DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16



Select a VPC above

Demo VPC is created

<https://www.youtube.com/@techwithtwinkle>

Adding Subnets



<https://www.youtube.com/@techwithtwinkle>

VPC – Subnet (IPv4)



- ▶ AWS reserves 5 IP addresses (first 4 & last 1) in each subnet
- ▶ These 5 IP addresses are not available for use and can't be assigned to an EC2 instance

Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:

- 10.0.0.0 – Network Address
- 10.0.0.1 – reserved by AWS for the VPC router
- 10.0.0.2 – reserved by AWS for mapping to Amazon-provided DNS
- 10.0.0.3 – reserved by AWS for future use
- 10.0.0.255 – Network Broadcast Address. AWS does not support broadcast in a VPC, therefore the address is reserved

Exam Tip, if you need 29 IP addresses for EC2 instances:

- ▶ You can't choose a subnet of size /27 (32 IP addresses, $32 - 5 = 27 < 29$)
- ▶ You need to choose a subnet of size /26 (64 IP addresses, $64 - 5 = 59 > 29$)

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Subnet Hands On

VPC dashboard < AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Your VPCs (2) [Info](#) Last updated 4 minutes ago **Actions** **Create VPC**

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR
☐	-	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16
☐	DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16

Select a VPC above

Click here

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard < Subnets (3) Info

Click on create subnet

Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC
☐	-	subnet-0fa41e55155c9f1a7	Available	vpc-0a506c34e326aff4f
☐	-	subnet-06b73b9c44595f643	Available	vpc-0a506c34e326aff4f
☐	-	subnet-0287ee3f45c829710	Available	vpc-0a506c34e326aff4f

Select a subnet

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

VPC > Subnets > Create subnet

Create subnet Info

VPC

VPC ID

Create subnets in this VPC.

Select a VPC

Click here

Q |

vpc-0a506c34e326aff4f (default)
172.31.0.0/16

vpc-06622afdfcf55aef7 (DemoVPC)
10.0.0.0/16

Select this

Select a VPC first to create new subnets.

Add new subnet

Cancel Create subnet

<https://www.youtube.com/@techwithtwinkle>

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

**Enter Subnet name****Select zone****Enter subnet value**IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

256 IPs

► Tags - optional

[Remove](#)[Add new subnet](#)**Click here add new subnet**[Cancel](#)[Create subnet](#)

PublicSubnetB

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

▼ **Tags - optional****Key**

Q Name

**Value - optional**

Q PublicSubnetB



Remove

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

► **Tags - optional**

Remove

Add new subnet

Click on New subnet

<https://www.youtube.com/@techwithtwinkle>

Cancel

Create subnet

Subnet 3 of 3

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

PrivateSubnetA

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.16.0/20

4,096

Enter Subnet name

Select A zone AZ

Enter private IP

<https://www.youtube.com/@techwithtwinkle>

IPv4 subnet CIDR block

10.0.16.0/20

4,096 IPs

Tags - optional

Key

Name

Value - optional

PrivateSubnetA

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Click on Add New Subnet

Cancel

Create subnet

Subnet 4 of 4

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

PrivateSubnetB

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.32.0/20

4,096 IPs

Enter name

Select AZ

Enter IP

<https://www.youtube.com/@techwithtwinkle>

IPv4 subnet CIDR block

10.0.32.0/20

4,096 IPs

▼ **Tags - optional**

Key

Q Name



Value - optional

Q PrivateSubnetB



Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Click on create Subnet

Create subnet

Menu icon VPC > Subnets

Subnets (7) Info Last updated 1 minute ago Actions Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC
☐	-	subnet-0287ee3f45c829710	Available	vpc-0a506c34e326aff4f
☐	PublicSubnetB	subnet-02cc21c0d1c1dcc87	Available	vpc-06622afdfcf55aef7 Demo...
☐	PrivateSubnetR	subnet-0dbe813169aab11ee	Available	vpc-06622afdfcf55aef7 Demo...

Select a subnet

4 subnets is created now

techwithtwinkle

<https://www.youtube.com/@techwithtwinkle>



Internet Gateway (IGW)

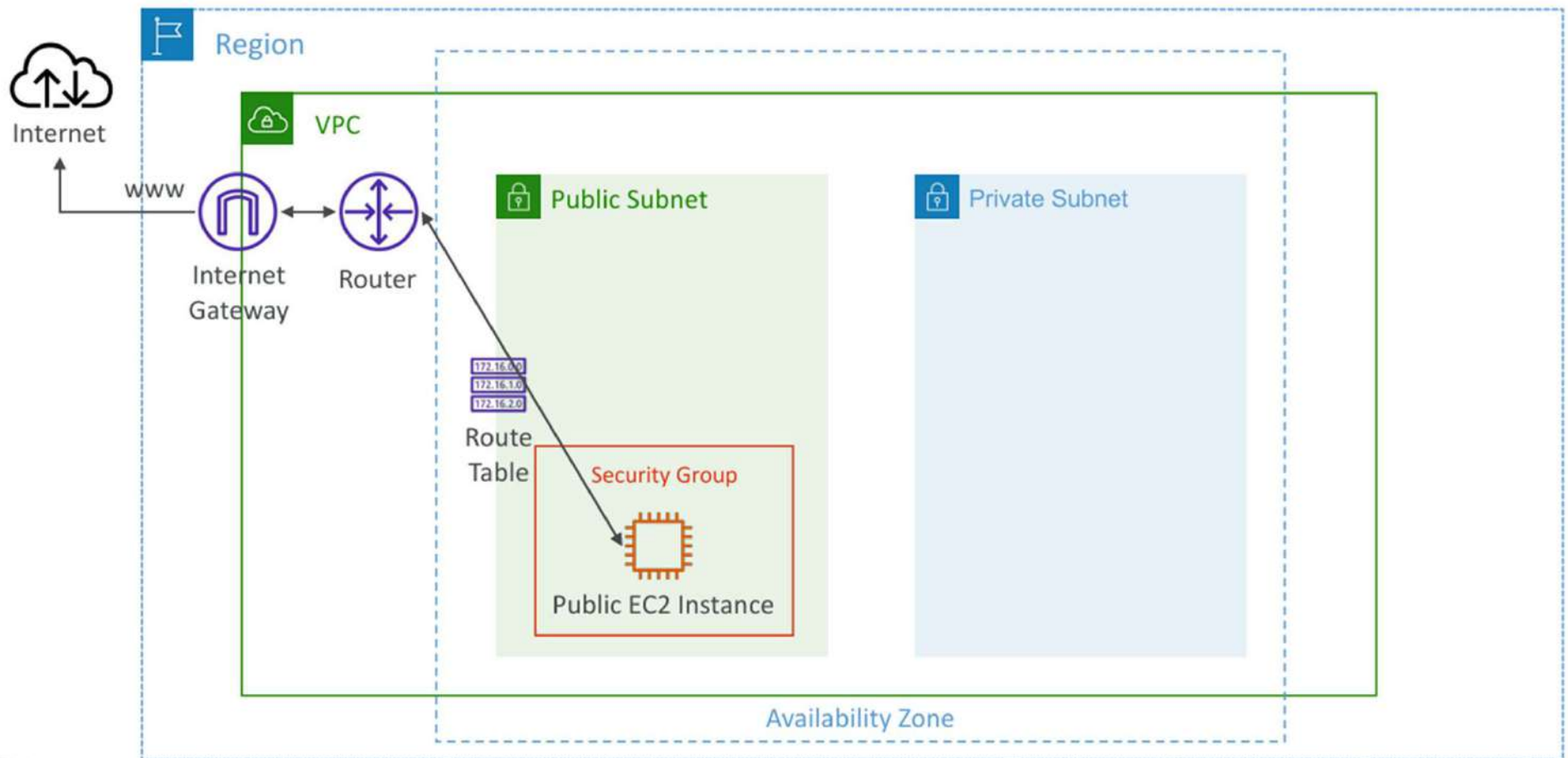
- ▶ Allows resources (e.g., EC2 instances) in a VPC connect to the Internet
- ▶ It scales horizontally and is highly available and redundant
- ▶ Must be created separately from a VPC
- ▶ One VPC can only be attached to one IGW and vice versa
- ▶ Internet Gateways on their own do not allow Internet access...
- ▶ Route tables must also be edited!

Adding Internet Gateway



<https://www.youtube.com/@techwithtwinkle>

Editing Route Tables



<https://www.youtube.com/@techwithtwinkle>

Internet Gateways & Route Tables Hands On

- ▶ Go to EC2 Console → Create Instance

The screenshot displays the AWS Management Console interface for the EC2 service. The top navigation bar includes the AWS logo, a search bar, and the current region (Asia Pacific (Mumbai)). The left sidebar shows the 'EC2' menu with options like Dashboard, AWS Global View, Events, and a list of instance-related services. The main content area is titled 'Instances' and features a search bar, a refresh button, and a 'Launch instances' button. An orange arrow points to the 'Launch instances' button, with the text 'Click on launch instance' above it. Below the search bar, there is a table header with columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. The table currently shows no instances, with a message 'No instances' and 'You do not have any instances in this region'.

<https://www.youtube.com/@techwithtwinkle>

you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value

Create new key pair

Network settings

Network

vpc-0a506c34e326aff4f

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Edit

Click on Edit

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

Preview code

<https://www.youtube.com/@techwithtwinkle>

Network settings

VPC - *required*

vpc-0a506c34e326aff4f
172.31.0.0/16

(default)

Q |

vpc-06622afdfcf55aef7 (DemoVPC)
10.0.0.0/16

vpc-0a506c34e326aff4f
172.31.0.0/16

(default)

Create new subnet

Enable additional zones

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Click here

Select this

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

Preview code

Network settings [Info](#)

VPC - required [Info](#)

vpc-06622afdcf55aef7 (DemoVPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0fb0e68a3ae9be50b PublicSubnetA
VPC: vpc-06622afdcf55aef7 Owner: 235562991793
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.0.0/24

Select subnet

Auto-assign public IP [Info](#)

Enable

Enable it

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

[Preview code](#)

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Add security group rule](#)

Advanced network configuration

Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

[Preview code](#)

Click on Launch Instance

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard < VPC > Subnets

✓ You have successfully changed subnet settings:
◦ Enable auto-assign public IPv4 address

Subnets (7) Info Last updated 4 minutes ago Actions Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC
☐	PrivateSubnetA	subnet-016e736c2ef7867b1	✓ Available	vpc-06622afdfcf55aef7 Demo...
☐	PublicSubnetA	subnet-0fb0e68a3ae9be50b	✓ Available	vpc-06622afdfcf55aef7 Demo...

Select a subnet

Click on internet gateways

VPC dashboard < VPC > Internet gateways

Internet gateways (1) Info

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID
☐	-	igw-098183a0345e4b2bf	✓ Attached	vpc-0a506c34e326aff4f

Select an internet gateway above

Click here Create internet gateway

<https://www.youtube.com/@techwithtwinkle>

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

DemoIGW



Enter name of internet gateway

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name



Value - optional

Q DemoIGW



Remove

Add new tag

You can add 49 more tags.

Click here



Create internet gateway

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

✓ The following internet gateway was created: igw-027195eb8ffafd531 - DemoIGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC



igw-027195eb8ffafd531 / DemoIGW

Details Info

Internet gateway ID

igw-027195eb8ffafd531

State

Detached

VPC ID

-

Owner

23556

Click here



Actions

Click here



Attach to VPC

Detach from VPC

Manage tags

Delete

Tags

Q Search tags

Key

Value

Name

DemoIGW

Manage tags

< 1 > Settings

Attach to VPC (igw-027195eb8ffafd531) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

vpc-06622afdfcf55aef7 - DemoVPC

AWS Command Line Interface Command

vpc-06622afdfcf55aef7 - DemoVPC

Click here

Select this

Cancel

Attach internet gateway

Then click here

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Internet gateway igw-027195eb8ffafd531 successfully attached to vpc-06622afdfcf55aef7

igw-027195eb8ffafd531 / DemoIGW

Actions

Details [Info](#)

Internet gateway ID

igw-027195eb8ffafd531

State

Attached

VPC ID

[vpc-06622afdfcf55aef7](#)
DemoVPC

Owner

235562991793

Tags

Search tags

Key | Value

Name | DemoIGW

Manage tags

< 1 > Settings

Click here

Attached now

<https://www.youtube.com/@techwithtwinkle>

VPC > Route tables

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Route tables (1) Info

Last updated about 1 hour ago

Actions

Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	-	rtb-062183f5cc8205cab	-		Yes

Select a route table

Click on create table

<https://www.youtube.com/@techwithtwinkle>

VPC > Route tables > Create route table

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PublicRouteTable

VPC
The VPC to use for this route table.

Select a VPC

vpc-0a506c34e326aff4f (default)

vpc-06622afdfcf55aef7 (DemoVPC)

Key **Value - optional**

Q Name X Q PublicRouteTable X Remove

Enter route table name

Select this

☰ VPC > Route tables > Create route table

Create a tag with a key of 'Name' and a value that you specify.

PublicRouteTable

VPC
The VPC to use for this route table.

vpc-06622afdcf55aef7 (DemoVPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional** Remove

Add new tag

You can add 49 more tags.

Click here  Create route table

☰ VPC > Route tables

VPC dashboard <

AWS Global View 

Filter by VPC ▼

▼ **Virtual private cloud**

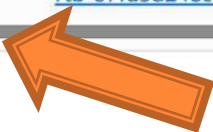
- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

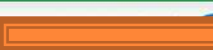
✓ Route table rtb-07fd3d246c91fa97b | PublicRouteTable was created successfully.

Route tables (3) [Info](#)

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	-	rtb-062183f5cc8205cab	-	-	Yes
☐	-	rtb-0f04c3dbde0d20eb8	-	-	Yes
☐	PublicRouteTable	rtb-07fd3d246c91fa97b	-	-	No

Select a route table

Created now 

Click here  Create route table

<https://www.youtube.com/@techwithtwinkle>

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

PrivateRouteTable

Enter name

VPC

The VPC to use for this route table.

Select a VPC

Q |

vpc-0a506c34e326aff4f

(default)

vpc-06622afdfcf55aef7 (DemoVPC)

Select VPC

Key

Q Name

Value - optional

Q PrivateRouteTable

Remove

Create a tag with a key of 'Name' and a value that you specify.

PrivateRouteTable

VPC

The VPC to use for this route table.

vpc-06622afdfcf55aef7 (DemoVPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q PrivateRouteTable

Remove

Add new tag

You can add 49 more tags.

Click on create Route table

Create route table

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard < AWS Global View [↗] Filter by VPC ▾

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Route table rtb-01e7a7e0497c8595c | PrivateRouteTable was created successfully. [X]

Route tables (1/4) Info Last updated less than a minute ago [↻] [Actions ▾] [Create route table]

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☒	PublicRouteTable	rtb-07fd3d246c91fa97b	-	-	No
☐	PrivateRouteTable	rtb-01e7a7e0497c8595c	-	-	No

rtb-07fd3d246c91fa97b / PublicRouteTable [⚙] ▾

Details Routes Subnet associations Edge associations Route propagation Tags

Details

Select this

Click here

VPC dashboard < AWS Global View [↗] Filter by VPC ▾

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Route table rtb-01e7a7e0497c8595c | PrivateRouteTable was created successfully. [X]

Route tables (1/4) Info Last updated 1 minute ago [↻] [Actions ▾] [Create route table]

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☒	PublicRouteTable	rtb-07fd3d246c91fa97b	-	-	No
☐	PrivateRouteTable	rtb-01e7a7e0497c8595c	-	-	No

rtb-07fd3d246c91fa97b / PublicRouteTable [⚙] ▾

Details Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (0)

Find subnet association

[Edit subnet associations]

Click here

<https://www.youtube.com/@techwithtwinkle>

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	PublicSubnetB	subnet-02cc21c0d1c1dcc87	10.0.1.0/24	-	Main (rtb-0f04c3dbde0d20eb8)
☐	PrivateSubnetB	subnet-0dbe813169aab11ee	10.0.32.0/20	-	Main (rtb-0f04c3dbde0d20eb8)
☐	PrivateSubnetA	subnet-016e736c2ef7867b1	10.0.16.0/20	-	Main (rtb-0f04c3dbde0d20eb8)
☒	PublicSubnetA	subnet-0fb0e68a3ae9be50b	10.0.0.0/24	-	Main (rtb-0f04c3dbde0d20eb8)

Select public subnet

Selected subnets

subnet-0fb0e68a3ae9be50b / PublicSubnetA subnet-02cc21c0d1c1dcc87 / PublicSubnetB

Click here

Save associations

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Select this

You have successfully updated subnet associations for rtb-07fd3d246c91fa97b / PublicRouteTable.

Route tables (1/4)

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets	-	No
☒	PrivateRouteTable	rtb-01e7a7e0497c8595c	-	-	No

rtb-01e7a7e0497c8595c / PrivateRouteTable

Click here

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Details

<https://www.youtube.com/@techwithtwinkle>

☰ VPC > Route tables

VPC dashboard <

AWS Global View

Filter by VPC

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

☑ You have successfully updated subnet associations for rtb-07fd3d246c91fa97b / PublicRouteTable.

Route tables (1/4) Info Last updated 1 minute ago Actions Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets	-	No
☒	PrivateRouteTable	rtb-01e7a7e0497c8595c	-	-	No

rtb-01e7a7e0497c8595c / PrivateRouteTable

Details Routes **Subnet associations** Edge associations Route propagation Tags

Explicit subnet associations (0)

Find subnet association

Click here

[Edit subnet associations](#)

<https://www.youtube.com/@techwithtwinkle>

☰ VPC > Route tables > [rtb-01e7a7e0497c8595c](#) > Edit subnet associations

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	PublicSubnetB	subnet-02cc21c0d1c1dcc87	10.0.1.0/24	-	rtb-07fd3d246c91fa97b / PublicRc
☒	PrivateSubnetB	subnet-0dbe813169aab11ee	10.0.32.0/20	-	Main (rtb-0f04c3dbde0d20eb8)
☒	PrivateSubnetA	subnet-016e736c2ef7867b1	10.0.16.0/20	-	Main (rtb-0f04c3dbde0d20eb8)
☐	PublicSubnetA	subnet-0fb0e68a3ae9be50b	10.0.0.0/24	-	rtb-07fd3d246c91fa97b / PublicRc

Selected subnets

[subnet-0dbe813169aab11ee / PrivateSubnetB](#) [subnet-016e736c2ef7867b1 / PrivateSubnetA](#)

Click here [Save associations](#)

VPC dashboard < AWS Global View Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Select this

Route tables (1/4) Info Last updated 11 minutes ago **Actions** **Create route table**

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☒	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets	-	No
☐	PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets	-	No

rtb-07fd3d246c91fa97b / PublicRouteTable

Click here

Details **Routes** Subnet associations Edge associations Route propagation Tags

Details <https://www.youtube.com/@techwithtwinkle>

VPC dashboard < AWS Global View Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Route tables (1/4) Info Last updated 12 minutes ago **Actions** **Create route table**

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☒	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets	-	No
☐	PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets	-	No

rtb-07fd3d246c91fa97b / PublicRouteTable

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (1) Filter routes

Click on Edit rules

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	✔ Active	No	CreateRouteTable
	🔍 local	✕		



Click on Add Route

Cancel Preview Save changes

<https://www.youtube.com/@techwithtwinkle>

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	✔ Active	No	CreateRouteTable
	🔍 local	✕		
🔍		-	No	CreateRoute
0.0.0.0/0				Remove
0.0.0.0/8				
0.0.0.0/16				
0.0.0.0/24				
0.0.0.0/32				
::/0				



select this

Cancel Preview Save changes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	local		No	CreateRoute
	Core Network			
	Egress Only Internet Gateway			
	Gateway Load Balancer Endpoint			
	Instance			
	Internet Gateway			
	local			

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#) [Remove](#)

Click here

Select this

Edit routes

<https://www.youtube.com/@techwithtwinkle>

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	igw-027195eb8ffafd531			
	Use: "igw-027195eb8ffafd531"			
	igw-027195eb8ffafd531 (DemoIGW)			

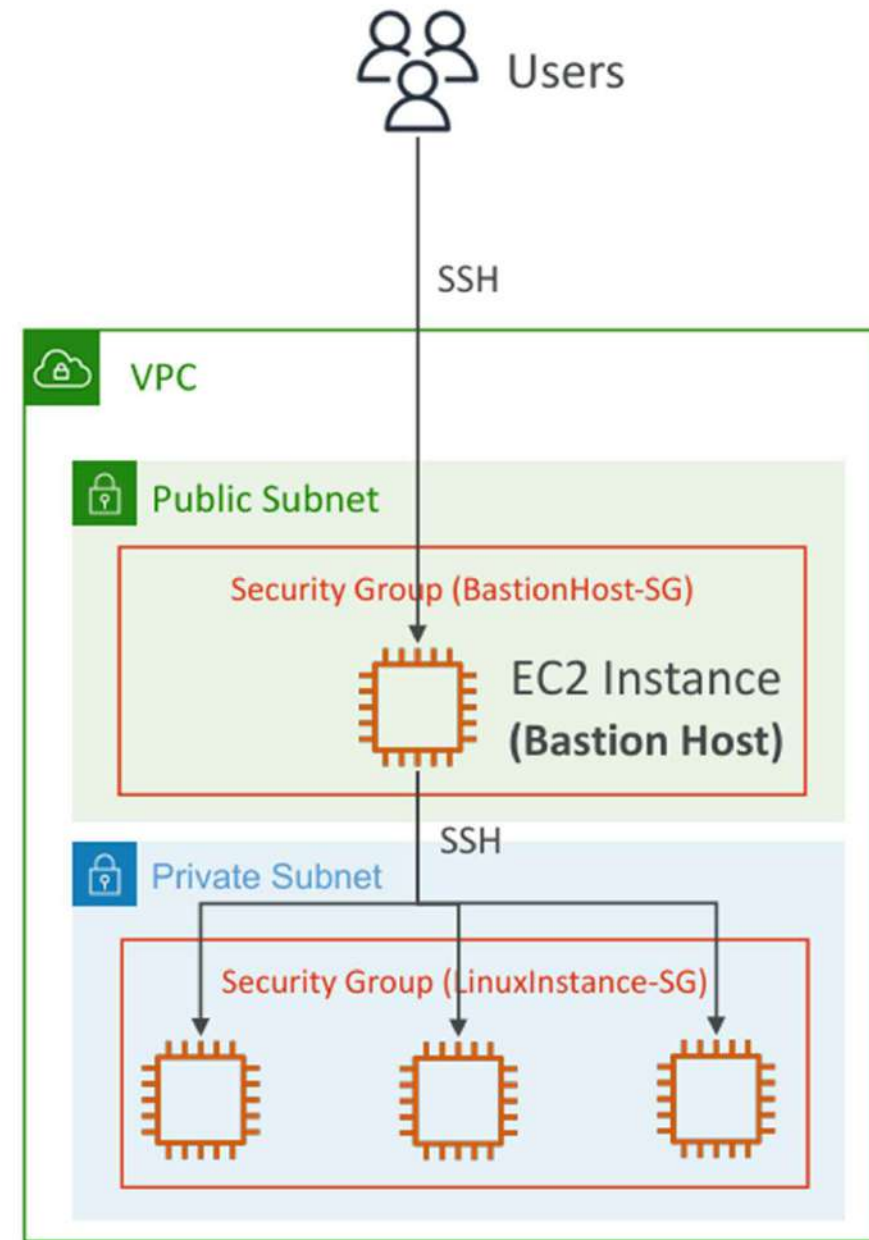
[Add route](#) [Cancel](#) [Preview](#) [Save changes](#) [Remove](#)

Select this

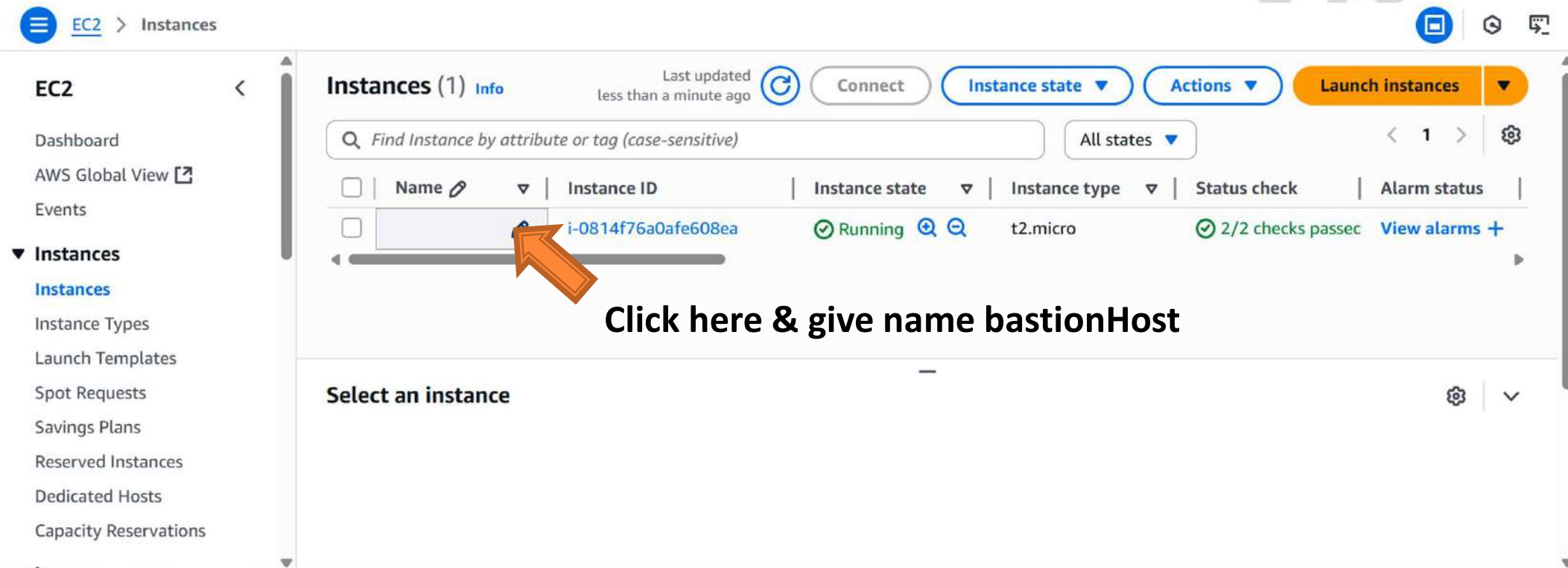
After that click here

Bastion Hosts

- ▶ We can use a Bastion Host to SSH into our private EC2 instances
- ▶ The bastion is in the public subnet which is then connected to all other private subnets
- ▶ **Bastion Host security group must allow** inbound from the internet on port 22 from restricted CIDR, for example the public CIDR of your corporation
- ▶ **Security Group of the EC2 Instances** must allow the Security Group of the Bastion Host, or the private IP of the Bastion host



Bastion Hosts Hands On



The screenshot shows the AWS Management Console interface for the EC2 Instances page. The left sidebar contains navigation links: EC2, Dashboard, AWS Global View, Events, Instances (selected), Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, and Capacity Reservations. The main content area displays a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. A single instance is listed with ID i-0814f76a0afe608ea, state Running, type t2.micro, and status check 2/2 checks passed. An orange arrow points to the 'Name' column header, and the text 'Click here & give name bastionHost' is overlaid on the image.

EC2 > Instances

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐		i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +

Select an instance

<https://www.youtube.com/@techwithtwinkle>

EC2 > Instances

Instances (1/1) Info Last updated about 1 hour ago Connect Instance state ▼ Actions ▼ Launch instances ▼

Find Instance by attribute or tag (case-sensitive) All states ▼ < 1 > ⚙

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	BastionHost	i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +

Click on Keypairs

i-0814f76a0afe608ea

[Details](#) Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
-------------	---------------------	------------------------

EC2 > Key pairs

Key pairs Info ⚙ 🔍 🗨

Find Key Pair by attribute or tag < 1 > ⚙

☐	Name	Type	Created	Fingerprint	ID
No key pairs to display					

Click here

[Create key pair](#)

<https://www.youtube.com/@techwithtwinkle>

A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name

DemoKeypair

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type [Info](#)

☒ RSA

☐ ED25519

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

Tags - optional

No tags associated with the resource.

[Add new tag](#)

Enter name

Select This

Select this

<https://www.youtube.com/@techwithtwinkle>

Key pair type [Info](#)

☒ RSA

☐ ED25519

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

Tags - optional

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Click here [Create key pair](#)

EC2 > Key pairs

Successfully created key pair

Key pairs (1) Info

Find Key Pair by attribute or tag

☐	Name	Type	Created	Fingerprint	ID
☐	DemoKeypair	rsa	2025/09/07 17:51 GMT+5:30	03:9c:85:77:f9:c8:f6:eb:8c:90:96:73:...	ke-

Click on instances

<https://www.youtube.com/@techwithtwinkle>

EC2 > Instances

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	BastionHost	i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +

Launch instances

Click on Launch instance

i-0814f76a0afe608ea (BastionHost)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID | Public IPv4 address | Private IPv4 address

aws twinkl

Search [Alt+S]

Asia Pacific (Mumbai)

EC2 > Instances > Launch an instance

you launch the instance.

Key pair name - required

DemoKeypair

Create new key pair

Network settings

Network

vpc-0a506c34e326aff4f

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Before this enter instance name

Select Keypair

Click on edit

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

you launch the instance.

Key pair name - required

DemoKeypair

Create new key pair

Network settings

Network

vpc-06622afdfcf55aef7 (DemoVPC) 10.0.0.0/16

vpc-0a506c34e326aff4f (default) 172.31.0.0/16

vpc-0a506c34e326aff4f (default) 172.31.0.0/16

Subnet

No preference

Create new subnet

Availability Zone

No preference

Enable additional zones

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

Preview code

Select this

<https://www.youtube.com/@techwithtwinkle>

vpc-06622afdcf55aef7 (DemoVPC)
10.0.0.0/16

Subnet Info

subnet-016e736c2ef7867b1 PrivateSubnetA
VPC: vpc-06622afdcf55aef7 Owner: 235562991793
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.16.0/20

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

PrivateSG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255

Select This

Enter Group name

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

Cancel

Launch instance

Preview code

<https://www.youtube.com/@techwithtwinkle>

Inbound Security Group Rules

Security group rule 1 (TCP, 22)

Remove

Type Info

ssh

Protocol Info

TCP

Port range Info

22

Source type Info

Custom

Source Info

Add CIDR, prefix list or security

Description - optional Info

e.g. SSH for admin desktop

Anywhere

Custom

My IP

Select this

Configure storage Info

Advanced

Summary

Number of instances Info

1

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22)

Type | Info
ssh

Source type | Info
Custom

Security groups

- launch-wizard-1
sg-0e31cbba3608766b2
- default
sg-090d87a6cd86d764a

Port range | Info
22

Description - optional | Info
SSH for admin desktop

Summary

Number of instances | Info
1

t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Select This

EC2 > Instances > Launch an instance

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-0e31cbba3608766b2, Allow SSH From Bastion Host)

Type | Info
ssh

Protocol | Info
TCP

Port range | Info
22

Source type | Info
Custom

Source | Info
Add CIDR, prefix list or security
sg-0e31cbba3608766b2

Description - optional | Info
Allow SSH From Bastion Host

Summary

Number of instances | Info
1

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where

Cancel Launch instance Preview code

Enter Description

Click on Launch Instance

<https://www.youtube.com/@techwithtwinkle>

EC2 > Instances

Click on connect

Instances (1/2) Info Last updated 1 minute ago Refresh Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	BastionHost	i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +
☐	PrivateInstance	i-09346fba51cd7665	Running	t2.micro	Initializing	View alarms +

Select This

i-0814f76a0afe608ea (BastionHost)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary Info

Instance ID: i-0814f76a0afe608ea

Public IPv4 address: 52.66.195.65

Private IPv4 address: 10.0.1.10

EC2 > Instances > i-0814f76a0afe608ea > Connect to instance

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
52.66.195.65

☐ IPv6 address

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Click on connect Connect

<https://www.youtube.com/@techwithtwinkle>


```

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CGMyHQZclwlllyQe98fxNdi7qJR5UoERIVzqawrStTmGWmUJylzhN
-----END RSA PRIVATE KEY-----
[ec2-user@ip-10-0-0-55 ~]$ chmod 0400DemoKeyPair.pem
chmod: missing operand after '0400DemoKeyPair.pem'
Try 'chmod --help' for more information.
[ec2-user@ip-10-0-0-55 ~]$ chmod 0400 DemoKeyPair.pem
[ec2-user@ip-10-0-0-55 ~]$

```

Enter this

<https://www.youtube.com/@techwithtwinkle>

```

[ec2-user@ip-10-0-0-55 ~]$ ssh ec2-user@10.0.21.198 -i DemoKeyPair.pem
Load key "DemoKeyPair.pem": error in libcrypto
ec2-user@10.0.21.198: Permission denied (publickey,gssapi-keyex,gss with-mic).
[ec2-user@ip-10-0-0-55 ~]$ rm DemoKeyPair.pem
rm: remove write-protected regular file 'DemoKeyPair.pem'? yes
[ec2-user@ip-10-0-0-55 ~]$

```

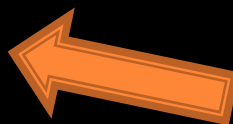
Enter This

After all these enter vi DemoKeypair.pem

```

LO9YznNU4psioYts06YZ1pHsrKi2XpdryRzRHtqbgswCD8Spq4N1/sPQctvVpvzM
nOOifDSGkUd00CehcqjQuTNJXHiq5yFZtpwtlFvPd7/oRPPOyTLuHJElNRMTBvFnd
wQ1b+2ECgYEA8vwd4lH0Zj0Yp7qOBS47C/uYdwY4duJ2pOWSsSLCVH7aPaKgEUGx
CJcmHrDiFmFUX+DkWcnamEGZkAC68j9mCtcWdBYKueVomxhfCXkZRHk9r6mioDyg
P6GwJ/Smd3rqm8bVZDyopjSBAWg3heGiVKxTgOdRDLY/lIuhVd+cb+UCgYEA8lU9
nIXnWS7n2tsddvA19tGtpXjbZhvuap0l7yrYNlhKfTEIg2VQ6YXCwL3noK0SJGw
GUz08d9+FxOEJYcX8W6Fg06aILw0QWA9QFZyNxnCsJMwLWisQvxxzunuKFof4IQ/u
GARwGKtMHGliaUSKowCLWZiihal+x0pB5ObiRlcGgYBp+HZ4C9IQ/LhR/lSOAdmS
okmeL0JU5vNjexP82PHzs9cuZkx0+jdhTbo8TFxt8ki//+7XEeeZ25JDcW4W1HuG
YsPSIQLQjCc6EOjx0rSoXk/+2naod4hT+ApBgrrPBHZpNdA/BQ1wXxmcKX19kCgh
0X4hi8CXUKxt9m+Iahtp2QKBgFQbVajvqJ4Bc6M4DX/sWy5xE82spqlH5HXHToc1
3ZjvB9v//OlmGzpBeRnvS7YOMzhEqFP/novAP+kOcvz7aEn5TAmIzGG0MIbIOjoM
RtPs8u+KBoC0taUCLcB5hDHRu0LuXnqlwTMusNnk9RUIJ9/6jocAzNeLP1843Knv
oHU9AoGBAio7OM75aDnK5vUr9/58L8WITdwcbnpQ2i5QxQHA3VnI8almVUBikHW0
H9pSmCm82t/DBRkCBLsa0LMGRdGKt13NGSv2nZOFXBERbHIYBxRssAlCBcxgkkt8
CGMyHQZclw1lyQe98fxNdi7qJR5UoERIVzqawrStTmGWmUJy1zhN
-----END RSA PRIVATE KEY-----

```



Paste Here

27,30

Bo

t

```

%
oHU9AoGBAio7OM75aDnK5vUr9/58L8WITdwcbnpQ2i5QxQHA3VnI8almVUBikHW0
%
H9pSmCm82t/DBRkCBLsa0LMGRdGKt13NGSv2nZOFXBERbHIYBxRssAlCBcxgkkt8
%
CGMyHQZclw1lyQe98fxNdi7qJR5UoERIVzqawrStTmGWmUJy1zhN
%
-----END RSA PRIVATE KEY-----
%

```

18,30

55

19,30

66

20,30

77

21,30

88

22,30

Bo

27,30

Bo

27,30

Bo

28,1

Bo

29,1

Bo

30,1

Bo



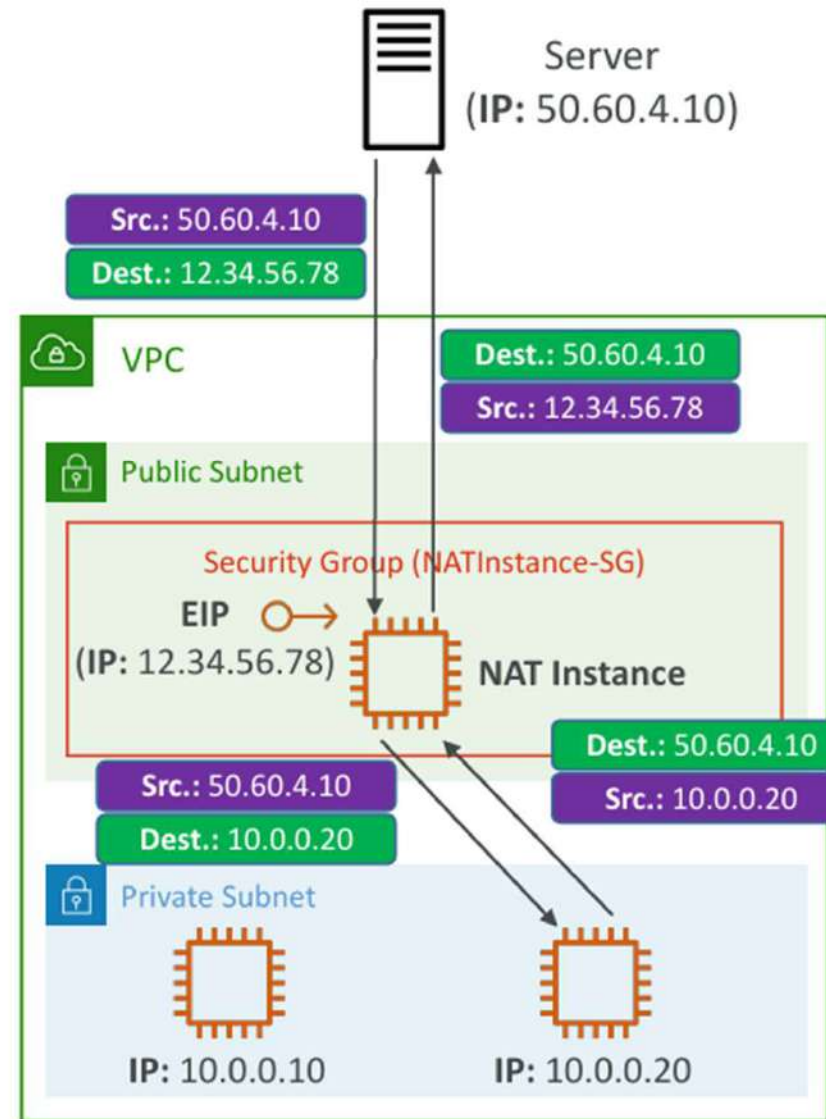
With this save it & Rewrite the previous SSH Comment

<https://www.youtube.com/@techwithtwinkle>

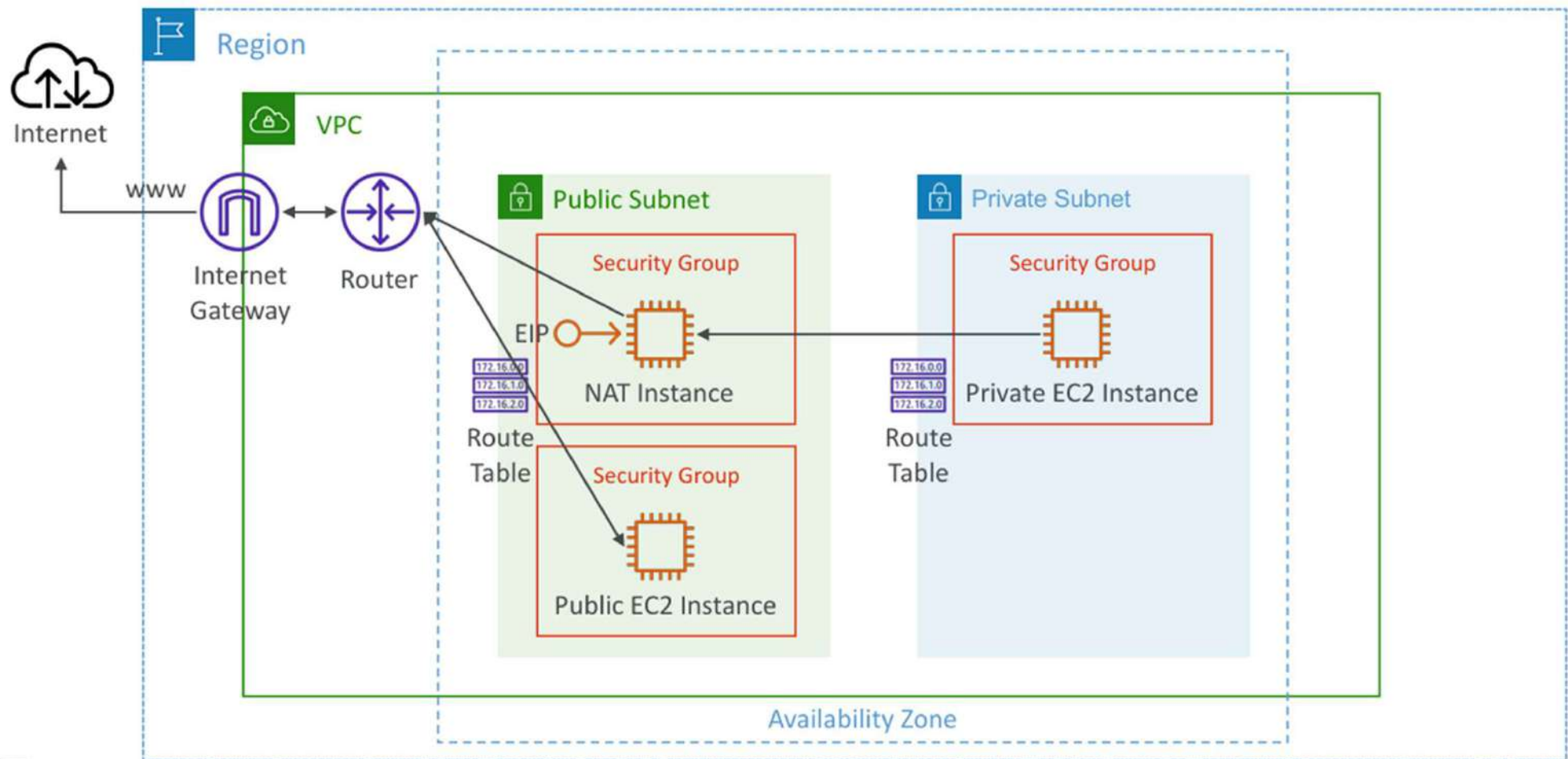
NAT Instance (outdated, but still at the exam)

- ▶ **NAT = Network Address Translation**
- ▶ Allows EC2 instances in private subnets to connect to the Internet
- ▶ Must be launched in a public subnet
- ▶ Must disable EC2 setting: Source / destination Check
- ▶ Must have Elastic IP attached to it
- ▶ Route Tables must be configured to route traffic from private subnets to the NAT Instance

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NAT Instance



<https://www.youtube.com/@techwithtwinkle>

NAT Instance – Comments

- ▶ Pre-configured Amazon Linux AMI is available
- Reached the end of standard support on December 31, 2020
- ▶ Not highly available / resilient setup out of the box
- You need to create an ASG in multi-AZ + resilient user-data script
- ▶ Internet traffic bandwidth depends on EC2 instance type
- ▶ You must manage Security Groups & rules:

Inbound:

- ▶ Allow HTTP / HTTPS traffic coming from Private Subnets
- ▶ Allow SSH from your home network (access is provided through Internet Gateway)

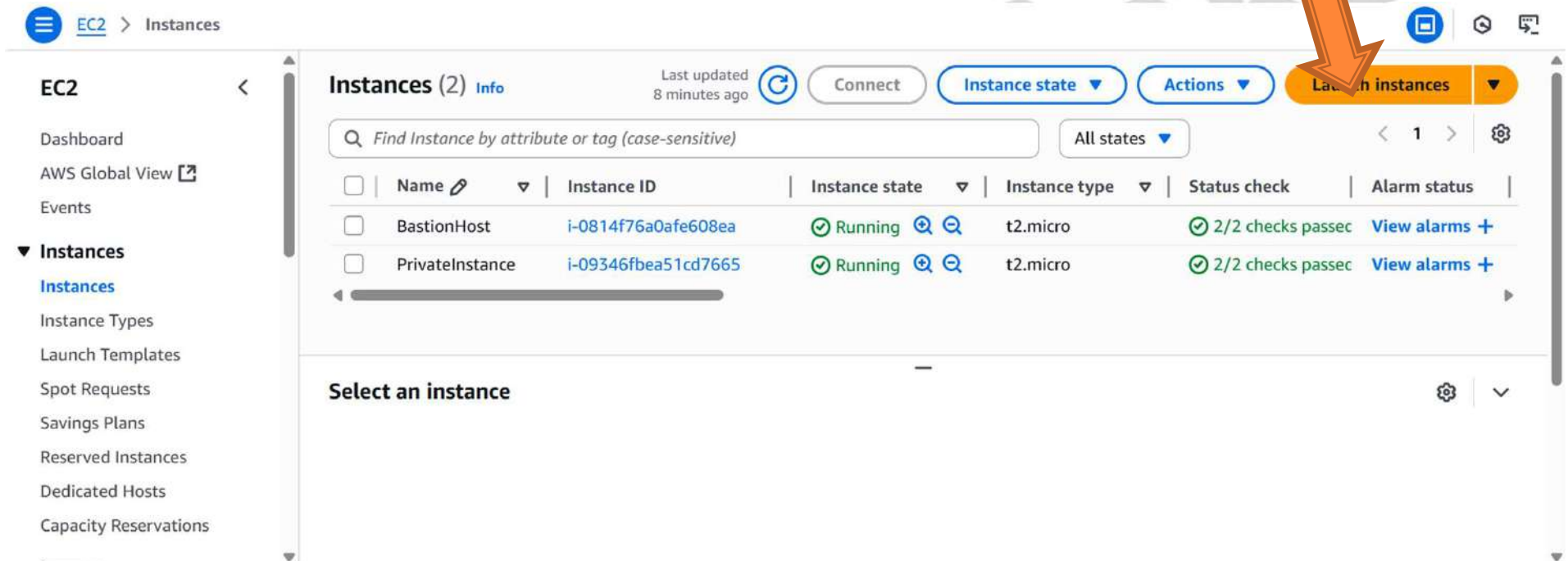
Outbound:

- ▶ Allow HTTP / HTTPS traffic to the Internet

<https://www.youtube.com/@techwithtwinkle>

NAT Instance Hands On

Click on Launch Instance



The screenshot shows the AWS Management Console for the EC2 service, specifically the 'Instances' page. The left-hand navigation pane lists various EC2-related options, with 'Instances' currently selected. The main content area displays a list of existing instances. At the top right of this area, there are several buttons: 'Connect', 'Instance state', 'Actions', and 'Launch instances'. An orange arrow points directly to the 'Launch instances' button. Below the buttons is a search bar and a filter dropdown set to 'All states'. The instance list table shows two instances, both in a 'Running' state.

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	BastionHost	i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +
☐	PrivateInstance	i-09346fbea51cd7665	Running	t2.micro	2/2 checks passed	View alarms +

<https://www.youtube.com/@techwithtwinkle>

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

Enter name of instance

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

▼ Summary

Number of instances [Info](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

[Cancel](#)

[Launch instance](#)

[Preview code](#)

<https://www.youtube.com/@techwithtwinkle>

Recents [Quick Start](#)

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Click here

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0861f4e788f5069dd (64-bit (x86), uefi-preferred) / ami-0fad8318b9405c6fb (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

▼ Summary

Number of instances [Info](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0861f4e788f5069dd

Virtual server type (instance type)

t2.micro

Firewall (security group)

[Cancel](#)

[Launch instance](#)

[Preview code](#)

nat

Search NAT

Quick Start AMIs (0)

Commonly used AMIs

My AMIs (0)

Created by me

AWS Marketplace AMIs (67)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone

☐ Free tier only [Info](#)

▼ OS category

☐ All Linux/Unix

☐ All Windows

▼ Architecture

☐ 64-bit (Arm)

☐ 32-bit (x86)

☐ 64-bit (x86)

☐ 64-bit (Mac)

☐ 64-bit (Mac-Arm)

Clear Filters

The following results for "nat" were found in other categories:

- 67 results in AWS Marketplace AMIs
AWS Marketplace AMIs are AMIs that are published by AWS & trusted third-parties
- 500+ results in Community AMIs
Community AMIs are AMIs that are shared by the general AWS community

Click here

<https://www.youtube.com/@techwithtwinkle>

nat

Quick Start AMIs (0)

Commonly used AMIs

My AMIs (0)

Created by me

AWS Marketplace AMIs (67)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone

▼ Linux/Unix

☐ All Linux/Unix

☐ Amazon Linux

☐ CentOS

☐ Debian

☐ Fedora

☐ Gentoo

☐ macOS

☐ openSUSE

☐ Other Linux

☐ Red Hat

☐ SUSE Linux

publisher before launching an AMI.



Verified provider

nats-2024-12-17 05_17_38-44c1b87b-481a-4caf-b423-40f20e907ac6

ami-00208c5511a9a31e2

NATS

OwnerAlias: aws-marketplace

Publish date: 2024-12-17

Platform: -

Root device type: ebs

Architecture: x86_64

Virtualization: hvm

Owner: 679593333241

ENA enabled: Yes

Select



vnscubed676-20250728-marketplace-nate-5mJD-de3c89e9-6794-4c6c-a933-3b7ae972f7c2

ami-00138d7d2a086c88d

This snapshot is registered as an EBS-backed Amazon Machine Instance, created from the

Select

Select This

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

Free tier eligible ▼

☐ All generations

[Compare instance types](#)

The AMI vendor recommends using a m5.large instance (or larger) for the best experience with this product.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select ▼

[Create new key pair](#)

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

nats-2024-12-17 05_17_38-44c1b...[read more](#)
ami-00208c5511a9a31e2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

The AMI vendor recommends using a m5.large instance (or larger) for the best experience with this product.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

DemoKeypair ▼

[Create new key pair](#)

▼ Network settings [Info](#)

Network [Info](#)

vpc-0a506c34e326aff4f

Subnet [Info](#)

No preference (Default subnet in any availability zone)

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

nats-2024-12-17 05_17_38-44c1b...[read more](#)
ami-00208c5511a9a31e2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB



Select Keypair



Click here

VPC - required [Info](#)

vpc-06622afdfcf55aef7 (DemoVPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0fb0e68a3ae9be50b PublicSubnetA
VPC: vpc-06622afdfcf55aef7 Owner: 235562991793
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 250 CIDR: 10.0.0.0/24

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

Select this

Select This

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

nats-2024-12-17 05_17_38-44c1b...[read more](#)
ami-00208c5511a9a31e2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

<https://www.youtube.com/@techwithtwinkle>

Security group name - required

NAT-Instance-sg

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ : / () # , @ [] + = & ; [] ! \$ *

Description - required [Info](#)

NATS.io Server, packaged by Synadia-v2.10.24-AutogenByAWSMP--1 create

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

🔍 Add CIDR, prefix list or security

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

Select security look

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

nats-2024-12-17 05_17_38-44c1b...[read more](#)
ami-00208c5511a9a31e2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

▼ Security group rule 2 (TCP, 80, 10.0.0.0/16) **Make new rule this**

Type [Info](#) Protocol [Info](#) Port range [Info](#)

HTTP TCP 80

Source type [Info](#) Source [Info](#) Description - optional [Info](#)

Custom Add CIDR, prefix list or security 10.0.0.0/16 X e.g. SSH for admin desktop

▼ Security group rule 3 (TCP, 443, 10.0.0.0/16) **Make new rule**

Type [Info](#) Protocol [Info](#) Port range [Info](#)

HTTPS TCP 443

Source type [Info](#) Source [Info](#) Description - optional [Info](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

nats-2024-12-17 05_17_38-44c1b...[read more](#)
ami-00208c5511a9a31e2

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

<https://www.youtube.com/@techwithtwinkle>

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

[Edit](#)

► **Advanced details** [Info](#)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are

[Cancel](#)

[Launch instance](#)

[Preview code](#)

Click on Launch Instance

Right Click on instance

Click here

Click on networking

Click here

Click on networking

Change Source / destination check

The source / destination check ensures that the instance is the source or destination of all the traffic it sends and receives. Each EC2 instance performs source and destination checks by default. [Learn more](#)

Instance ID
i-07a819ffb793d2eb3 (NAT Instance)

Network interface
eni-06c0ee56bd261673b

Source / destination checking
Stop to allow your instance to send and receive traffic when the source or destination is not itself.

☒ Stop

Select this

Click on save

Save

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Updated routes for rtb-07fd3d246c91fa97b / PublicRouteTable successfully

Route tables (1/4)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets	-	No

Select Privateroute table

Click on Routes

rtb-01e7a7e0497c8595c / PrivateRouteTable

Details Routes Subnet associations Edge associations Route propagation Tags

Details

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard

AWS Global View

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DHCP option sets

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Managed prefix lists

NAT gateways

Updated routes for rtb-07fd3d246c91fa97b / PublicRouteTable successfully

Route tables (1/4)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets	-	No

rtb-01e7a7e0497c8595c / PrivateRouteTable

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (1)

Click on Edit Route

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local 🔍 local ✕	✔ Active	No	CreateRouteTable
Add route				

Cancel

Preview

Save changes

Click on Add Route

<https://www.youtube.com/@techwithtwinkle>

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local 🔍 local ✕	✔ Active	No	CreateRouteTable	
🔍 0.0.0.0/0 ✕	Core Network Egress Only Internet Gateway Gateway Load Balancer Endpoint Instance Internet Gateway local	-	No	CreateRoute	Remove
Add route					

Cancel

Preview

Save changes

Click on instance

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Instance	-	No	CreateRoute

Q 0.0.0.0/0 X

Q i-

Use: "i-"

i-0814f76a0afe608ea (BastionHost)

i-09346fbea51cd7665 (PrivateInstance)

i-07a819ffb793d2eb3 (NAT Instance)

Select NAT Instance

Cancel Preview Save changes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Instance	-	No	CreateRoute

Click on Save Changes

EC2 > Instances

Instances (1/3) Info Last updated 11 minutes ago Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	BastionHost	i-0814f76a0afe608ea	Running	t2.micro	2/2 checks passed	View alarms +
☐	PrivateInstance	i-09346f6ea51cd7665	Running	t2.micro	2/2 checks passed	View alarms +
☒	NAT Instance	i-07a819ffb793d2eb3	Running	t2.micro	2/2 checks passed	View alarms +

Select This

i-07a819ffb793d2eb3 (NAT Instance)

Details Status and alarms Monitoring Security Networking Storage Tags

Click on security

▼ Instance summary Info

<https://www.youtube.com/@techwithtwinkle>

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	NAT Instance	i-07a819ffb793d2eb3	Running	t2.micro	2/2 checks passed	View alarms +

i-07a819ffb793d2eb3 (NAT Instance)

sg-047bc7cfa878f959c (NAT-Instance-sg)

Click on sg

▼ Inbound rules

EC2 > Security Groups > sg-047bc7cfa878f959c - NAT-Instance-sg

Owner: 235562991793 Inbound rules count: 3 Permission entries Outbound rules count: 1 Permission entry

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

Click on edit

Inbound rules (3)

Name	Security group rule ID	IP version	Type	Protocol
-	sgr-050c810fa08c0d9e9	IPv4	HTTP	TCP
-	sgr-0a3e106ad36183c96	IPv4	HTTPS	TCP
-	sgr-011d8d6765f5dd1b2	IPv4	SSH	TCP

EC2 > Security Groups > sg-047bc7cfa878f959c - NAT-Instance-sg > Edit inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-050c810fa08c0d9e9	HTTP	TCP	80	Cus...	10.0.0.0/16	Delete
sgr-0a3e106ad36183c96	HTTPS	TCP	443	Cus...	10.0.0.0/16	Delete
sgr-011d8d6765f5dd1b2	SSH	TCP	22	Cus...	0.0.0.0/0	Delete

Click on Add Rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

EC2 > Security Groups > sg-047bc7cfa878f959c - NAT-Instance-sg > Edit inbound rules

Security Group ID	Protocol	Port Range	Source	Action
sgr-050c810fa08c0d9e9	HTTP	TCP 80	10.0.0.0/16	Delete
sgr-0a3e106ad36183c96	HTTPS	TCP 443	10.0.0.0/16	Delete
sgr-011d8d6765f5dd1b2	SSH	TCP 22	0.0.0.0/0	Delete
-	Custom TCP	TCP 0		Delete

Select this →

Q icmp
Custom ICMP - IPv4
All ICMP - IPv4
All ICMP - IPv6
Custom TCP

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes **Save rules**

EC2 > Security Groups > sg-047bc7cfa878f959c - NAT-Instance-sg > Edit inbound rules

Security Group ID	Protocol	Port Range	Source	Action
sgr-0a3e106ad36183c96	HTTPS	TCP 443	10.0.0.0/16	Delete
sgr-011d8d6765f5dd1b2	SSH	TCP 22	10.0.0.0/16	Delete
-	All ICMP - IPv4	ICMP All	0.0.0.0/0	Delete
-			10.0.0.0/16	Delete

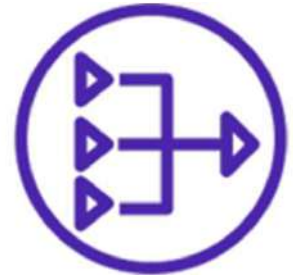
Click on save rules →

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Save rules

<https://www.youtube.com/@techwithtwinkle>

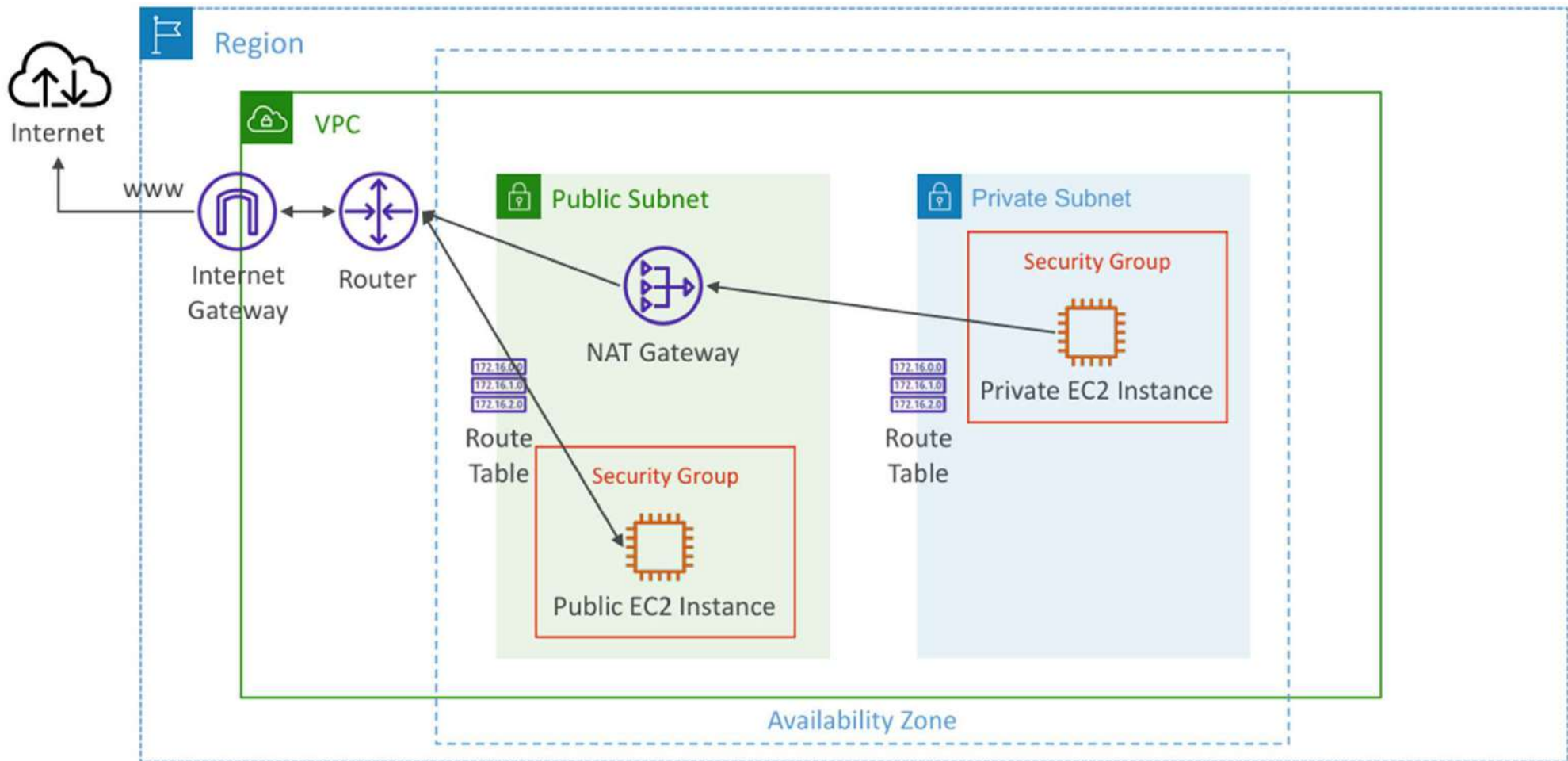
NAT Gateway



- ▶ AWS-managed NAT, higher bandwidth, high availability, no administration
- ▶ Pay per hour for usage and bandwidth
- ▶ NATGW is created in a specific Availability Zone, uses an Elastic IP
- ▶ Can't be used by EC2 instance in the same subnet (only from other subnets)
- ▶ Requires an IGW (Private Subnet => NATGW => IGW)
- ▶ 5 Gbps of bandwidth with automatic scaling up to 100 Gbps
- ▶ No Security Groups to manage / required

<https://www.youtube.com/@techwithtwinkle>

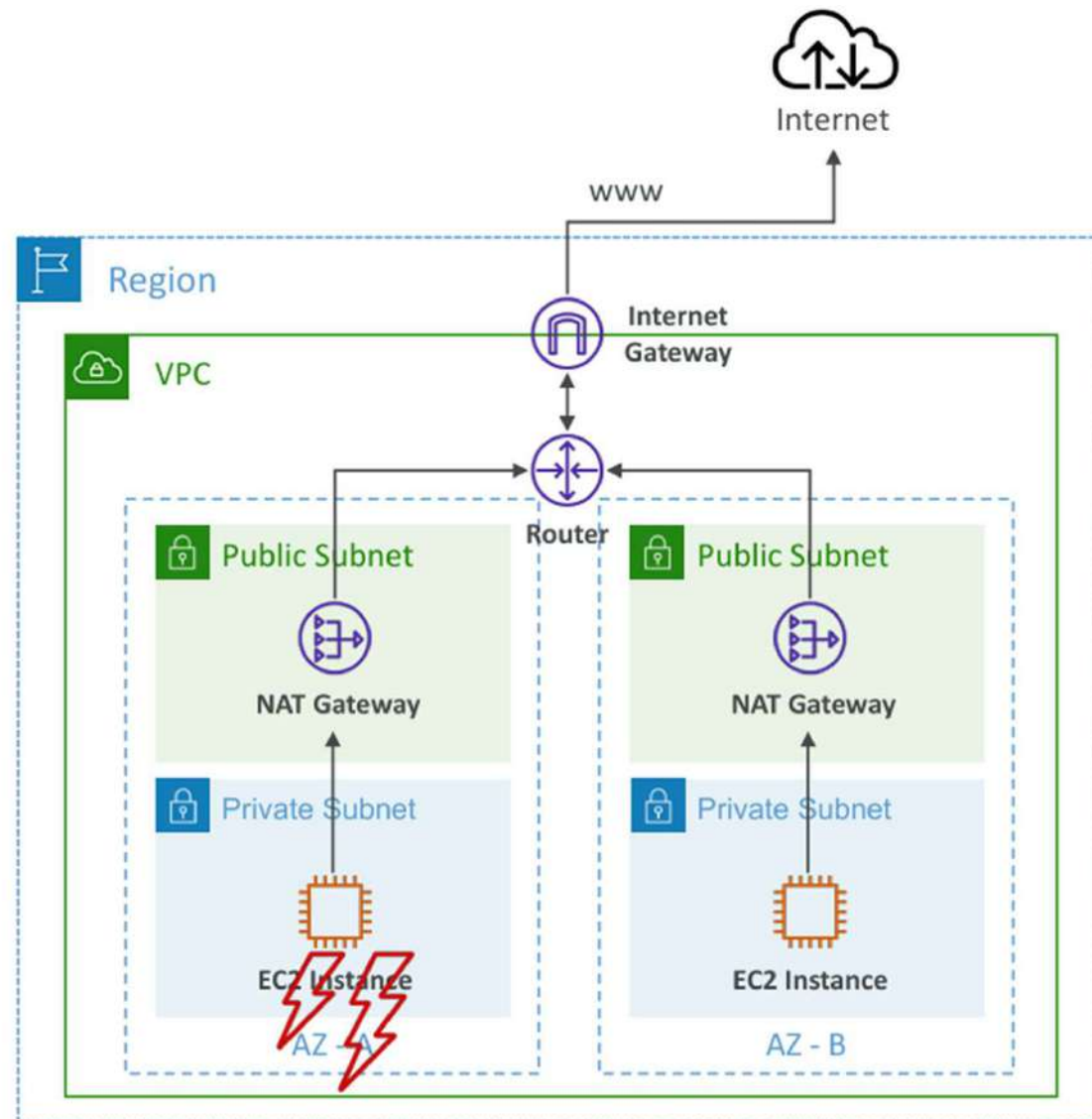
NAT Gateway



<https://www.youtube.com/@techwithtwinkle>

NAT Gateway with High Availability

- ▶ NAT Gateway is resilient within a single Availability Zone
- ▶ Must create multiple NAT Gateways in multiple AZs for fault-tolerance
- ▶ There is no cross-AZ failover needed because if an AZ goes down it doesn't need NAT



NAT Gateway vs. NAT Instance

	NAT Gateway	NAT Instance
Availability	Highly available within AZ (create in another AZ)	Use a script to manage failover between instances
Bandwidth	Up to 100 Gbps	Depends on EC2 instance type
Maintenance	Managed by AWS	Managed by you (e.g., software, OS patches, ...)
Cost	Per hour & amount of data transferred	Per hour, EC2 instance type and size, + network \$
Public IPv4	✓	✓
Private IPv4	✓	✓
Security Groups	✗	✓
Use as Bastion Host?	✗	✓

More at: <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html>

<https://www.youtube.com/@techwithtwinkle>

NAT Gateways Hands On

VPC > Route tables > rtb-01e7a7e0497c8595c

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

▼ **Security**

Network ACLs

Security groups

▼ **PrivateLink and Lattice**

Getting started [Updated](#)

Details intro

Route table ID
rtb-01e7a7e0497c8595c

VPC
vpc-06622afdfcf55aef7 | DemoVPC

Main
No

Owner ID
235562991793

Explicit subnet associations
2 subnets

Edge associations
-

Routes Subnet associations Edge associations Route propagation

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	eni-06c0ee56bd2616...	Blackhole	No	Create Route
0.0.0.0/16	local	Active	No	Create Route Table

Both Edit routes

After Terminate instance it will show this here

Click on NAT Gateways

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VPC

>

NAT gateways

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started Updated

NAT gateways Info

Click on create NAT Gateways

Find NAT gateways by attribute or tag

< 1 > ⚙

Name	NAT gateway ID	Connectivity...	State	State message	Primary
No NAT gateways found					

Select a NAT gateway

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VPC

>

NAT gateways

>

Create NAT gateway

Create NAT gateway Info

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

DemoNATGW

The name can be up to 256 characters long.

Subnet

Select a subnet in which to create the NAT gateway.

subnet-0fb0e68a3ae9be50b (PublicSubnetA)

Connectivity type

Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID Info

Subnet

Select a subnet in which to create the NAT gateway.

subnet-0fb0e68a3ae9be50b (PublicSubnetA) ▼

Connectivity type

Select a connectivity type for the NAT gateway.

- ☒ Public
☐ Private

Elastic IP allocation ID [Info](#)

Assign an Elastic IP address to the NAT gateway.

Select an Elastic IP ▼

Allocate Elastic IP

Click on Allocate Elastic IP



► Additional settings [Info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

✓ Elastic IP address 65.2.157.235 (eipalloc-082db154358f9dd64) allocated.



Assign an Elastic IP address to the NAT gateway.

eipalloc-082db154358f9dd64 ▼

Allocate Elastic IP

► Additional settings [Info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

🔍 Name



🔍 DemoNATGW



Remove

Add new tag

You can add 49 more tags.

Click on create NAT Gateway



Create NAT gateway

<https://www.youtube.com/@techwithtwinkle>

VPC > NAT gateways > nat-0e20925a019598c68

NAT gateway nat-0e20925a019598c68 | DemoNATGW was created successfully.

nat-0e20925a019598c68 / DemoNATGW

Details

NAT gateway ID nat-0e20925a019598c68	Connectivity type Public	State Pending	State message Info -
NAT gateway ARN arn:aws:ec2:ap-south-1:235562991793:natgateway/nat-0e20925a019598c68	Primary public IPv4 address -	Primary private IPv4 address -	Primary network interface ID -
VPC vpc-06622afdfcf55aef7 / DemoVPC	Subnet subnet-0fb0e68a3ae9be50b / PublicSubnetA	Created Sunday, September 7, 2025 at 20:40:55 GMT+5:30	Deleted -

Secondary IPv4 addresses | **Monitoring** | **Tags**

Click on route tables

VPC > Route tables

Route tables (1/4) Info

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ.
☒	PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets
☐	-	rtb-0f04c3dbde0d20eb8	-
☐	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets
☐	-	rtb-062183f5cc8205cab	-

rtb-01e7a7e0497c8595c / PrivateRouteTable

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Click here

Select Route Table

Click on Edit Routes

Click on Edit Routes

<https://www.youtube.com/@techwithtwinkle>

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway		No	CreateRoute

Select NAT Gateway

Click on save changes

<https://www.youtube.com/@techwithtwinkle>

VPC dashboard

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Updated routes for rtb-01e7a7e0497c8595c / PrivateRouteTable successfully

VPC	Owner ID
vpc-06622afdfcf55aef7 DemoVPC	235562991793

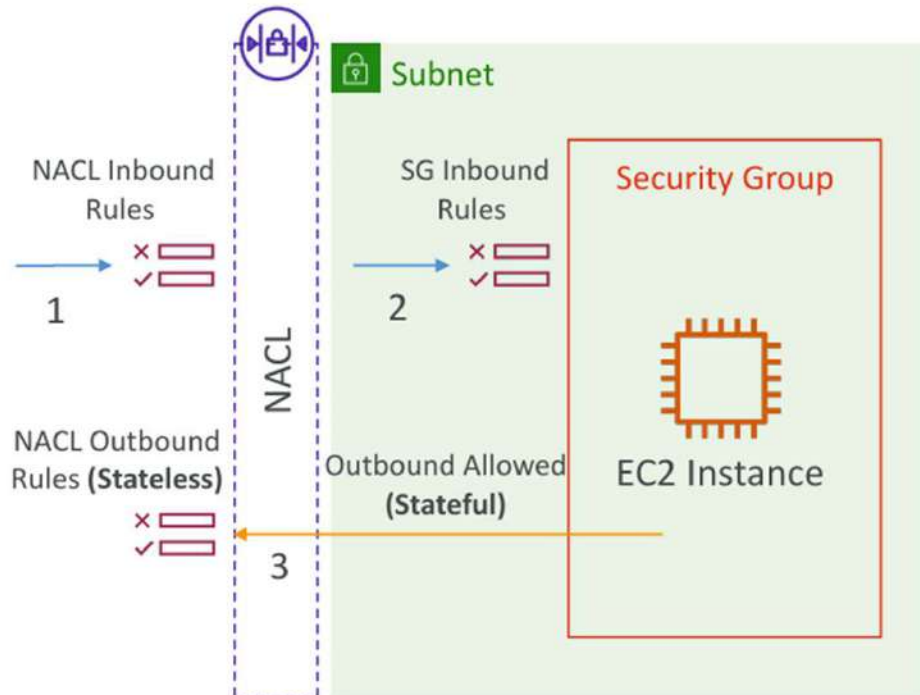
Routes Subnet associations Edge associations Route propagation Tags

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0e20925a019598...	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

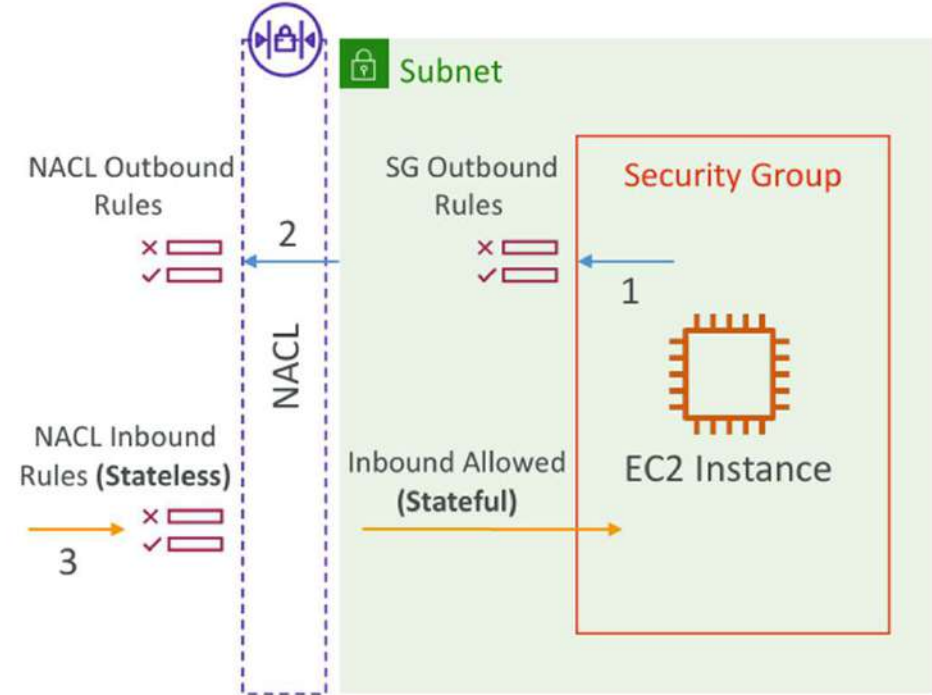
Active Now

Security Groups & NACLs

Incoming Request



Outgoing Request



<https://www.youtube.com/@techwithtwinkle>

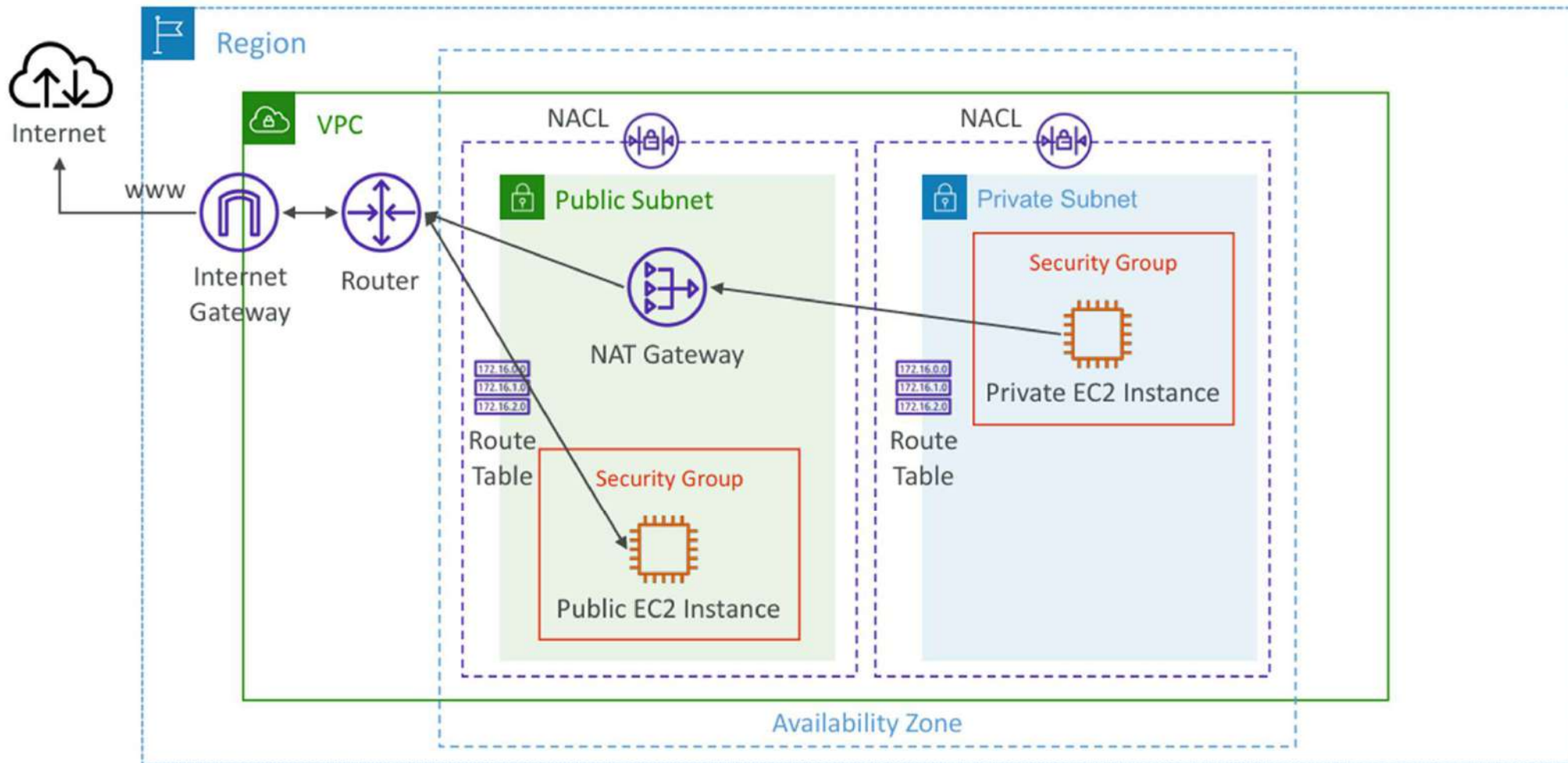
Network Access Control List (NACL)

- ▶ NACL are like a firewall which control traffic from and to subnets
- ▶ **One NACL per subnet**, new subnets are assigned the **Default NACL**
- ▶ **You define NACL Rules:**
 - Rules have a number (1-32766), higher precedence with a lower number
 - First rule match will drive the decision
 - Example: if you define #100 ALLOW 10.0.0.10/32 and #200 DENY 10.0.0.10/32, the IP address will be allowed because 100 has a higher precedence over 200
 - The last rule is an asterisk (*) and denies a request in case of no rule match
 - AWS recommends adding rules by increment of 100
- ▶ Newly created NACLs will deny everything
- ▶ NACL are a great way of blocking a specific IP address at the subnet level



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NACLs



<https://www.youtube.com/@techwithtwinkle>

Default NACL

- ▶ Accepts everything inbound/outbound with the subnets it's associated with
- ▶ Do NOT modify the Default NACL, instead create custom NACLs



Default NACL for a VPC that supports IPv4

Inbound Rules

Rule #	Type	Protocol	Port Range	Source	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

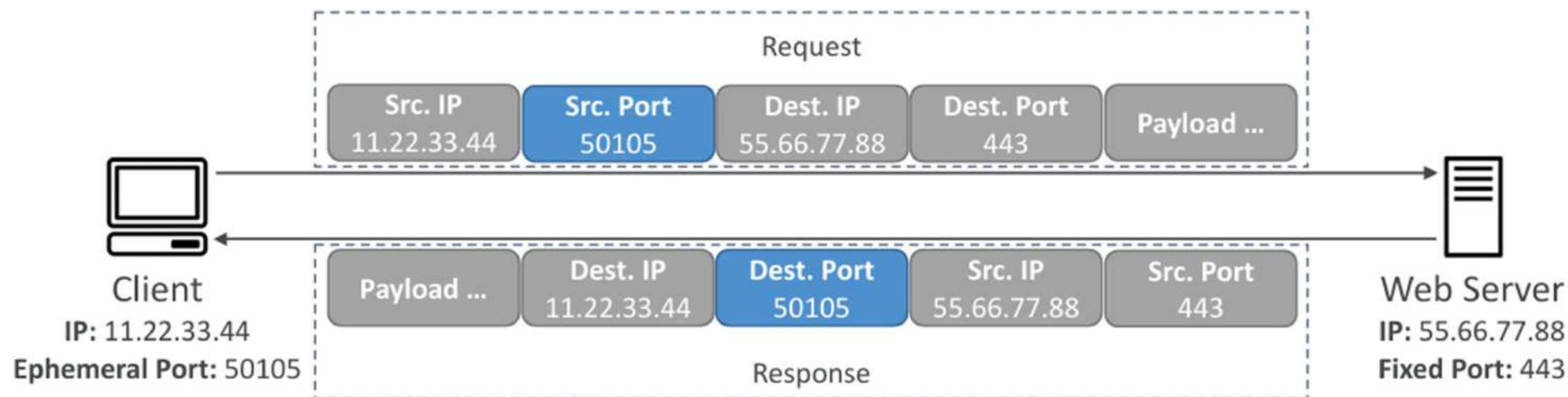
Outbound Rules

Rule #	Type	Protocol	Port Range	Destination	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

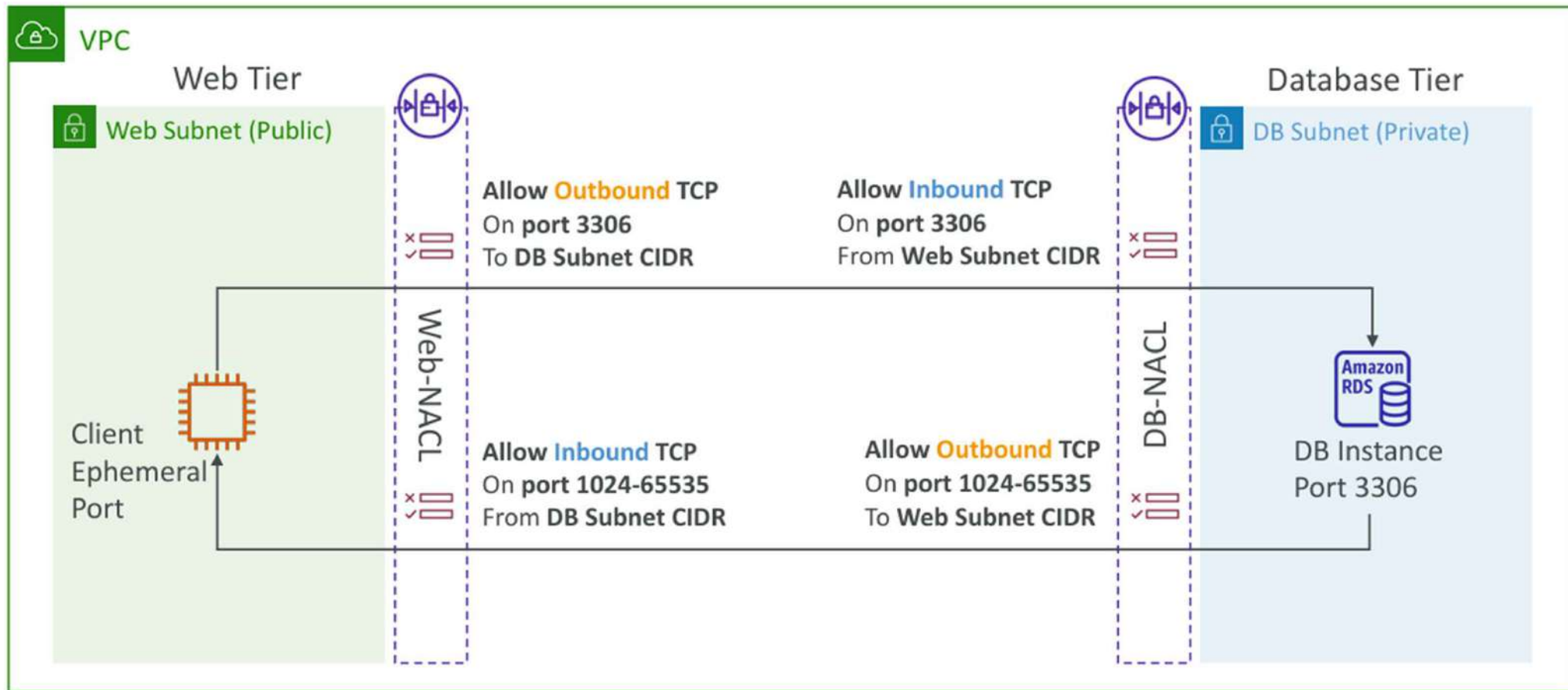
<https://www.youtube.com/@techwithtwinkle>

Ephemeral Ports

- ▶ For any two endpoints to establish a connection, they must use ports
- ▶ Clients connect to a defined port, and expect a response on an ephemeral port
- ▶ Different Operating Systems use different port ranges, examples:
 - IANA & MS Windows 10 → 49152 – 65535
 - Many Linux Kernels → 32768 – 60999



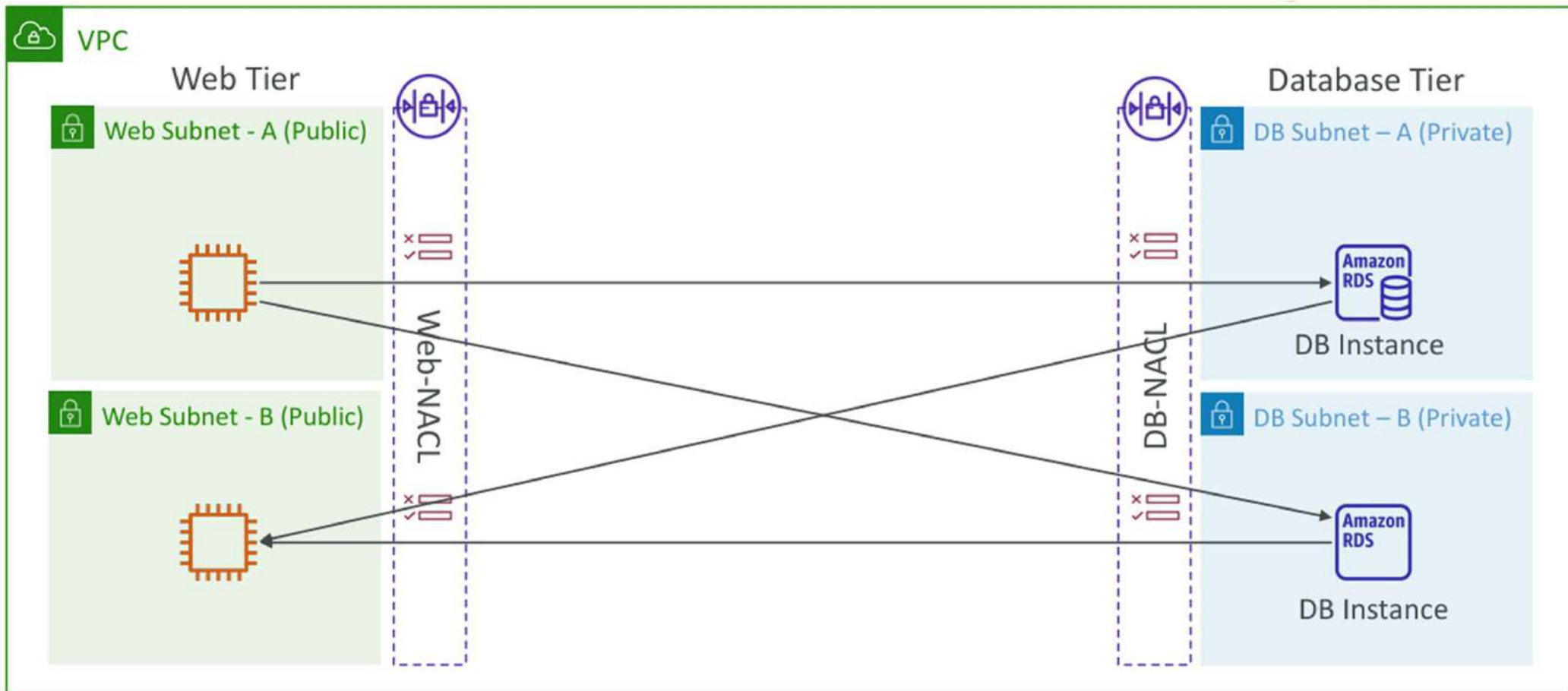
NACL with Ephemeral Ports



Control subnet traffic with network access control lists - Amazon Virtual Private Cloud

<https://www.youtube.com/@techwithtwinkle>

Create NACL rules for each target subnets CIDR



<https://www.youtube.com/@techwithtwinkle>

Security Group vs. NACLs

Security Group	NACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Stateful: return traffic is automatically allowed, regardless of any rules	Stateless: return traffic must be explicitly allowed by rules (think of ephemeral ports)
All rules are evaluated before deciding whether to allow traffic	Rules are evaluated in order (lowest to highest) when deciding whether to allow traffic, first match wins
Applies to an EC2 instance when specified by someone	Automatically applies to all EC2 instances in the subnet that it's associated with

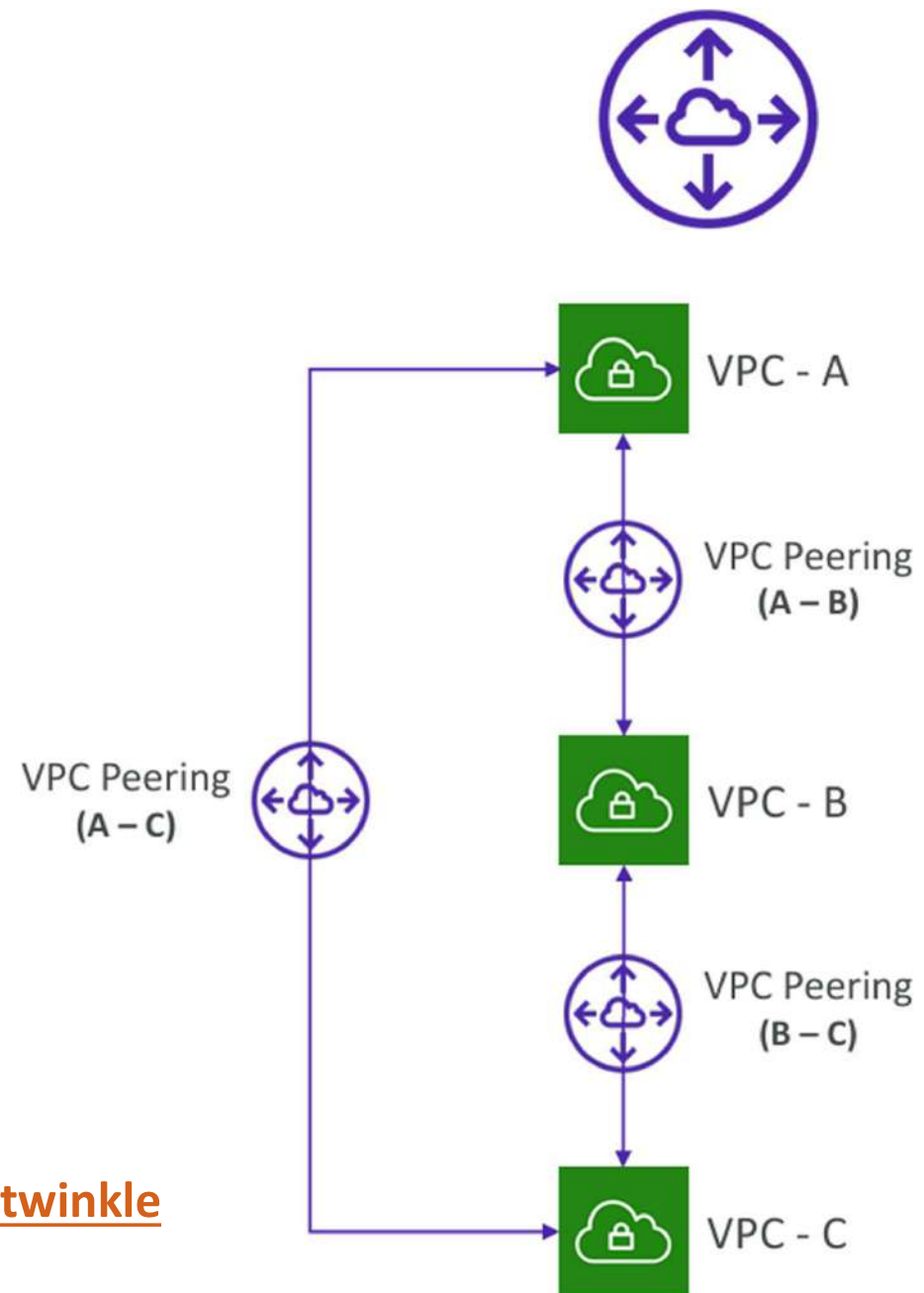
NACL Examples: <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

<https://www.youtube.com/@techwithtwinkle>

VPC Peering

- ▶ Privately connect two VPCs using AWS' network
- ▶ Make them behave as if they were in the same network
- ▶ Must not have overlapping CIDRs
- ▶ VPC Peering connection is **NOT transitive** (must be established for each VPC that need to communicate with one another)
- ▶ **You must update route tables in VPC's subnets each to ensure EC2 instances can communicate with each other**

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VPC Peering – Good to know

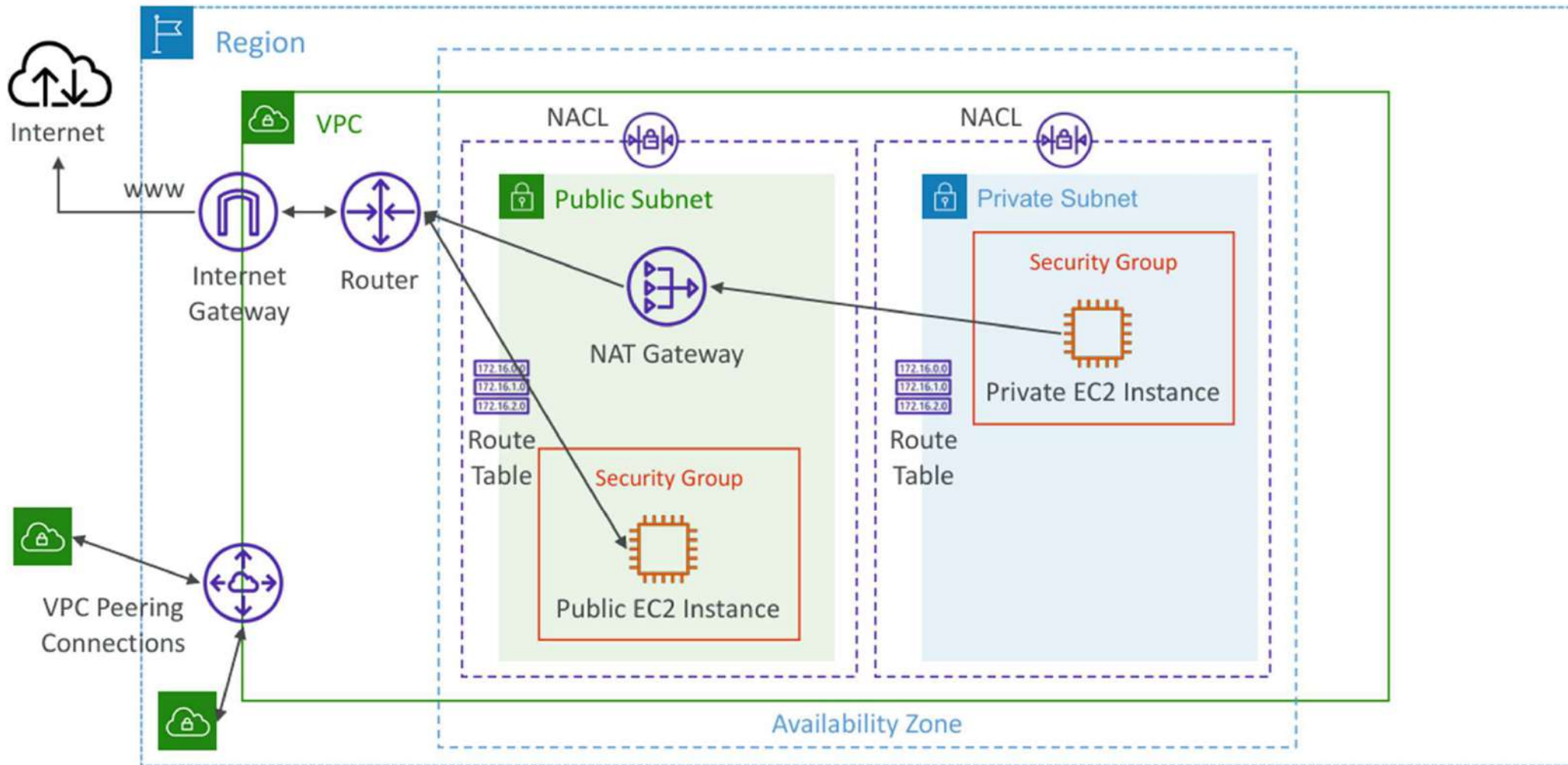
- ▶ You can create VPC Peering connection between VPCs in different AWS accounts/regions
- ▶ You can reference a security group in a peered VPC (works cross accounts – same region)

Type ▾	Protocol ▾	Port range ▾	Source ▾
HTTP	TCP	80	sg-04991f9af3473b939 / default
HTTP	TCP	80	/ sg-027ad1f7865d4be76

↑
Account ID

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VPC Peering



<https://www.youtube.com/@techwithtwinkle>

VPC Peering Hands On

VPC dashboard <

AWS Global View

Filter by VPC ▾

Virtual private cloud

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Your VPCs (2) [Info](#)

Last updated 4 minutes ago **Actions** ▾ **Create VPC**

Find VPCs by attribute or tag

☐	Name ▾	VPC ID ▾	State ▾	Block Public... ▾	IPv4 CIDR
☐	DemoVPC	vpc-06622afdfcf55aef7	✓ Available	⊖ Off	10.0.0.0/16
☐	-	vpc-0a506c34e326aff4f	✓ Available	⊖ Off	172.31.0.0/16

Select a VPC above

<https://www.youtube.com/@techwithtwinkle>

VPC > Your VPCs

Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections

Your VPCs (2) Info

Last updated 10 minutes ago

Actions Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR
☐	DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16
☐	-	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16

Click on peering connections

VPC > Peering connections

Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections

Peering connections Info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC
No peering connection found			

Create this

Create peering connection

Select a peering connection above

<https://www.youtube.com/@techwithtwinkle>

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Enter peering name

Select a local VPC to peer with

VPC ID (Requester)

Select Demo VPC

VPC CIDRs for vpc-06622afdfcf55aef7 (DemoVPC)

CIDR	Status	Status reason
10.0.0.0/16	✓ Associated	-

- ☒ My account
- ☐ Another account

Region

- ☒ This Region (ap-south-1)
- ☐ Another Region

VPC ID (Accepter)

Click here

Select This

Key

Value - optional

Remove

Add new tag

You can add 49 more tags.

<https://www.youtube.com/@techwithtwinkle>

vpc-0a506c34e326aff4f

VPC CIDRs for vpc-0a506c34e326aff4f

CIDR	Status	Status reason
172.31.0.0/16	✓ Associated	-

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X Q DemoPeeringConnection X Remove

Add new tag

You can add 49 more tags.

Click here

Create peering connection

<https://www.youtube.com/@techwithtwinkle>

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started Updated

✓ A VPC peering connection pcx-017810e9c3c8ebcb2 / DemoPeeringConnection has been requested.

pcx-017810e9c3c8ebcb2 / DemoPeeringConnection

ⓘ Pending acceptance

You can accept or reject this peering connection request using the 'Actions' menu. You have until Sunday, September 14, 2025 at 5:30:43 GMT+5:30 to accept or reject the request, otherwise it expires.

Details Info

Requester owner ID

235562991793

Peering connection ID

pcx-017810e9c3c8ebcb2

Status

⌚ Pending Acceptance by 235562991793

Expiration time

Sunday, September 14, 2025 at 21:05:43

Acceptor owner ID

235562991793

Requester VPC

vpc-06622afdfcf55aef7 / DemoVPC

Requester CIDRs

10.0.0.0/16

Requester Region

Mumbai (ap-south-1)

VPC Peering connection ARN

arn:aws:ec2:ap-south-1:235562991793:vpc-peering-connection/pcx-017810e9c3c8ebcb2

Acceptor VPC

vpc-0a506c34e326aff4f

Acceptor CIDRs

-

Click here

Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections

▼ **Security**
Network ACLs
Security groups

▼ **PrivateLink and Lattice**
Getting started **Updated**

VPC > Peering connections > pcx-017810e9c3c8ebcb2

✓ A VPC peering connection pcx-017810e9c3c8ebcb2 / DemoPeeringConnection has been requested.

pcx-017810e9c3c8ebcb2 / DemoPeeringConnection

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Sunday, September 14, 2025 at 21:05:43 GMT+5:30 to accept or reject the request, otherwise it expires.

Details Info

Requester owner ID 235562991793	Accepter owner ID 235562991793
Peering connection ID pcx-017810e9c3c8ebcb2	Requester VPC vpc-06622afdfcf55aef7 / DemoVPC
Status Pending Acceptance by 235562991793	Requester CIDRs 10.0.0.0/16
Expiration time Sunday, September 14, 2025 at 21:05:43	Requester Region Mumbai (ap-south-1)

Accepter VPC
vpc-0a506c34e326aff4f

Accepter CIDRs
-

Actions

- Accept request
- Reject request
- Edit DNS settings
- Manage tags
- Delete peering connection

Click here

Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections

▼ **Security**
Network ACLs
Security groups

▼ **PrivateLink and Lattice**
Getting started **Updated**

VPC > Peering connections > pcx-017810e9c3c8ebcb2

✓ A VPC peering connection pcx-017810e9c3c8ebcb2 / DemoPeeringConnection has been requested.

pcx-017810e9c3c8ebcb2 / DemoPeeringConnection

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Sunday, September 14, 2025 at 21:05:43 GMT+5:30 to accept or reject the request, otherwise it expires.

Details Info

Requester VPC vpc-06622afdfcf55aef7 / DemoVPC	Accepter VPC vpc-0a506c34e326aff4f
Accepter CIDRs -	Requester Region Mumbai (ap-south-1)
Requester owner ID 235562991793(This account)	Accepter owner ID 235562991793(This account)

Requester CIDRs
10.0.0.0/16

Accepter Region
Mumbai (ap-south-1)

Status
Pending Acceptance by 235562991793

Expiration time
Sunday, September 14, 2025 at 21:05:43

Requester Region
Mumbai (ap-south-1)

Accepter CIDRs
-

Accept VPC peering connection request Info

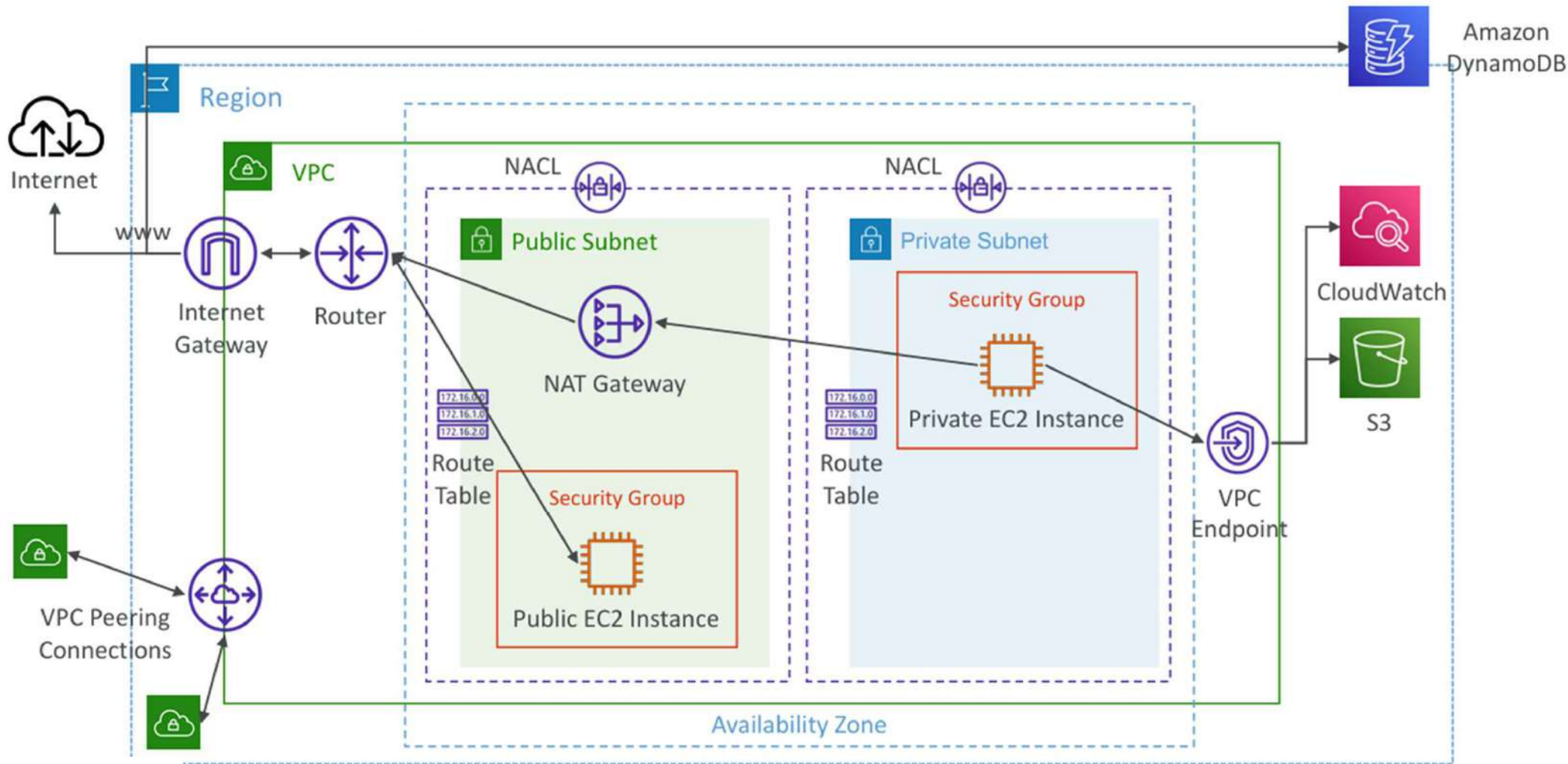
Are you sure you want to accept this VPC peering connection request? (pcx-017810e9c3c8ebcb2 / DemoPeeringConnection)

Click here

Accept request

<https://www.youtube.com/@techwithtwinkle>

VPC Endpoints

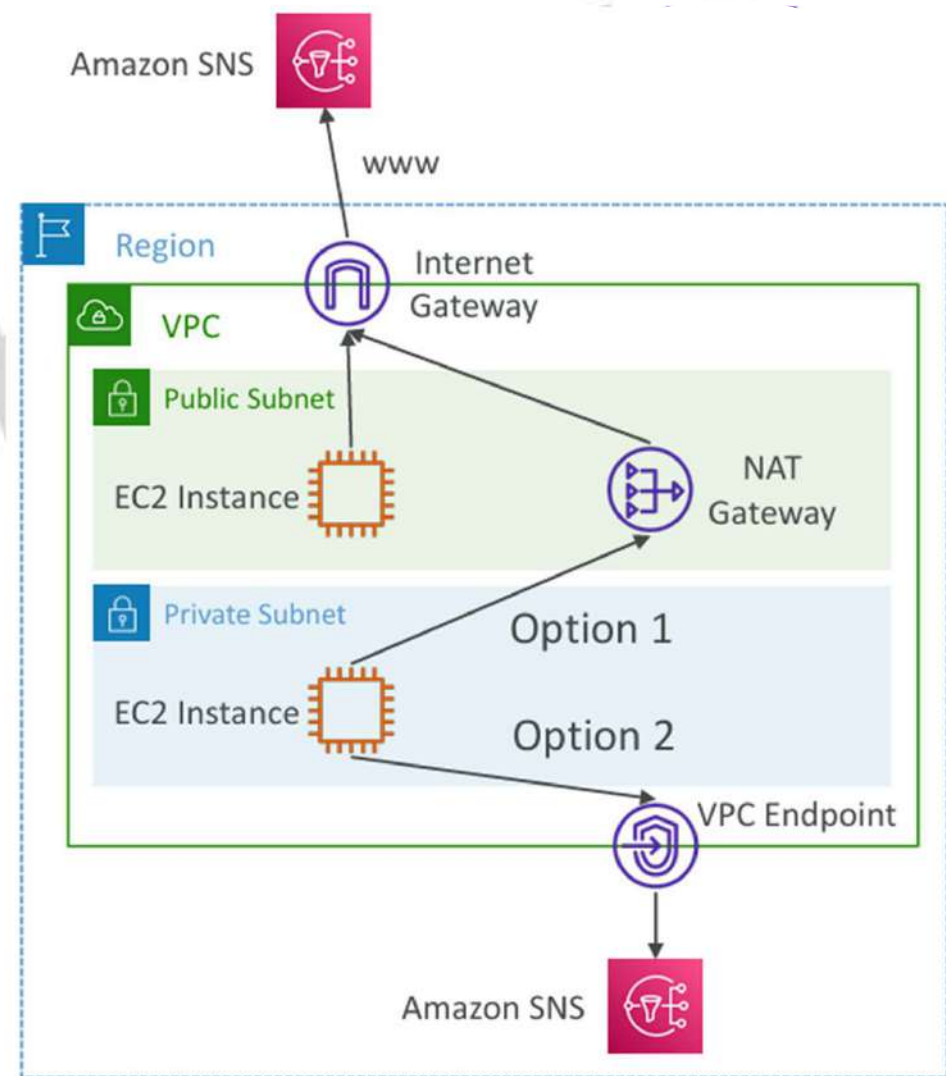


<https://www.youtube.com/@techwithtwinkle>

VPC Endpoints (AWS PrivateLink)



- ▶ Every AWS service is publicly exposed (public URL)
- ▶ VPC Endpoints (powered by AWS PrivateLink) allows you to connect to AWS services using a private network instead of using the public Internet
- ▶ They're redundant and scale horizontally
- ▶ They remove the need of IGW, NATGW, ... to access AWS Services
- ▶ In case of issues:
 - Check DNS Setting Resolution in your VPC
 - Check Route Tables



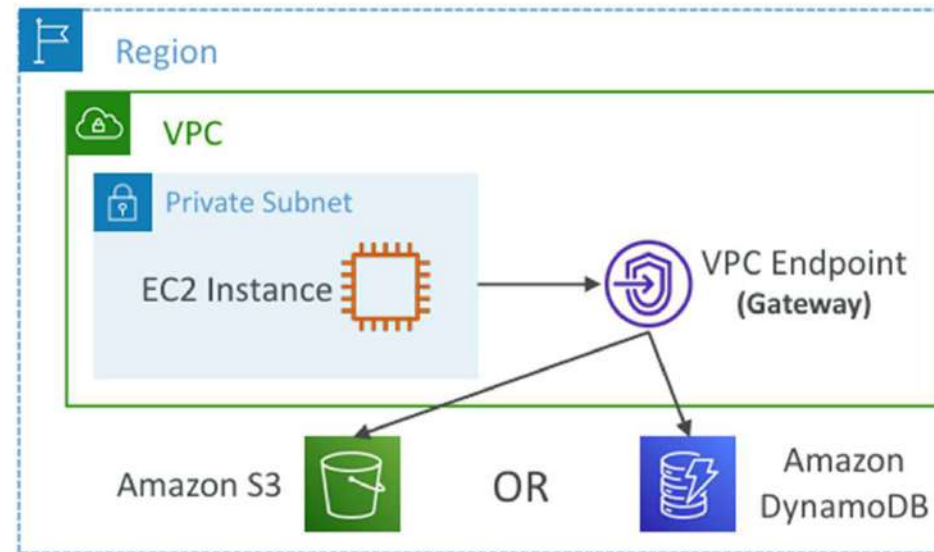
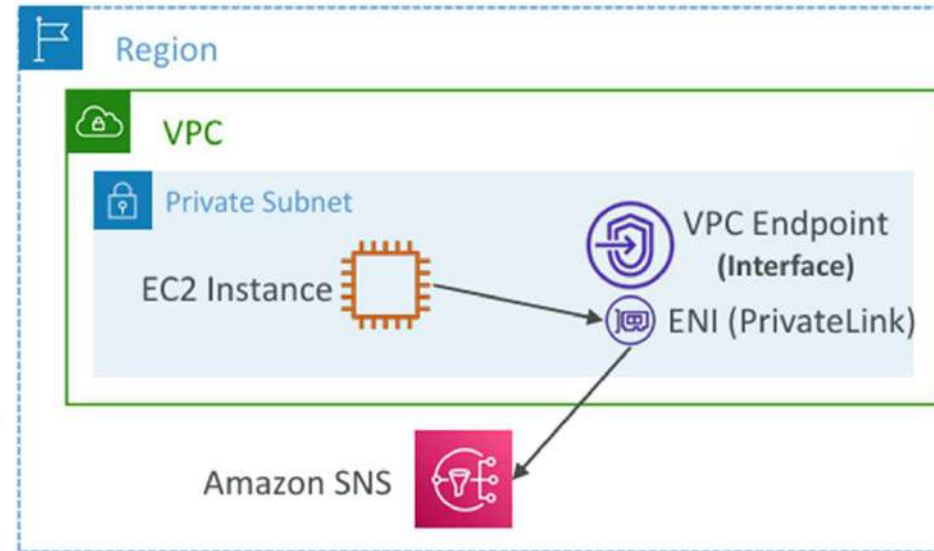
Types of Endpoints

Interface Endpoints (powered by PrivateLink)

- ▶ Provisions an ENI (private IP address) as an entry point (must attach a Security Group)
- ▶ Supports most AWS services
- ▶ \$ per hour + \$ per GB of data processed

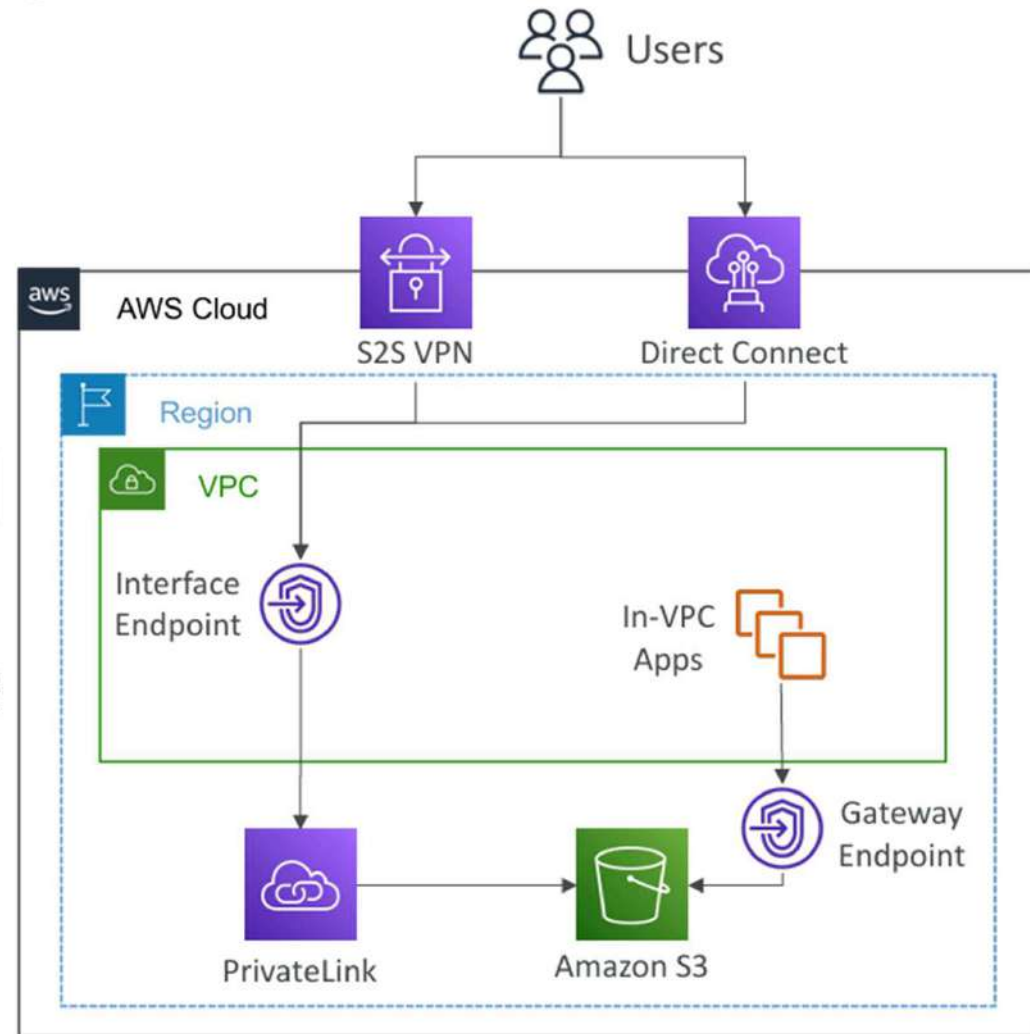
Gateway Endpoints

- ▶ Provisions a gateway and must be used as a target in a route table (does not use security groups)
- ▶ Supports both S3 and DynamoDB
- ▶ Free



Gateway or Interface Endpoint for S3?

- ▶ **Gateway is most likely going to be preferred all the time at the exam**
- ▶ **Cost: free for Gateway, \$ for interface endpoint**
- ▶ **Interface Endpoint is preferred access is required from on premises (Site to Site VPN or Direct Connect), a different VPC or a different region**



VPC Endpoints Hands On

VPC > Peering connections > pcx-017810e9c3c8ebcb2

✔ Your VPC peering connection (pcx-017810e9c3c8ebcb2 | DemoPeeringConnection) has been established.
To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.
[Info](#) [Modify my route tables now](#) ✕

pcx-017810e9c3c8ebcb2 / DemoPeeringConnection [Actions](#) ▼

Details [Info](#)

Requester owner ID 235562991793	Acceptor owner ID 235562991793	VPC Peering connection ARN arn:aws:ec2:ap-south-1:235562991793:vpc-peering-connection/pcx-017810e9c3c8ebcb2
Peering connection ID pcx-017810e9c3c8ebcb2	Requester VPC vpc-06622afdfcf55aef7 / DemoVPC	Acceptor VPC vpc-0a506c34e326aff4f
Status ✔ Active	Requester CIDRs 10.0.0.0/16	Acceptor CIDRs 172.31.0.0/16
Expiration time	Requester Region us-east-1	

Click on Endpoints

<https://www.youtube.com/@techwithtwinkle>

VPC > Endpoints

NAT gateways

Peering connections

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started [Updated](#)

[Endpoints](#) [Updated](#)

Endpoint services

Service networks [Updated](#)

Lattice services

Resource configurations [New](#)

Resource gateways [New](#)

Target groups

Endpoints [Info](#)

Find endpoints by attribute or tag

☐	Name	VPC endpoint ID	Endpoint type
No endpoint found			

←

Actions [Create endpoint](#)

1 > ⚙

Click on create Endpoint

Select an endpoint

<https://www.youtube.com/@techwithtwinkle>

VPC > [Endpoints](#) > Create endpoint

Create endpoint [Info](#)

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

Type [Info](#)
Select a category

Select This

☒ **AWS services**
Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint

☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint

☐ **EC2 Instance Connect Endpoint**
An elastic network interface that allows you to connect to resources in a private subnet

☐ **Resources**
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS

☐ **Service networks**
Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink

Enter Endpoint name

Network LoadBalancer (NLB) service with an Interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint

Services (1/263)

Search

1 2 3 4 5 6 7 ... 27

Service Name	Owner	Type	Service Region
☒ api.aws.ap-south-1.cassandra-streams	amazon	Interface	ap-south-1
☐ aws.api.ap-south-1.emr-service-cell01	amazon	Interface	ap-south-1
☐ aws.api.ap-south-1.kendra-ranking	amazon	Interface	ap-south-1
☐ aws.api.ap-south-1.resource-explorer-2	amazon	Interface	ap-south-1
☐ aws.sagemaker.ap-south-1.experiments	amazon	Interface	ap-south-1
☐ aws.sagemaker.ap-south-1.notebook	amazon	Interface	ap-south-1
☐ aws.sagemaker.ap-south-1.partner-app	amazon	Interface	ap-south-1

Select this

Network settings

Select the VPC in which to create the endpoint

VPC

Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

Select a VPC

Q

vpc-06622afdfcf55aef7 (DemoVPC)

10.0.0.0/16

vpc-0a506c34e326aff4f

172.31.0.0/16

Select this

DNS record IP type

- ☒ IPv4
- ☐ IPv6
- ☐ Dualstack

<https://www.youtube.com/@techwithtwinkle>

- Dualstack
- Service defined

Select AZ & Private subnets



Subnets (2/3) [Info](#)

☒	Availability Zone	Subnet ID	Designate IP addresses	IPv4 address	IPv6 address
☒	aps1-az1 (ap-south-1a)	subnet-016e736c2ef7867b1	☐		
☒	aps1-az3 (ap-south-1b)	subnet-0dbe813169aab11ee	☐		
☐	aps1-az2 (ap-south-1c)	No subnet available	☐		

IP address type

- ☒ IPv4
- ☐ IPv6
- ☐ Dualstack

<https://www.youtube.com/@techwithtwinkle>

JSON Line 1, Column 1 ✖ Errors: 0 ⚠ Warnings: 0

Code blocks in square or curly brackets can be expanded and collapsed with CTRL+Enter or Command+Enter.

Tags

No tags associated with the resource.

[Add new tag](#)

You can add 50 more tags.

Click here



Create endpoint

Lambda in VPC accessing DynamoDB

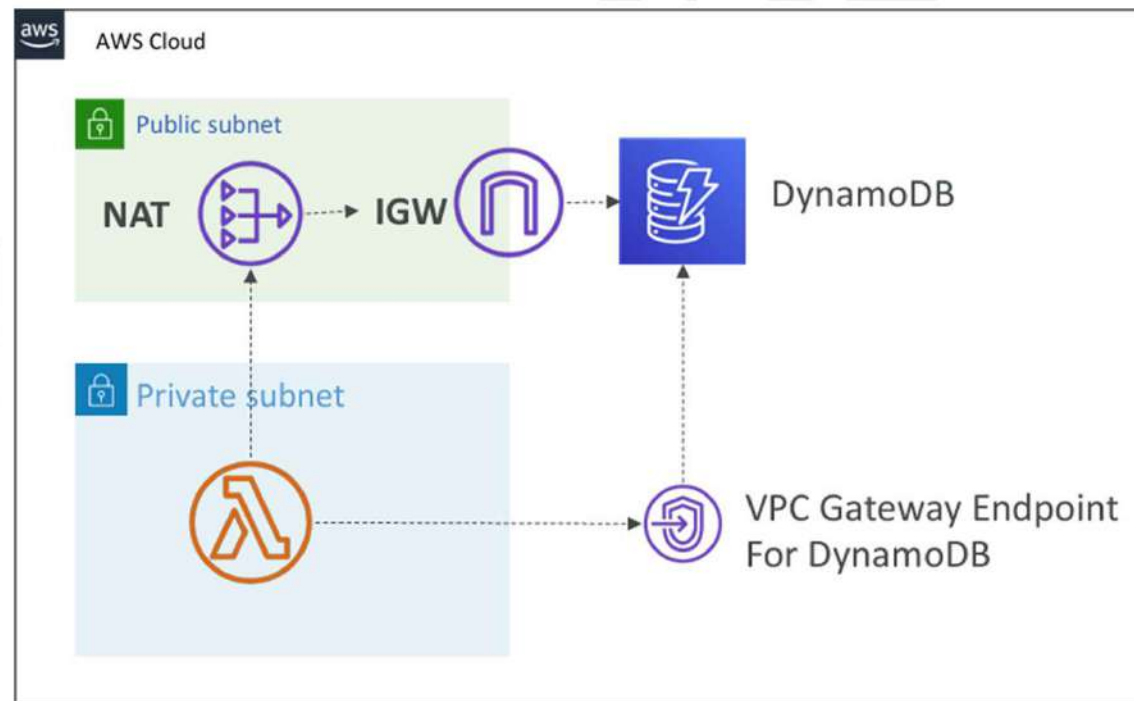
- ▶ DynamoDB is a public service from AWS

Option 1: Access from the public internet

- Because Lambda is in a VPC, it needs a NAT Gateway in a public subnet and an internet gateway

- ▶ **Option 2 (better & free): Access from the private VPC network**

- Deploy a VPC Gateway endpoint for DynamoDB
- Change the Route Tables



<https://www.youtube.com/@techwithtwinkle>

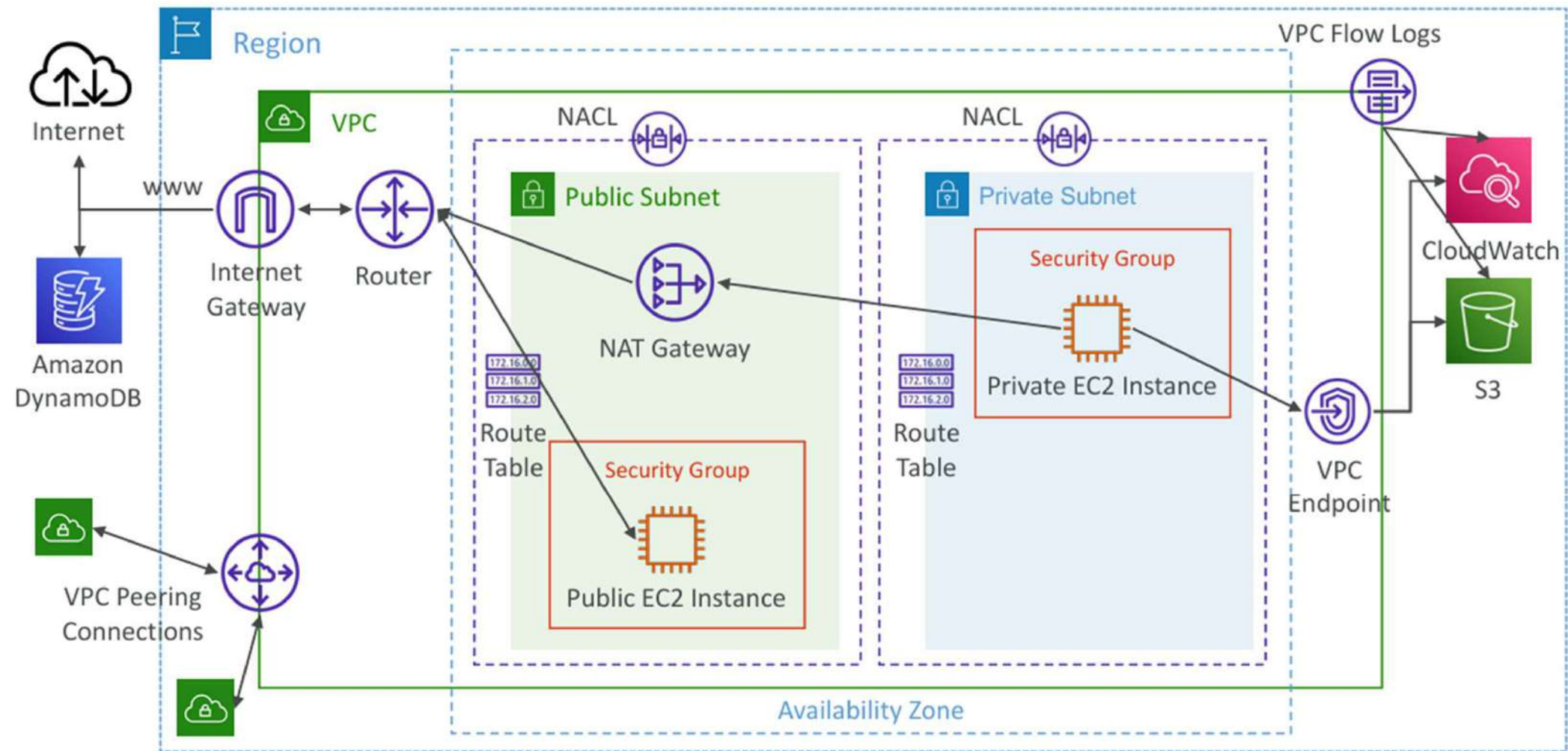
VPC Flow Logs



- ▶ Capture information about IP traffic going into your interfaces:
 - VPC Flow Logs
 - Subnet Flow Logs
 - Elastic Network Interface (ENI) Flow Logs
- ▶ Helps to monitor & troubleshoot connectivity issues
- ▶ Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose
- ▶ Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW, Transit Gateway...

<https://www.youtube.com/@techwithtwinkle>

VPC Flow Logs



<https://www.youtube.com/@techwithtwinkle>

VPC Flow Logs Syntax

version	interface-id	dstaddr	dstport	packets	start	action	account-id	srcaddr	srcport	protocol	bytes	end	log-status
2	123456789010	eni-1235b8ca123456789	172.31.16.139	172.31.16.21	20641	22	6	20	4249	1418530010	1418530070	ACCEPT	OK
2	123456789010	eni-1235b8ca123456789	172.31.9.69	172.31.9.12	49761	3389	6	20	4249	1418530010	1418530070	REJECT	OK

- ▶ srcaddr & dstaddr – help identify problematic IP protocol
- ▶ srcport & dstport – help identify problematic ports bytes end
- ▶ Action – success or failure of the request due to Security Group / NACL
- ▶ Can be used for analytics on usage patterns, or malicious behavior
- ▶ Query VPC flow logs using Athena on S3 or CloudWatch Logs Insights

- ▶ **Flow Logs examples:**

<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html>

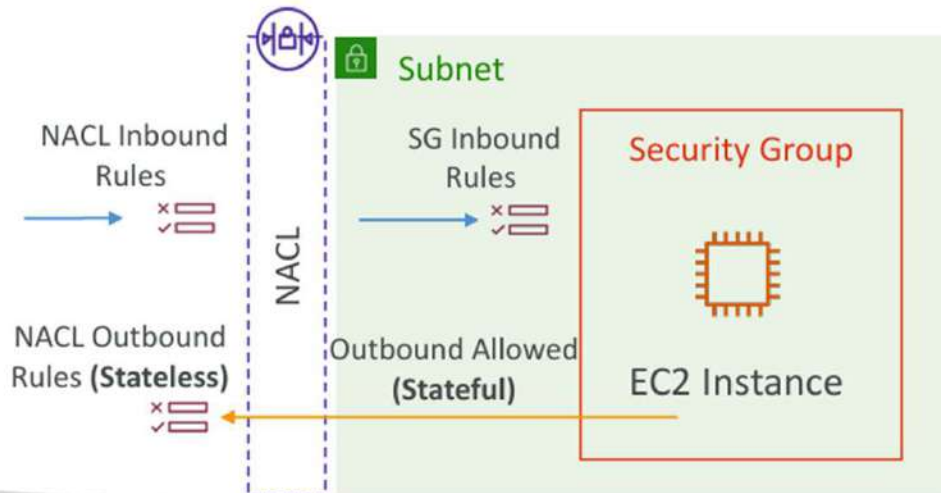
<https://www.youtube.com/@techwithtwinkle>

VPC Flow Logs – Troubleshoot SG & NACL issues

Look at the “ACTION” field

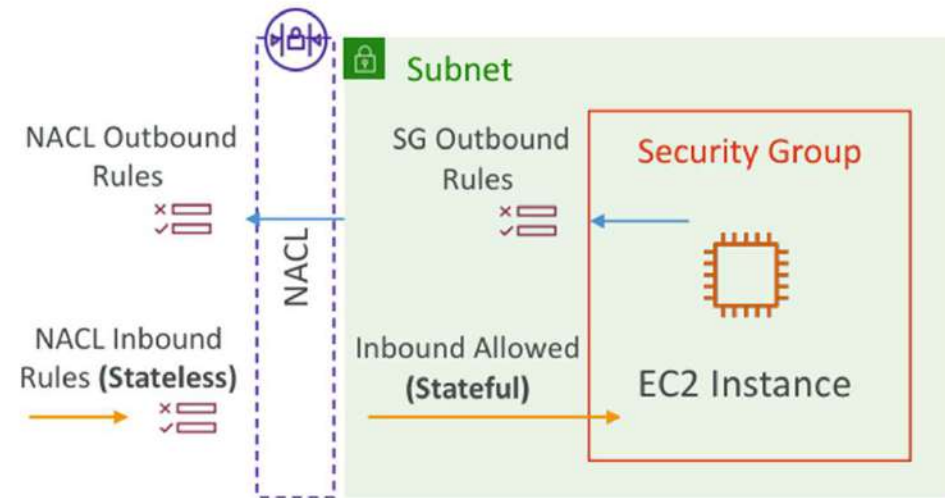
Incoming Requests

- Inbound REJECT => NACL or SG
- Inbound ACCEPT, Outbound REJECT => NACL



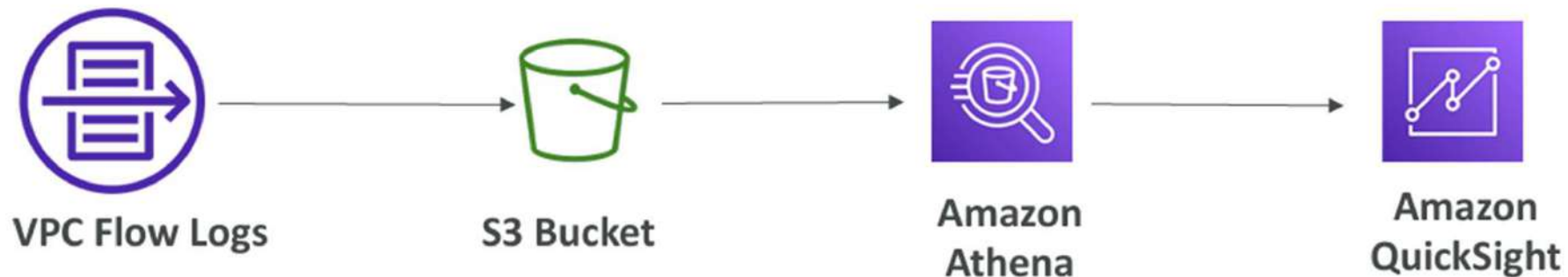
Outgoing Requests

- Outbound REJECT => NACL or SG
- Outbound ACCEPT, Inbound REJECT => NACL



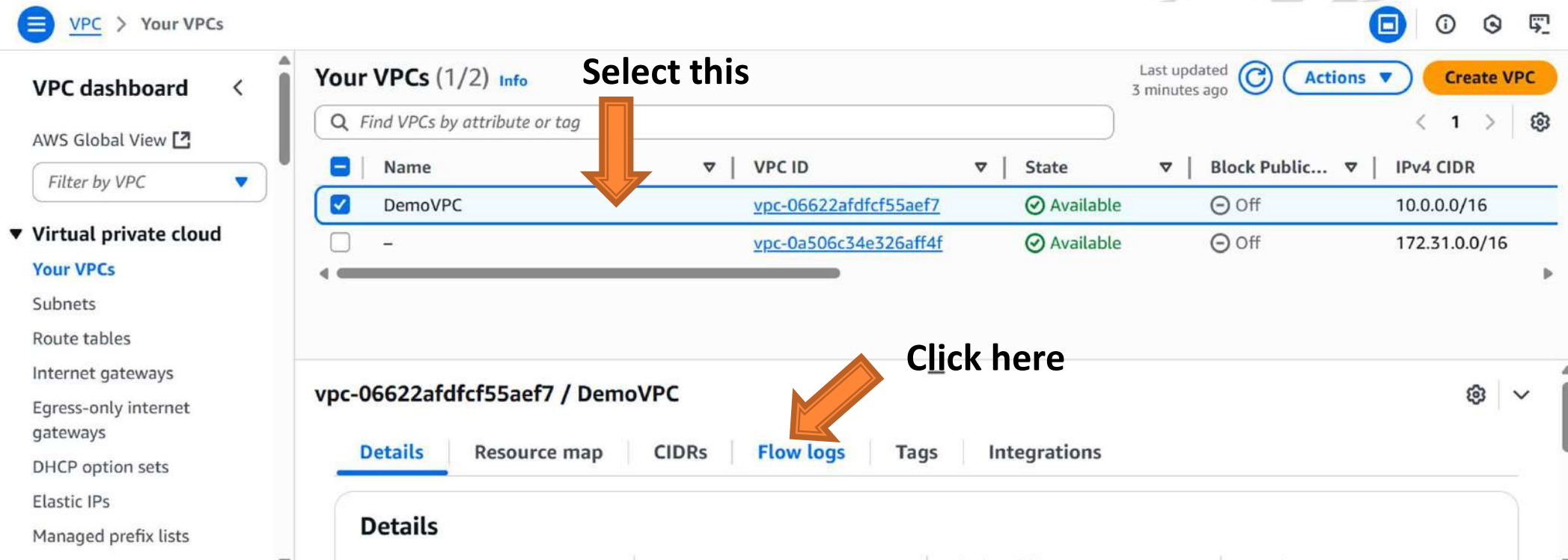
<https://www.youtube.com/@techwithtwinkle>

VPC Flow Logs – Architectures



<https://www.youtube.com/@techwithtwinkle>

VPC Flow LOGS – Hands On



The screenshot displays the AWS Management Console interface for VPCs. On the left, the 'VPC dashboard' sidebar is visible, with 'Your VPCs' selected. The main content area shows a list of VPCs under the heading 'Your VPCs (1/2)'. An orange arrow points to the 'DemoVPC' entry in the table. Below the table, the 'vpc-06622afdfcf55aef7 / DemoVPC' details page is shown, with the 'Flow logs' tab selected. Another orange arrow points to the 'Flow logs' tab.

Select this

Name	VPC ID	State	Block Public...	IPv4 CIDR
☒ DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16
☐ -	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16

Click here

Flow logs

<https://www.youtube.com/@techwithtwinkle>

VPC > Your VPCs

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Your VPCs (1/2) Info

Last updated 4 minutes ago

Find VPCs by attribute or tag

	Name	VPC ID	State	Block Public...	IPv4 CIDR
☒	DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16
☐	-	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16

vpc-06622afdfcf55aef7 / DemoVPC

Details | Resource map | CIDRs | **Flow logs** | Tags | Integrations

Flow logs

Actions | Create flow log

Click here

VPC > Your VPCs > Create flow logs

Flow log settings

Name - optional

DemoS3Flow Logs

Enter name

Filter

The type of traffic to capture (accepted traffic only, rejected traffic only, or all traffic).

☐ Accept

☐ Reject

☒ All

Select this

Maximum aggregation interval | Info

The maximum interval of time during which a flow of packets is captured and aggregated into a flow log record.

☒ 10 minutes

☐ 1 minute

Select this

Destination

The destination to which to publish the flow log data.

☒ Send to CloudWatch Logs

<https://www.youtube.com/@techwithtwinkle>

☒ 10 minutes

☐ 1 minute

Destination

The destination to which to publish the flow log data.

☐ Send to CloudWatch Logs

☒ Send to an Amazon S3 bucket

☐ Send to Amazon Data Firehose in the same account

☐ Send to Amazon Data Firehose in a different account

Select this

Select Bucket

S3 bucket ARN

The ARN of the Amazon S3 bucket to which the flow log is published. You can specify a specific folder in the bucket using the `arn:aws:s3:::[bucket-name]/[folder_name]/` format.

Create S3 bucket [↗](#)

Must be in the format: `arn:aws:s3:::[bucket-name]`

ⓘ Please note, a resource-based policy will be created for you and attached to the target bucket.

Log record format

Specify the fields to include in the flow log record.

☒ AWS default format

☐ Enable

<https://www.youtube.com/@techwithtwinkle>

Partition logs by time [Info](#)

Partition your logs per hour to reduce your query costs and get faster response if you have a large volume of logs and typically run queries targeted to a specific hour timeframe.

☒ Every 24 hours (default)

☐ Every 1 hour (60 minutes)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

×

Value - optional

×

Remove tag

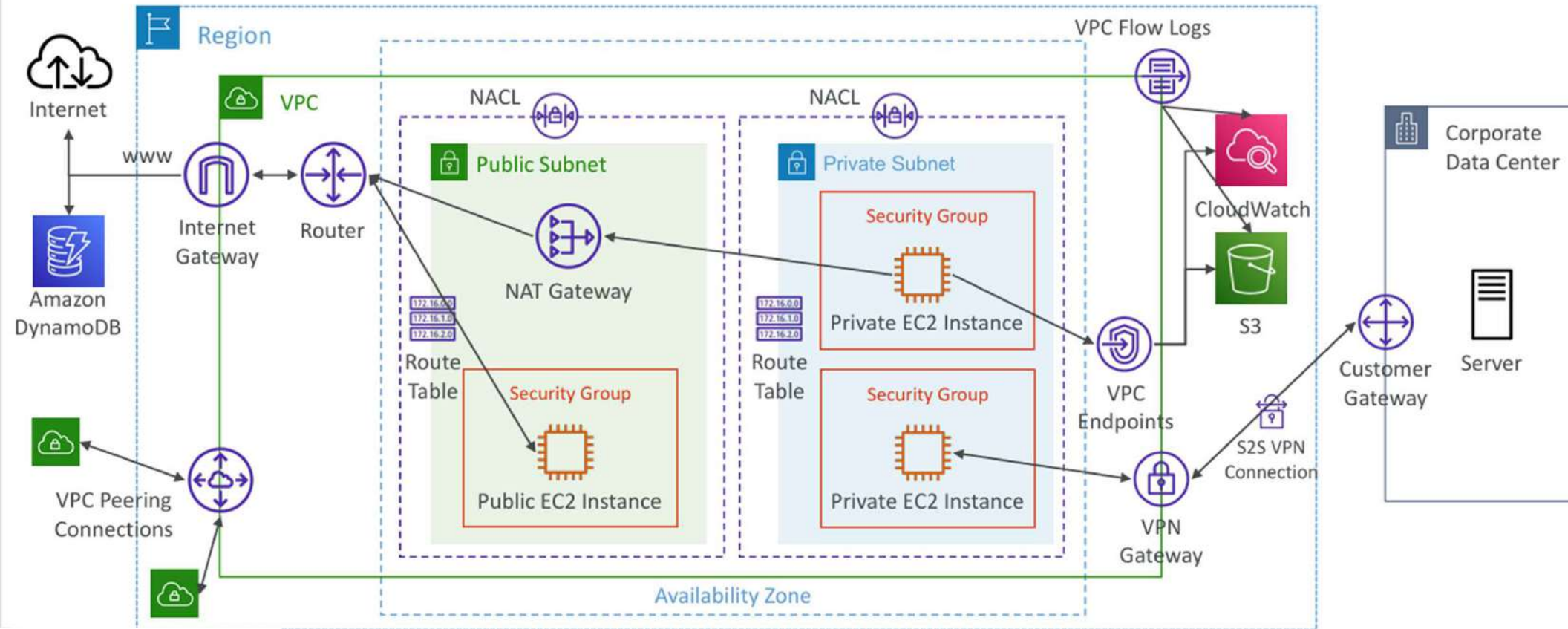
Add tag

You can add 49 more tags

Click here

Create flow log

AWS Site-to-Site VPN



<https://www.youtube.com/@techwithtwinkle>

AWS Site-to-Site VPN



Virtual Private Gateway (VGW)

- ▶ VPN concentrator on the AWS side of the VPN connection
- ▶ VGW is created and attached to the VPC from which you want to create the Site-to-Site VPN connection
- ▶ Possibility to customize the ASN (Autonomous System Number)

Customer Gateway (CGW)

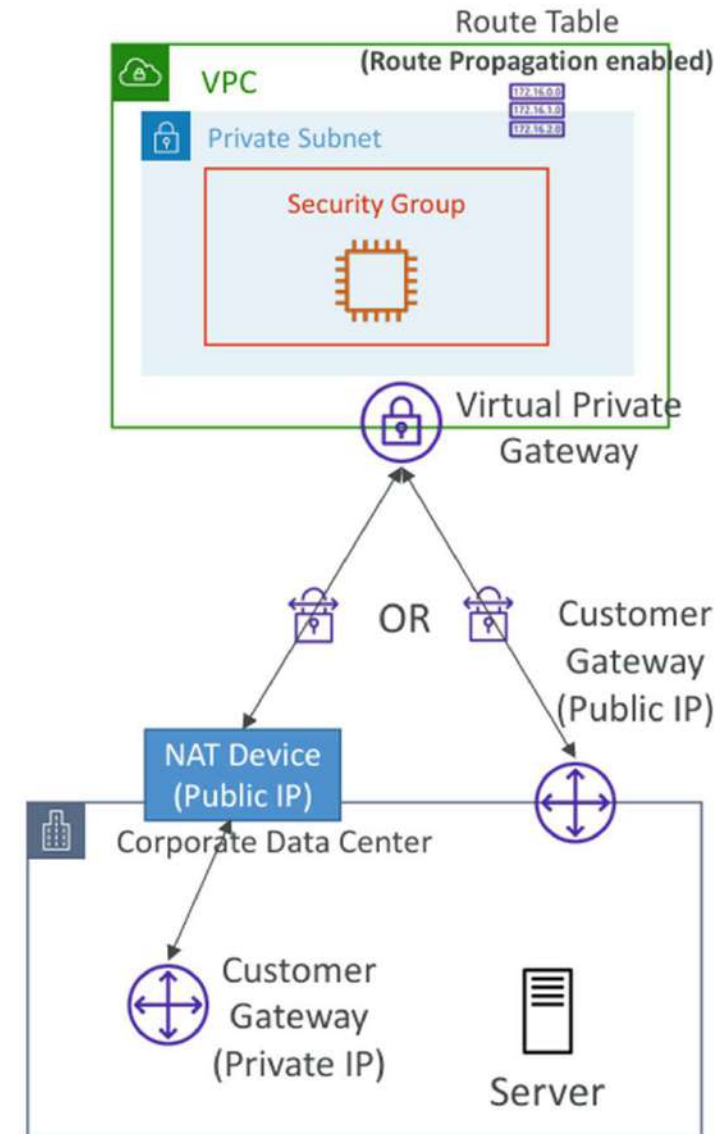
- ▶ Software application or physical device on customer side of the VPN connection
- ▶ <https://docs.aws.amazon.com/vpn/latest/s2svpn/your-cgw.html#DevicesTested>

<https://www.youtube.com/@techwithtwinkle>

Site-to-Site VPN Connections

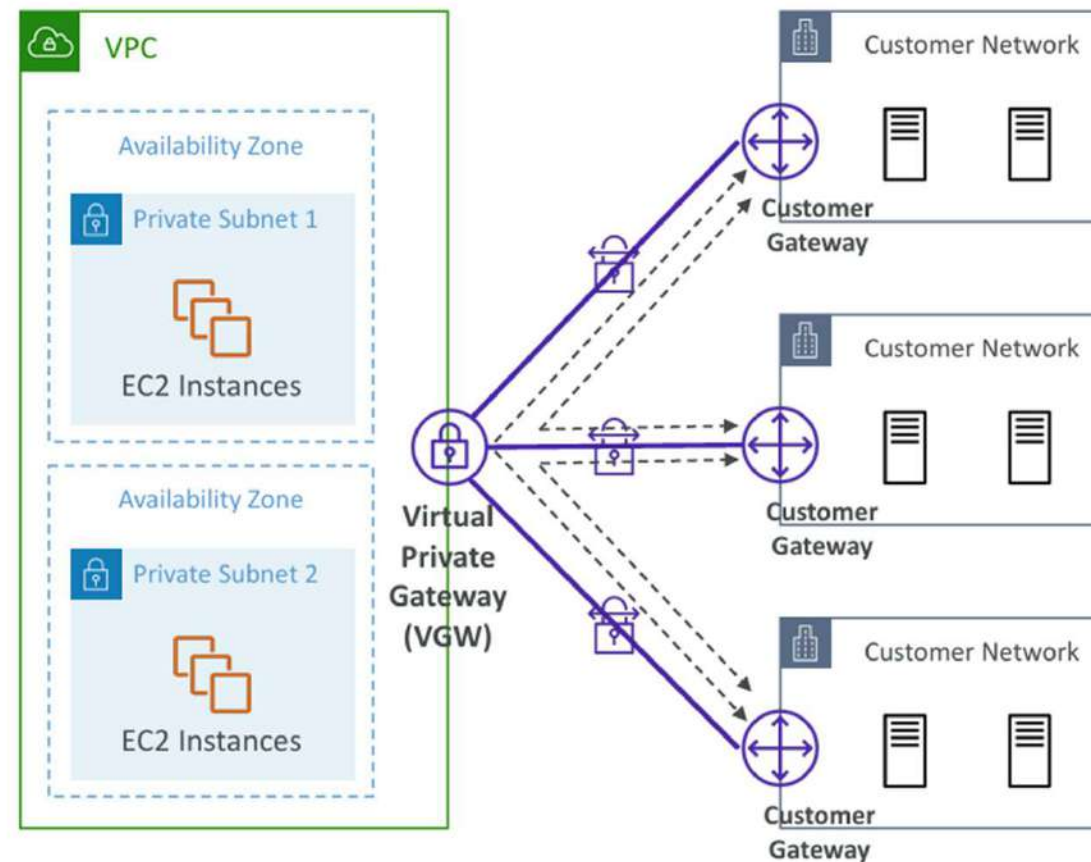
Customer Gateway Device (On-premises)

- ▶ **What IP address to use?**
 - Public Internet-routable IP address for your Customer Gateway device
 - If it's behind a NAT device that's enabled for NAT traversal (NAT-T), use the public IP address of the NAT device
- ▶ Important step: enable Route Propagation for the Virtual Private Gateway in the route table that is associated with your subnets
- ▶ If you need to ping your EC2 instances from on-premises, make sure you add the ICMP protocol on the inbound of your security groups



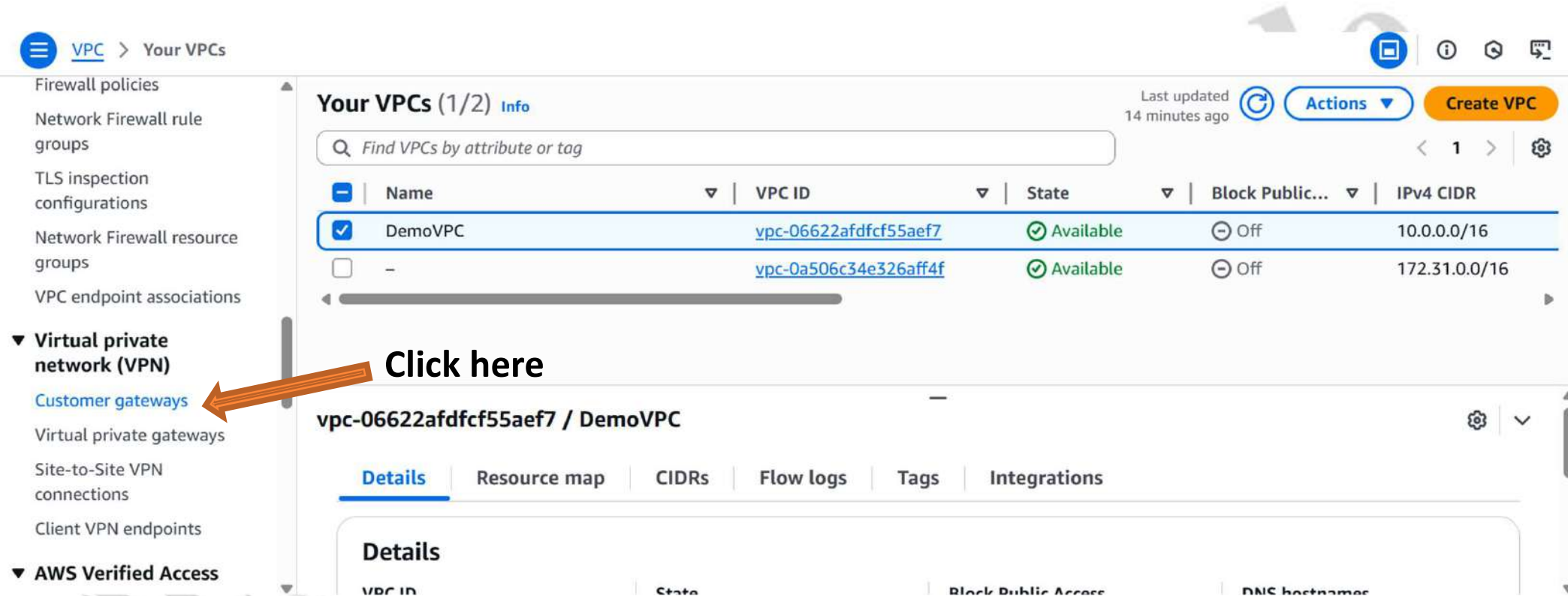
AWS VPN CloudHub

- ▶ Provide secure communication between multiple sites, if you have multiple VPN connections
- ▶ Low-cost hub-and-spoke model for primary or secondary network connectivity between different locations (VPN only)
- ▶ It's a VPN connection so it goes over the public Internet
- ▶ To set it up, connect multiple VPN connections on the same VGW, setup dynamic routing and configure route tables



<https://www.youtube.com/@techwithtwinkle>

Site to site VPN, Virtual Private Gateway, Customer Gateway Hands on



The screenshot shows the AWS Management Console interface. On the left sidebar, under the 'Virtual private network (VPN)' section, the 'Customer gateways' link is highlighted with an orange arrow pointing to it. The main content area displays 'Your VPCs (1/2)' with a table listing two VPCs. Below the table, the details for 'vpc-06622afdfcf55aef7 / DemoVPC' are shown, including tabs for Details, Resource map, CIDRs, Flow logs, Tags, and Integrations.

Virtual private network (VPN)

- Customer gateways
- Virtual private gateways
- Site-to-Site VPN connections
- Client VPN endpoints

AWS Verified Access

Your VPCs (1/2)

Find VPCs by attribute or tag

	Name	VPC ID	State	Block Public...	IPv4 CIDR
☒	DemoVPC	vpc-06622afdfcf55aef7	Available	Off	10.0.0.0/16
☐	-	vpc-0a506c34e326aff4f	Available	Off	172.31.0.0/16

vpc-06622afdfcf55aef7 / DemoVPC

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

Details

VPC ID	State	Block Public Access	DNS hostnames
--------	-------	---------------------	---------------

<https://www.youtube.com/@techwithtwinkle>

Navigation: VPC > Customer gateways

Left sidebar menu:

- Firewall policies
- Network Firewall rule groups
- TLS inspection configurations
- Network Firewall resource groups
- VPC endpoint associations
- Virtual private network (VPN)
 - Customer gateways**
 - Virtual private gateways
 - Site-to-Site VPN connections
 - Client VPN endpoints
- AWS Verified Access

Customer gateways info

Find resource by attribute or tag

Name	Customer gateway ID	State	BGP ASN	IP address
No customer gateways				

You do not have any customer gateways in this region

Create customer gateway

Click here

<https://www.youtube.com/@techwithtwinkle>

Navigation: VPC > Customer gateways > Create customer gateway

Details

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

customer-gateway-01

Value must be 256 characters or less in length.

BGP ASN Info
The ASN of your customer gateway device.

65000

Value must be in 1 - 4294967294 range.

IP address Info
Specify the IP address for your customer gateway device's external interface.

192.0.2.1

Certificate ARN - optional
The ARN of a private certificate provisioned in AWS Certificate Manager (ACM).

Select certificate ARN

Device - optional

Enter name

Enter IP Address

The ARN of a private certificate provisioned in AWS Certificate Manager (ACM).

Select certificate ARN

Device - optional

Enter a name for the customer gateway device.

Enter device name

Tags

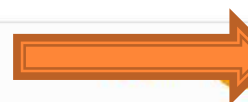
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Click here



Create customer gateway

<https://www.youtube.com/@techwithtwinkle>

configurations

Network Firewall resource groups

VPC endpoint associations

Virtual private network (VPN)

Customer gateways

Virtual private gateway

Site-to-Site VPN connections

Client VPN endpoints

AWS Verified Access

Verified Access instances

Verified Access trust providers

Verified Access groups

Virtual private gateways info

Find resource by attribute or tag



Actions

Create virtual private gateway

Name	Virtual private gateway ID	State	VPC attachment	Type
------	----------------------------	-------	----------------	------

No virtual private gateways

You do not have any virtual private gateways in this region

Create virtual private gateway

Select a virtual private gateway

Click here



After that Click here



Create virtual private gateway [Info](#)

A virtual private gateway is the VPN concentrator on the Amazon side of the site-to-site VPN connection.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

Value must be 256 characters or less in length.

Autonomous System Number (ASN)

- ☒ Amazon default ASN
☐ Custom ASN

Enter Gateway name



Select this



Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

No tags associated with the resource.

Value must be 256 characters or less in length.

Autonomous System Number (ASN)

- ☒ Amazon default ASN
☐ Custom ASN

<https://www.youtube.com/@techwithtwinkle>

Tags

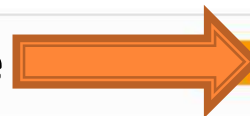
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Click here



Create virtual private gateway

VPN connections Info

Find resource by attribute or tag

Name	VPN ID	State	Virtual private gateway	Transit gateway
No VPN connections You do not have any VPN connections in this region				

[Create VPN connection](#)

Virtual private network (VPN)

- Customer gateways
- Virtual private gateways
- Site-to-Site VPN connections**
- Client VPN endpoints

AWS Verified Access

- Verified Access instances
- Verified Access trust providers
- Verified Access groups

Click here

After that click here

Create VPN connection

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpn-01

Value must be 256 characters or less in length.

Target gateway type Info

☒ Virtual private gateway **Select this**

☐ Transit gateway

☐ Not associated

Virtual private gateway

Select a virtual private gateway **Select gateway created before**

Customer gateway Info

☒ Existing **Select this**

☐ New

Customer gateway ID

Select a customer gateway

Enter name

<https://www.youtube.com/@techwithtwinkle>

Customer gateway [Info](#)

- ☒ Existing
☐ New

Customer gateway ID

Select a customer gateway



Select customer gateway created before

Routing options [Info](#)

- ☒ Dynamic (requires BGP)
☐ Static

Pre-shared key storage [Info](#)

- ☒ Standard
☐ Secrets Manager

Local IPv4 network CIDR - optional

The IPv4 CIDR range on the customer gateway (on-premises) side that is allowed to communicate over the VPN tunnels. The default is 0.0.0.0/0.

0.0.0.0/0

Remote IPv4 network CIDR - optional

▶ **Tunnel 1 options - optional** [Info](#)

Customize tunnel inside CIDR and pre-shared keys for your VPN tunnels. Unspecified tunnel options will be randomly generated by Amazon.

▶ **Tunnel 2 options - optional** [Info](#)

Customize tunnel inside CIDR and pre-shared keys for your VPN tunnels. Unspecified tunnel options will be randomly generated by Amazon.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Click here



Create VPN connection

Direct Connect (DX)



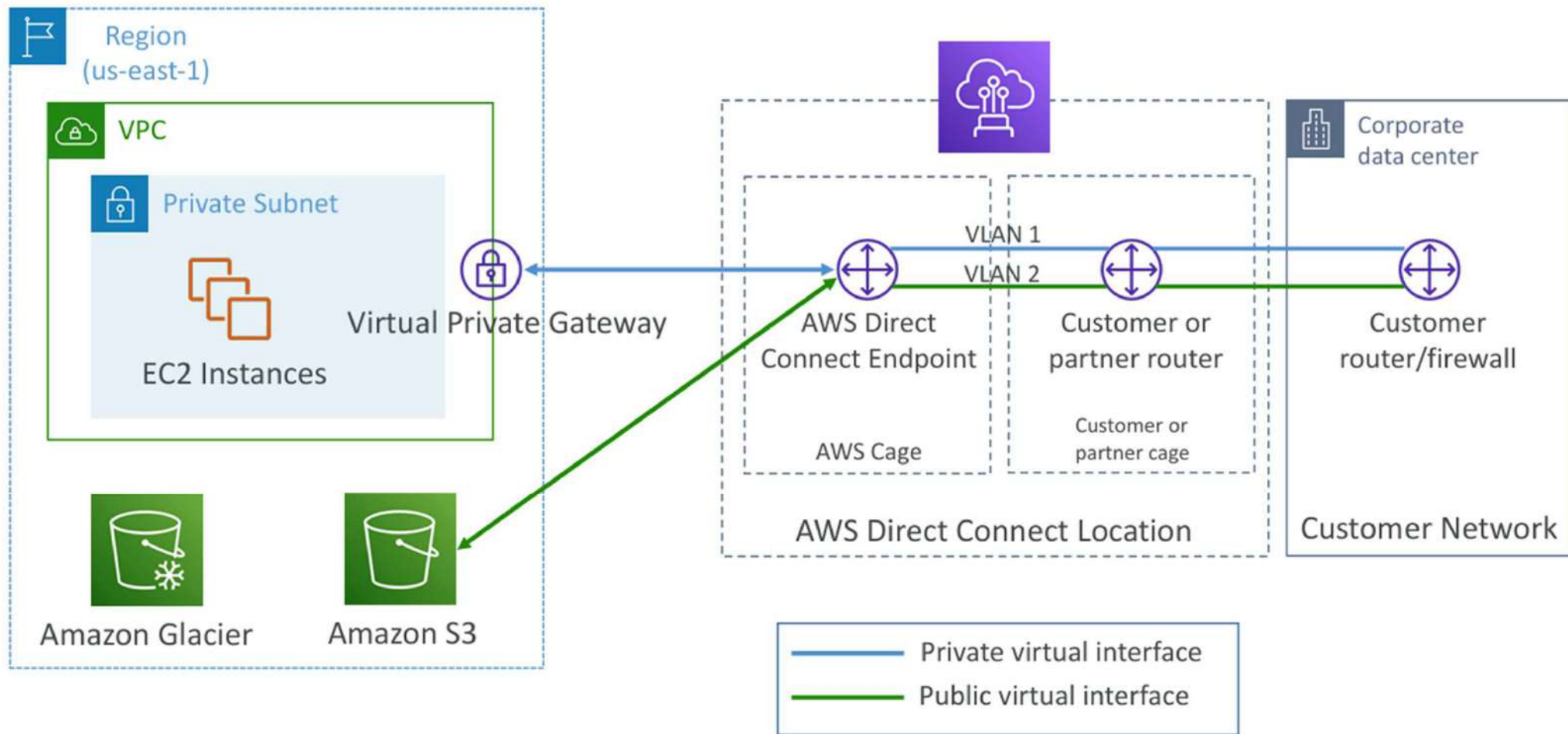
- ▶ Provides a dedicated private VPC connection from a remote network to your
- ▶ Dedicated connection must be setup between your DC and AWS Direct Connect locations
- ▶ You need to setup a Virtual Private Gateway on your VPC
- ▶ Access public resources (S3) and private (EC2) on same connection

Use Cases:

- Increase bandwidth throughput - working with large data sets – lower cost
- More consistent network experience - applications using real-time data feeds
- Hybrid Environments (on prem + cloud)
- ▶ Supports both IPv4 and IPv6

<https://www.youtube.com/@techwithtwinkle>

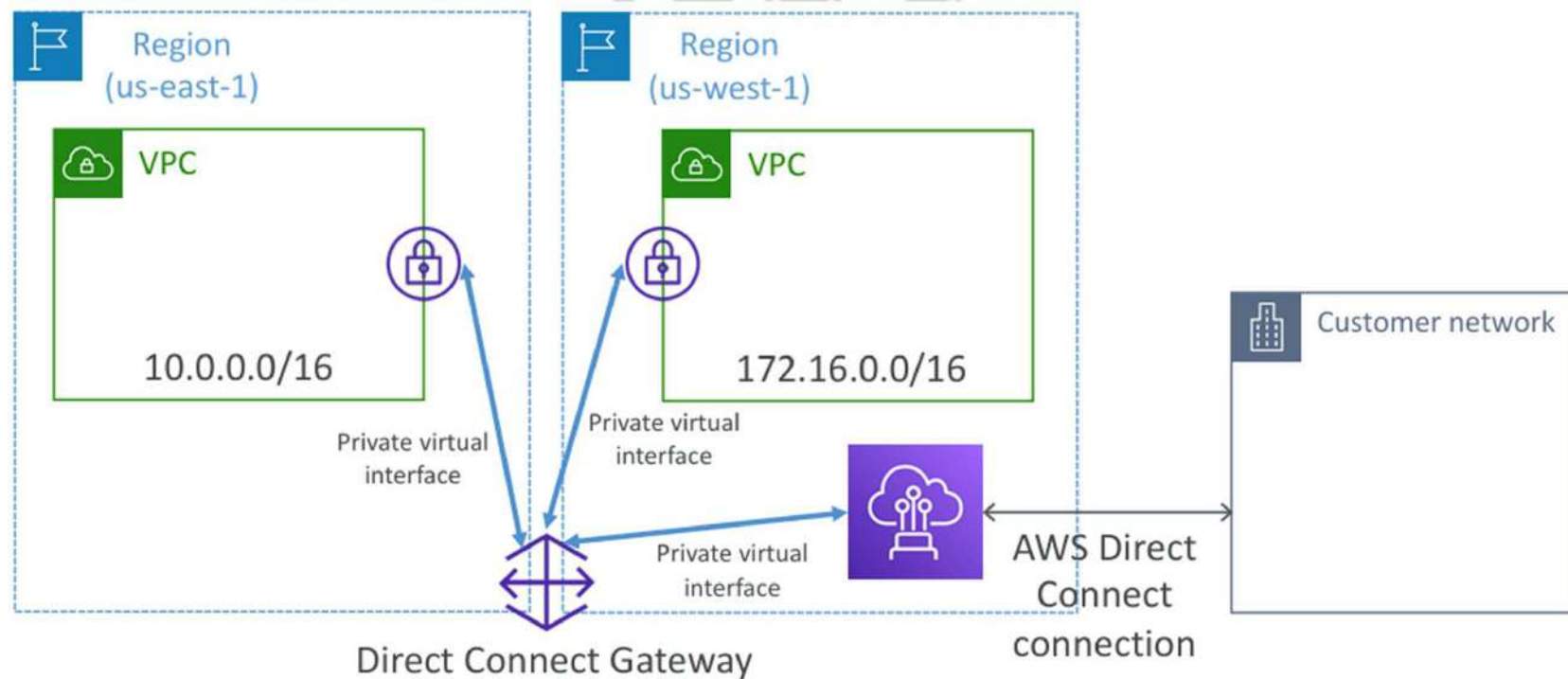
Direct Connect Diagram



<https://www.youtube.com/@techwithtwinkle>

Direct Connect Gateway

- ▶ If you want to setup a Direct Connect to one or more VPC in many different regions (same account), you must use a Direct Connect Gateway



<https://www.youtube.com/@techwithtwinkle>

Direct Connect – Connection Types

Dedicated Connections: 1Gbps, 10 Gbps and 100 Gbps capacity

- ▶ Physical ethernet port dedicated to a customer
- ▶ Request made to AWS first, then completed by AWS Direct Connect Partners

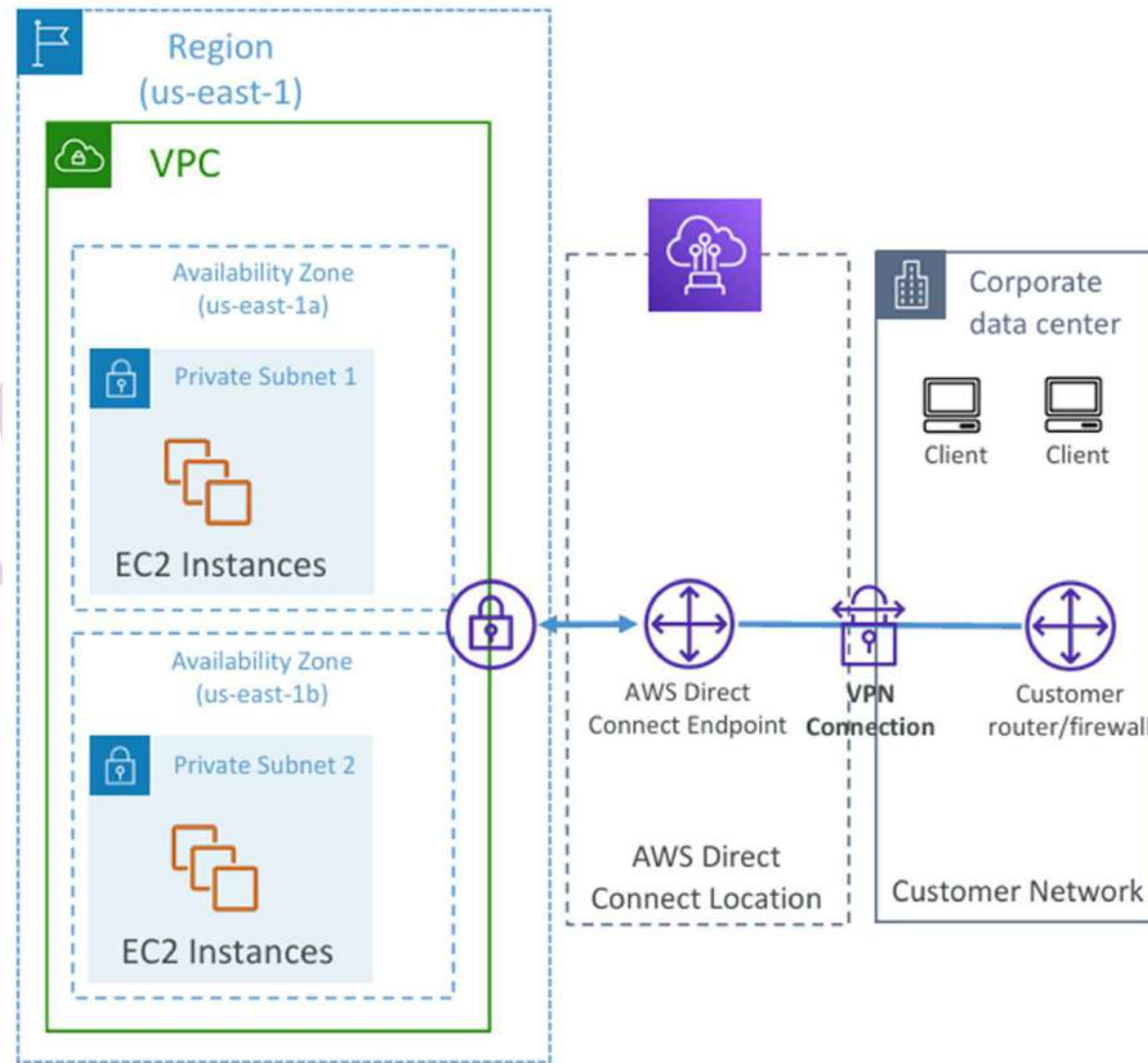
Hosted Connections: 50Mbps, 500 Mbps, to 10 Gbps

- ▶ Connection requests are made via AWS Direct Connect Partners
- ▶ Capacity can be added or removed on demand
- ▶ 1, 2, 5, 10 Gbps available at select AWS Direct Connect Partners

Lead times are often longer than 1 month to establish a new connection

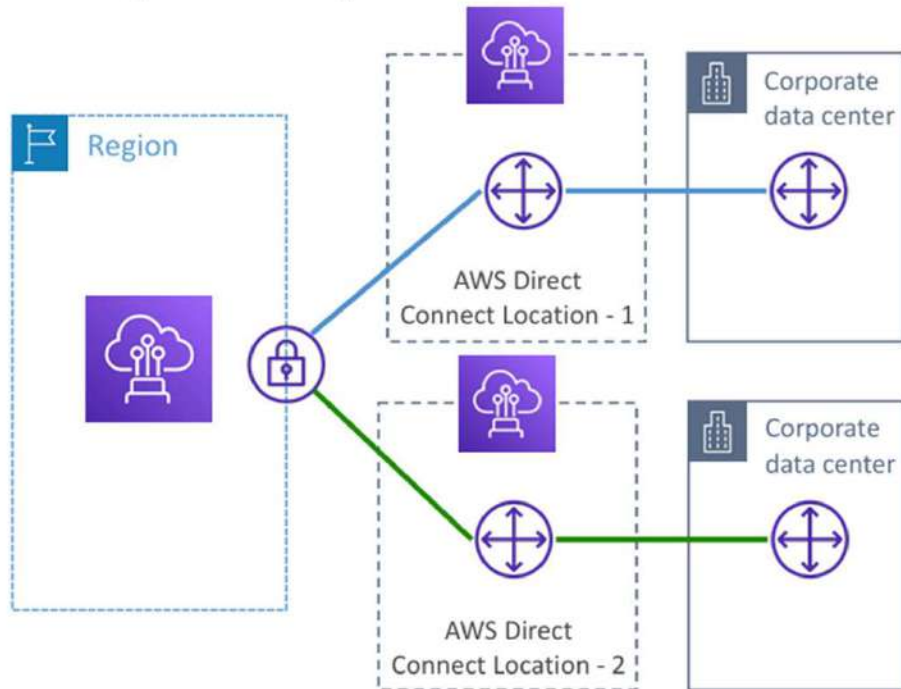
Direct Connect – Encryption

- ▶ Data in transit is but is not encrypted private
- ▶ AWS Direct Connect + VPN provides an IPsec-encrypted private connection
- ▶ Good for an extra level of security, but slightly more complex to put in place



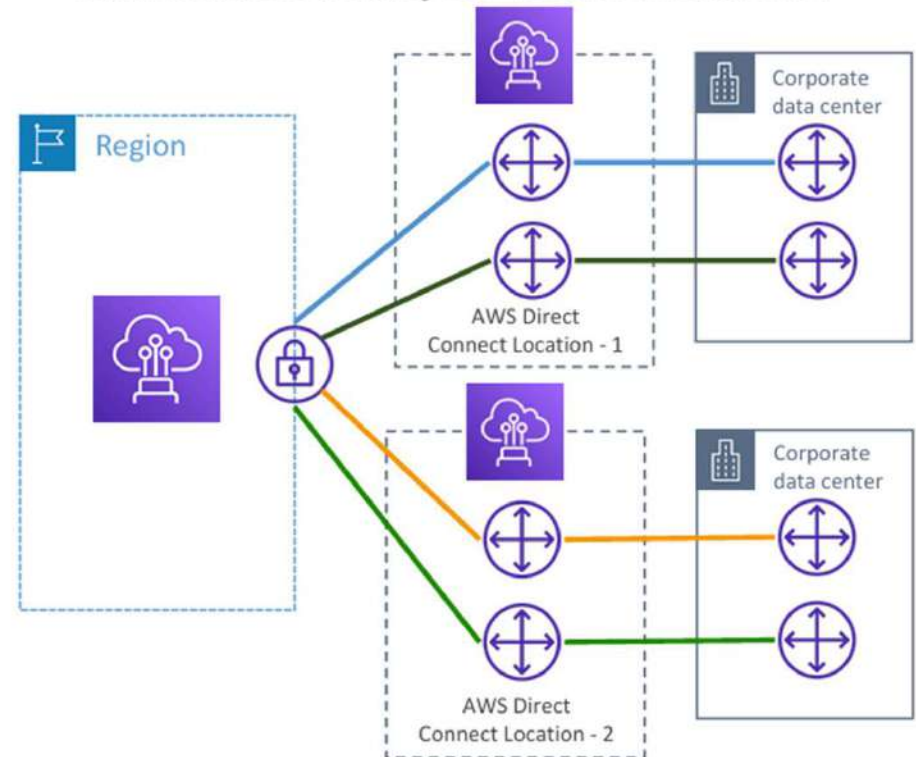
Direct Connect - Resiliency

High Resiliency for Critical Workloads



One connection at multiple locations

Maximum Resiliency for Critical Workloads

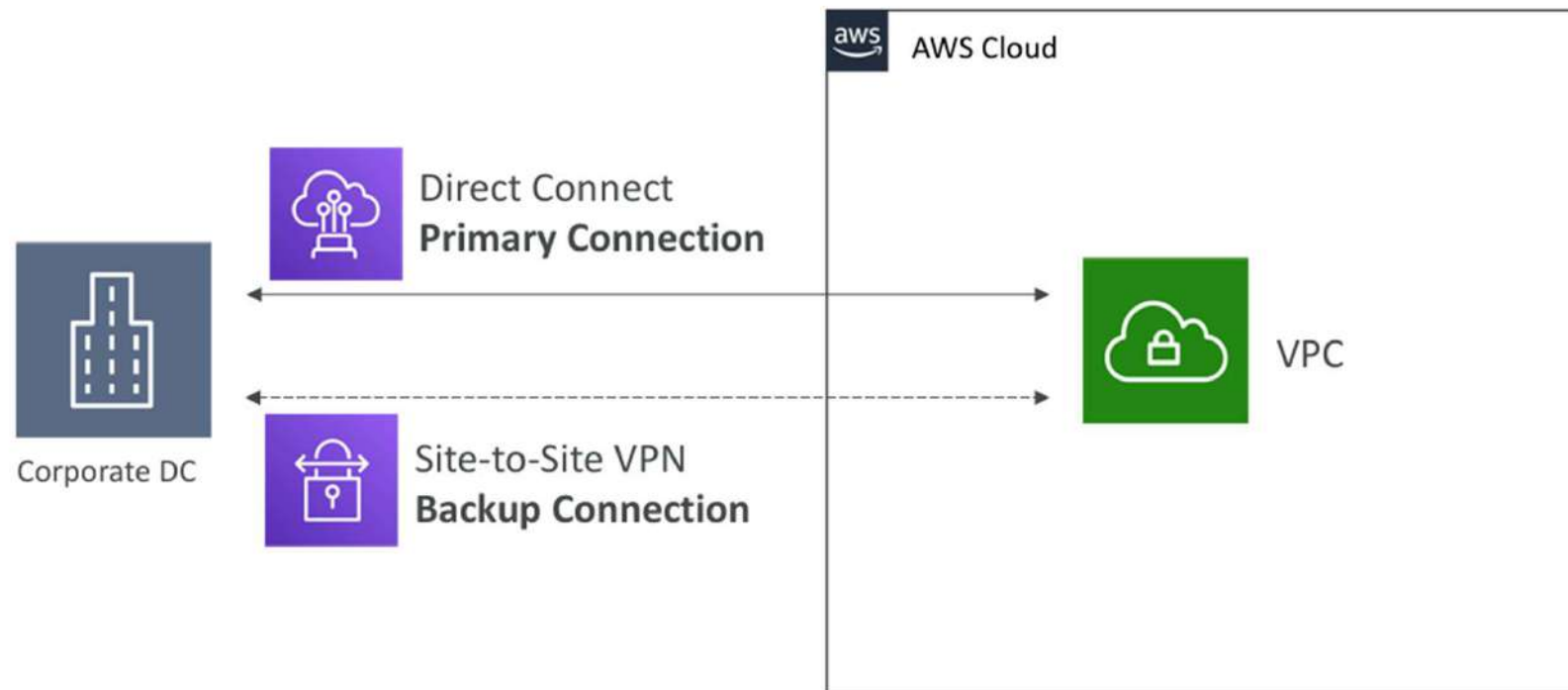


Maximum resilience is achieved by separate connections terminating on separate devices in more than one location.

<https://www.youtube.com/@techwithtwinkle>

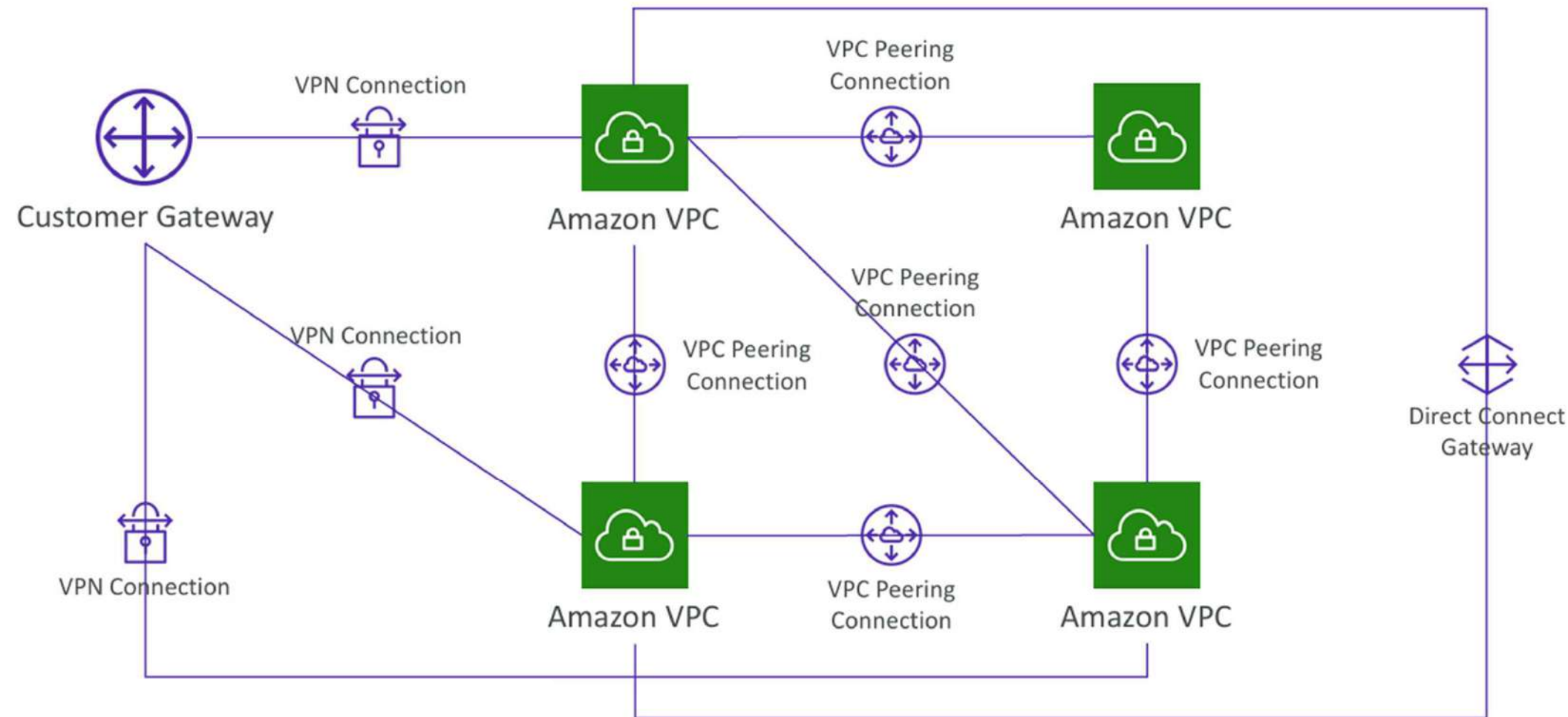
Site-to-Site VPN connection as a backup

- ▶ In case Direct Connect fails, you can set up a backup Direct Connect connection (expensive), or a Site-to-Site VPN connection



<https://www.youtube.com/@techwithtwinkle>

Network topologies can become complicated

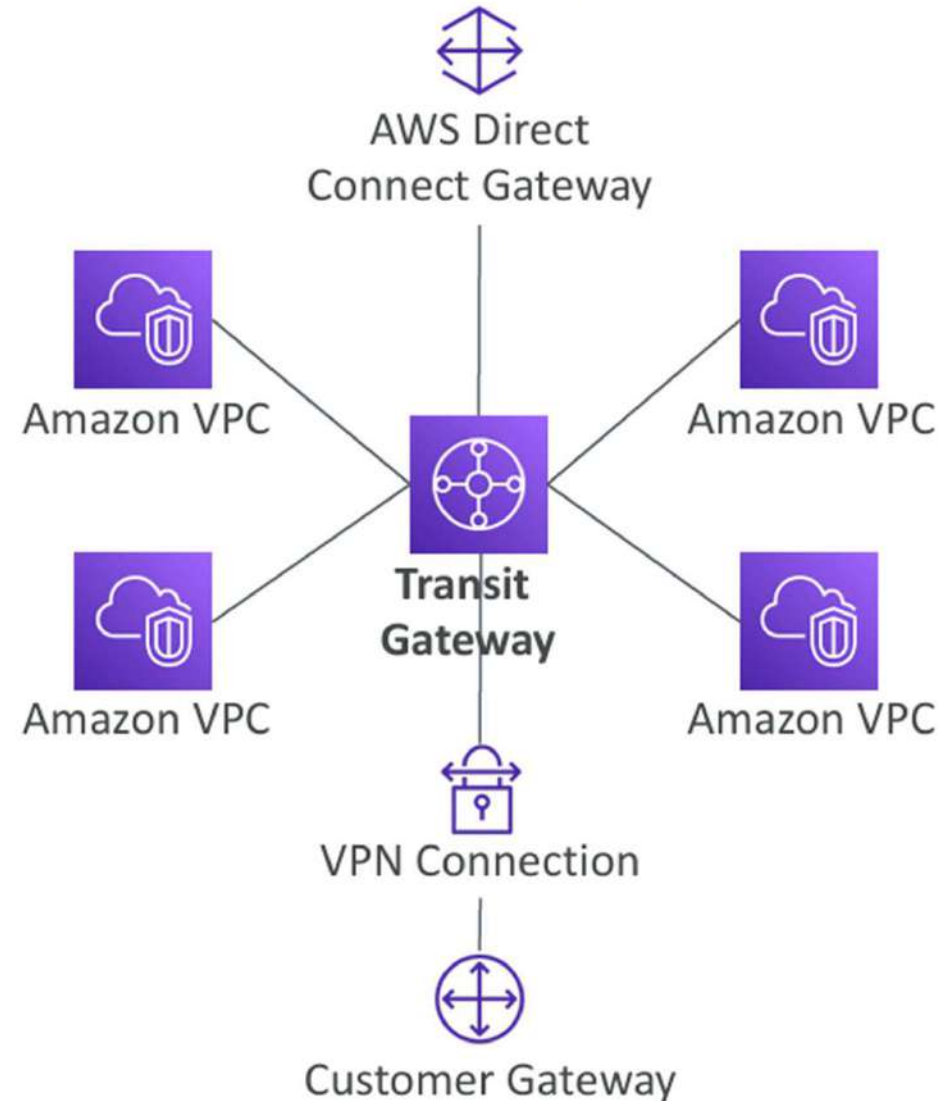


<https://www.youtube.com/@techwithtwinkle>

Transit Gateway

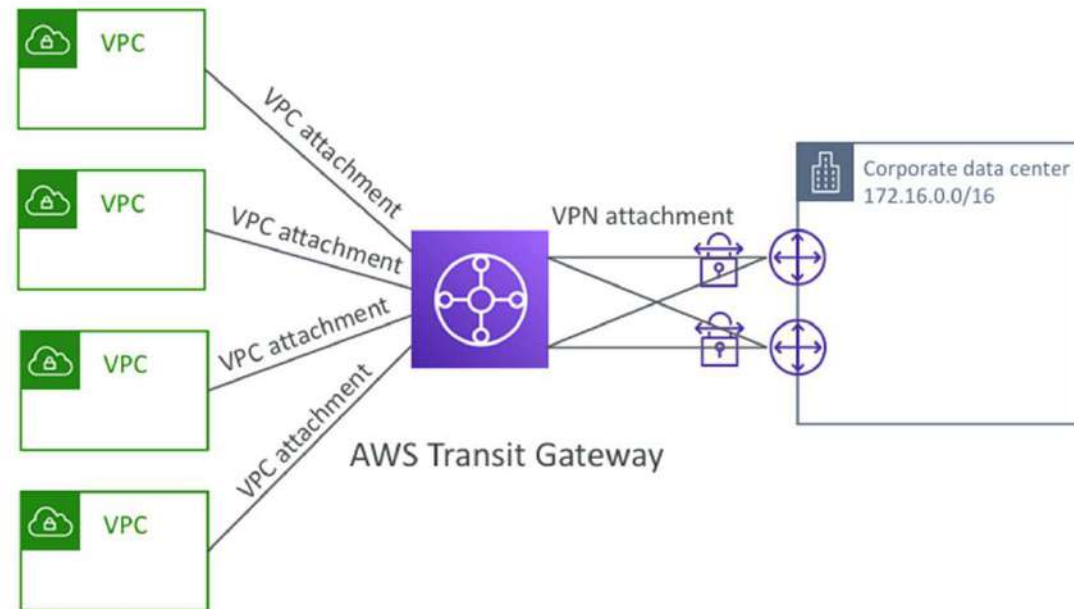
<https://www.youtube.com/@techwithtwinkle>

- ▶ For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- ▶ Regional resource, can work cross-region
- ▶ Share cross-account using Resource Access Manager (RAM)
- ▶ You can peer Transit Gateways across regions
- ▶ Route Tables: limit which VPC can talk with other VPC
- ▶ Works with Direct Connect Gateway, VPN connections
- ▶ Supports IP Multicast (not supported by any other AWS service)



Transit Gateway: Site-to-Site VPN ECMP

- ▶ ECMP = Equal-cost multi-path routing
- ▶ Routing strategy to allow to forward a packet over multiple best path
- ▶ Use case: create multiple Site to-Site VPN connections to increase the bandwidth of your connection to AWS



Transit Gateway: throughput with ECMP

 VPN to virtual private gateway

$$1x \text{  } = 1x \text{  VPC}$$

$$1x \text{  } = 1.25 \text{ Gbps}$$

 VPN connection
(2 tunnels)

 VPN to transit gateway

$$1x \text{  } = 1x \text{   VPC  VPC  VPC  VPC$$

$$1x \text{  } = 2.5 \text{ Gbps (ECMP) – 2 tunnels used}$$

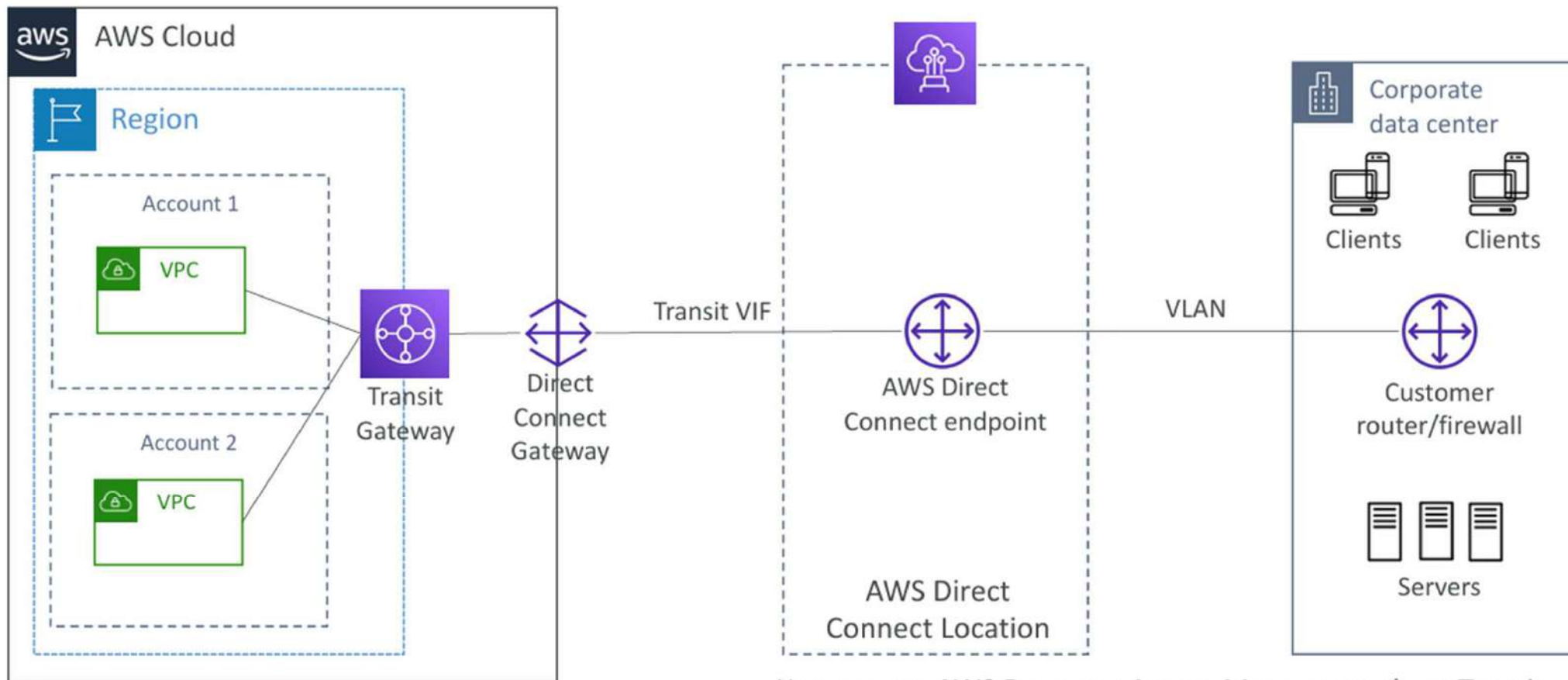
$$2x \text{  } = 5.0 \text{ Gbps (ECMP)}$$

$$3x \text{  } = 7.5 \text{ Gbps (ECMP)}$$

+\$ per GB of TGW
processed data

<https://www.youtube.com/@techwithtwinkle>

Transit Gateway – Share Direct Connect between multiple accounts

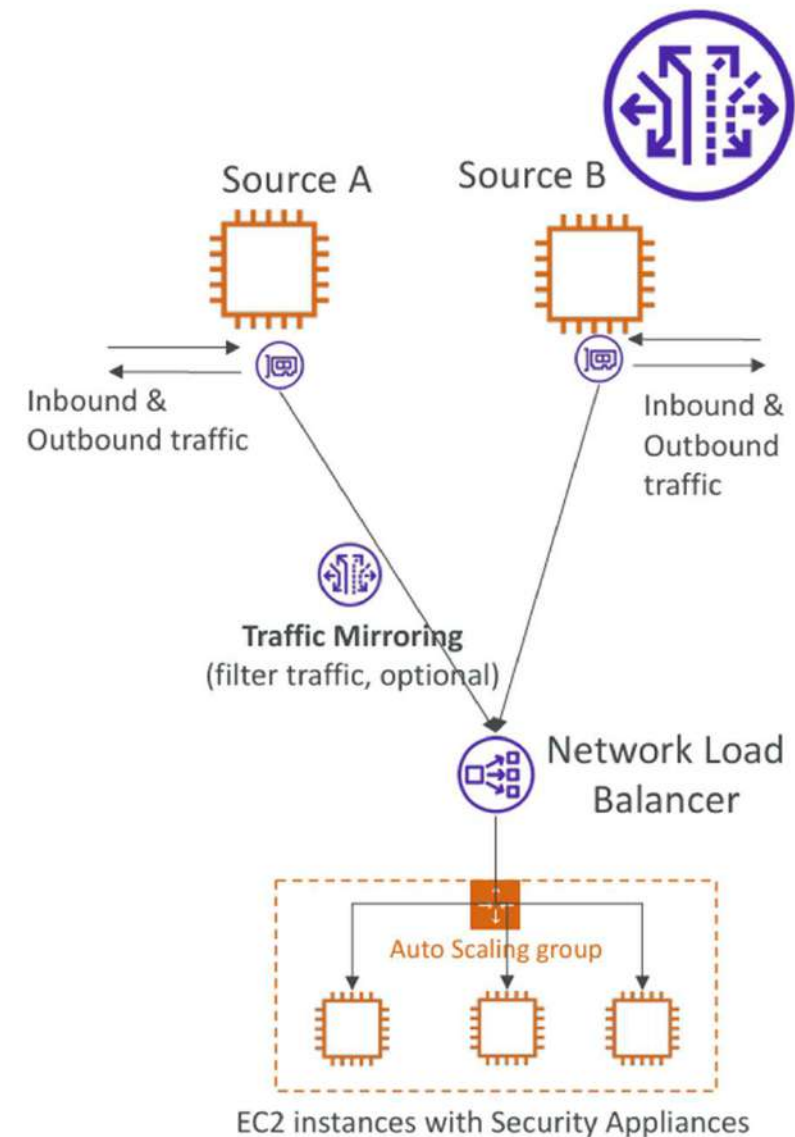


You can use AWS Resource Access Manager to share Transit Gateway with other accounts.

<https://www.youtube.com/@techwithtwinkle>

VPC – Traffic Mirroring

- ▶ Allows you to capture and inspect network traffic in your VPC
- ▶ Route the traffic to security appliances that you manage
- ▶ Capture the traffic
 - **From (Source)** – ENIs
 - **To (Targets)** – an ENI or a Network Load
- ▶ Capture all packets or capture the packets of your interest (optionally, truncate packets)
- ▶ Source and Target can be in the same VPC or different VPCs (VPC Peering)
- ▶ Use cases: content inspection, threat troubleshooting, ...

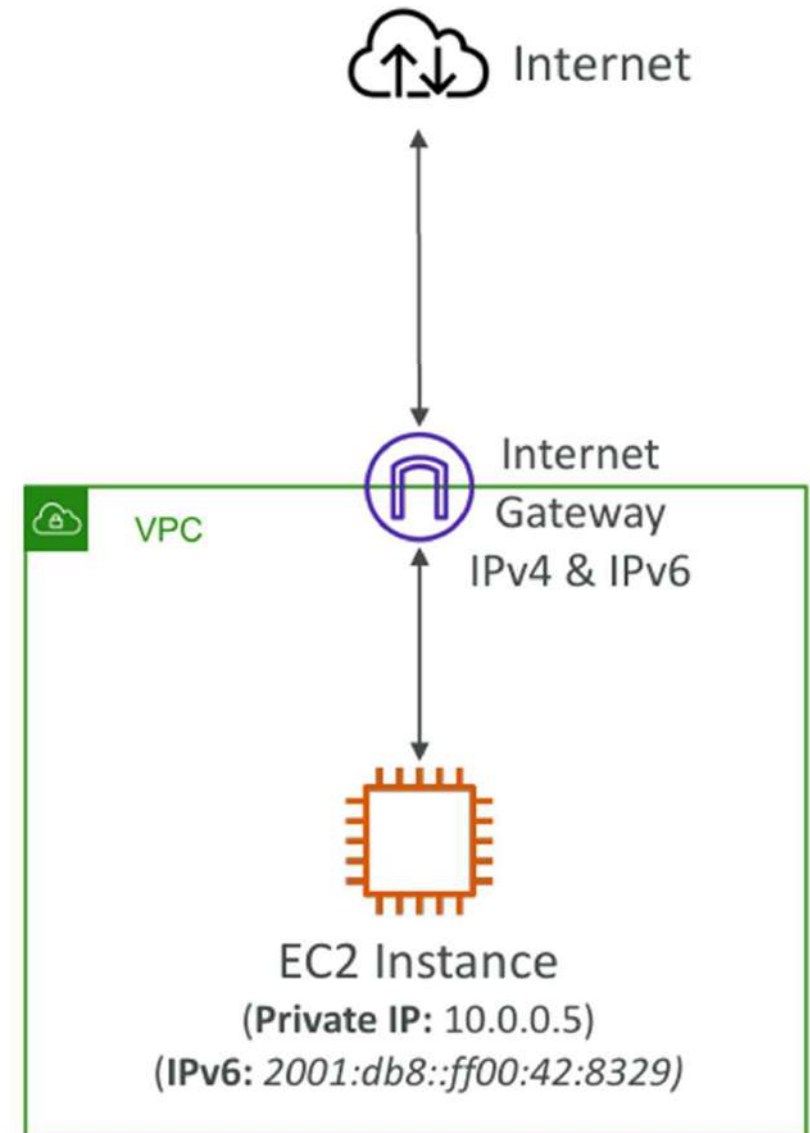


What is IPv6?

- ▶ IPv4 designed to provide 4.3 Billion addresses (they'll be exhausted soon)
- ▶ IPv6 is the successor of IPv4
- ▶ IPv6 is designed to provide 3.4×10^{38} unique IP addresses
- ▶ Every IPv6 address in AWS is public and Internet-routable (no private range)
- ▶ Format → x.x.x.x.x.x.x.x (x is hexadecimal, range can be from 0000 to ffff)
- ▶ Examples:
 - 2001:db8:3333:4444:5555:6666:7777:8888
 - 2001:db8:3333:4444:cccc:dddd:eeee:ffff
 - :: → all 8 segments are zero
 - 2001:db8:: → the last 6 segments are zero
 - ::1234:5678 → the first 6 segments are zero
 - 2001:db8::1234:5678 → the middle 4 segments are zero

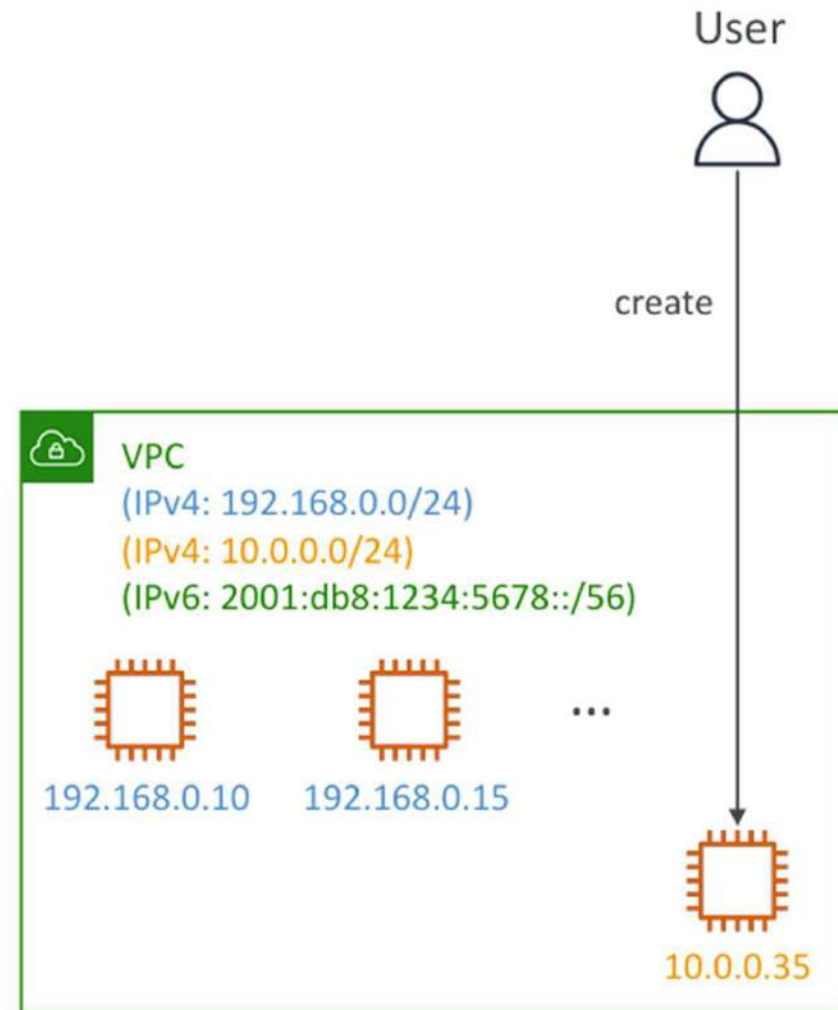
IPv6 in VPC

- ▶ **IPv4 cannot be disabled for your VPC and subnets**
- ▶ You can enable IPv6 (they're public IP addresses) to operate in dual-stack mode
- ▶ Your EC2 instances will get at least a private internal IPv4 and a public IPv6
- ▶ They can communicate using either IPv4 or IPv6 to the internet through an Internet Gateway



IPv4 Troubleshooting

- ▶ **IPv4 cannot be disabled for your VPC and subnets**
- ▶ So, if you cannot launch an EC2 instance in your subnet
- ▶ It's not because it cannot acquire an IPv6 (the space is very large)
- ▶ It's because there are no available IPv4 in your subnet
- ▶ **Solution:** create a new IPv4 CIDR in your subnet



IPv6 for VPC – Hands On

VPC dashboard < **Your VPCs**

AWS Global View

Filter by VPC

▼ **Virtual private cloud**

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Your VPCs (1/2) [Info](#)

Find VPCs by attribute or tag

☐	Name	VPC ID	State
☒	DemoVPC	vpc-06622afdfcf55aef7	Available
☐	-	vpc-0a506c34e326aff4f	Available

Last updated 27 minutes ago

Actions

- Create default VPC
- Create flow log
- Edit VPC settings
- Edit CIDRs**
- Manage middlebox routes
- Manage tags
- Delete VPC

vpc-06622afdfcf55aef7 / DemoVPC

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Details

VPC ID	State	Block Public Access	DNS hostnames
--------	-------	---------------------	---------------

<https://www.youtube.com/@techwithtwinkle>

IPv4 CIDRs Info

CIDR	Status	
10.0.0.0/16	✓ Associated	<button>Remove</button>

Add new IPv4 CIDR

IPv6 CIDRs Info

CIDR (Network border group)	Pool	Address attribute	Status	
-----------------------------	------	-------------------	--------	--

You have no IPv6 CIDR blocks associated with your VPC.

Add new IPv6 CIDR

Click on Add

Close

CIDR	Status	
10.0.0.0/16	✓ Associated	

Add new IPv4 CIDR

IPv6 CIDRs Info

CIDR (Network border group)	Pool	Address attribute	Status	
-----------------------------	------	-------------------	--------	--

Add new IPv6 CIDR

Add IPv6 CIDR

IPv6 CIDR block

- ☐ IPAM-allocated IPv6 CIDR block
- ☒ Amazon-provided IPv6 CIDR block
- ☐ IPv6 CIDR owned by me

Select this

Network border group

A network border group is a unique group of Zones from where IPv4 and IPv6 IP addresses are advertised. All Availability Zones in this VPC will use this network border group.

ap-south-1

Click here

Select CIDR

Close

☰ VPC > Your VPCs > vpc-06622afdcf55aef7 / DemoVPC > Edit CIDRs

✔ Succeeded associating IPv6 CIDR

10.0.0.0/16 ✔ Associated Remove

Add new IPv4 CIDR

IPv6 CIDRs Info

CIDR (Network border group)	Pool	Address attribute	Status	
2406:da1a:c2e:7500::/56 (ap-south-1)	Amazon	Public	✔ Associated	Remove

Add new IPv6 CIDR

<https://www.youtube.com/@techwithtwinkle> Close

☰ VPC > Subnets

VPC dashboard <

AWS Global View

Filter by VPC ▼

▼ Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Subnets (1/7) Info

Find subnets by attribute or tag

	Name	Subnet ID	
☐	PrivateSubnetA	subnet-016e736c2ef7867b1	
☒	PublicSubnetA	subnet-0fb0e68a3ae9be50b	
☐	-	subnet-0287ee3f45c829710	

subnet-0fb0e68a3ae9be50b / PublicSubnetA

Details Flow logs Route table Network ACL CIDR reservati

Details

Subnet ID	Subnet ARN	State	Block Public Access
-----------	------------	-------	---------------------

Click here → Actions ▲

- View details
- Create flow log
- Edit subnet settings
- Edit IPv6 CIDRs**
- Edit network ACL association
- Edit route table association
- Edit CIDR reservations
- Share subnet
- Manage tags
- Delete subnet

Click here →

Select this →

Edit IPv6 CIDRs [Info](#)

VPC details

VPC ID
 vpc-06622afdfcf55aef7

VPC CIDR block
2406:da1a:c2e:7500::/56

Subnet ID
 subnet-0fb0e68a3ae9be50b

Network border group
 ap-south-1

Subnet CIDR block

You have no IPv6 CIDR blocks associated with your subnet

[Add IPv6 CIDR](#)

1 remaining

Click here

[Cancel](#)

[Save](#)

Subnet CIDR block

VPC CIDR block

2406:da1a:c2e:7500::/56

Subnet CIDR block

2406:da1a:c2e:7500::/56

4,722.3Q IPs

[Remove](#)

[Add IPv6 CIDR](#)

0 remaining

<https://www.youtube.com/@techwithtwinkle>

Click on save

[Save](#)

VPC > Subnets

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Successfully edited IPv6 CIDR for subnet (subnet-0fb0e68a3ae9be50b / PublicSubnetA).

Subnets (1/7)

Find subnets by attribute or tag

Name	Subnet ID
PublicSubnetA	subnet-0fb0e68a3ae9be50b
	subnet-0287ee3f45c82971

Subnet-0fb0e68a3ae9be50b / PublicSubnetA

Details

Flow logs

Route table

Network ACL

CIDR reservations

Details

Subnet ID

Subnet ARN

State

Block Public Access

View details

Create flow log

Edit subnet settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Edit CIDR reservations

Share subnet

Manage tags

Delete subnet

1

55aef7 | Demo...

326aff4f

Create subnet

VPC > Subnets > subnet-0fb0e68a3ae9be50b > Edit subnet settings

Subnet ID

subnet-0fb0e68a3ae9be50b

Name

PublicSubnetA

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign IPv6 address

☒ Enable auto-assign public IPv4 address

☐ Enable auto-assign customer-owned IPv4 address

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch

☐ Enable resource name DNS AAAA record on launch

Click here

Select this

Enable it

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Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

- ☐ Enable resource name DNS A record on launch [Info](#)
- ☐ Enable resource name DNS AAAA record on launch [Info](#)

Hostname type [Info](#)

- ☐ Resource name
- ☒ IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

- ☐ Enable DNS64 [Info](#)

Click here  **Save**

Assign it to Instance you want in inbound rule

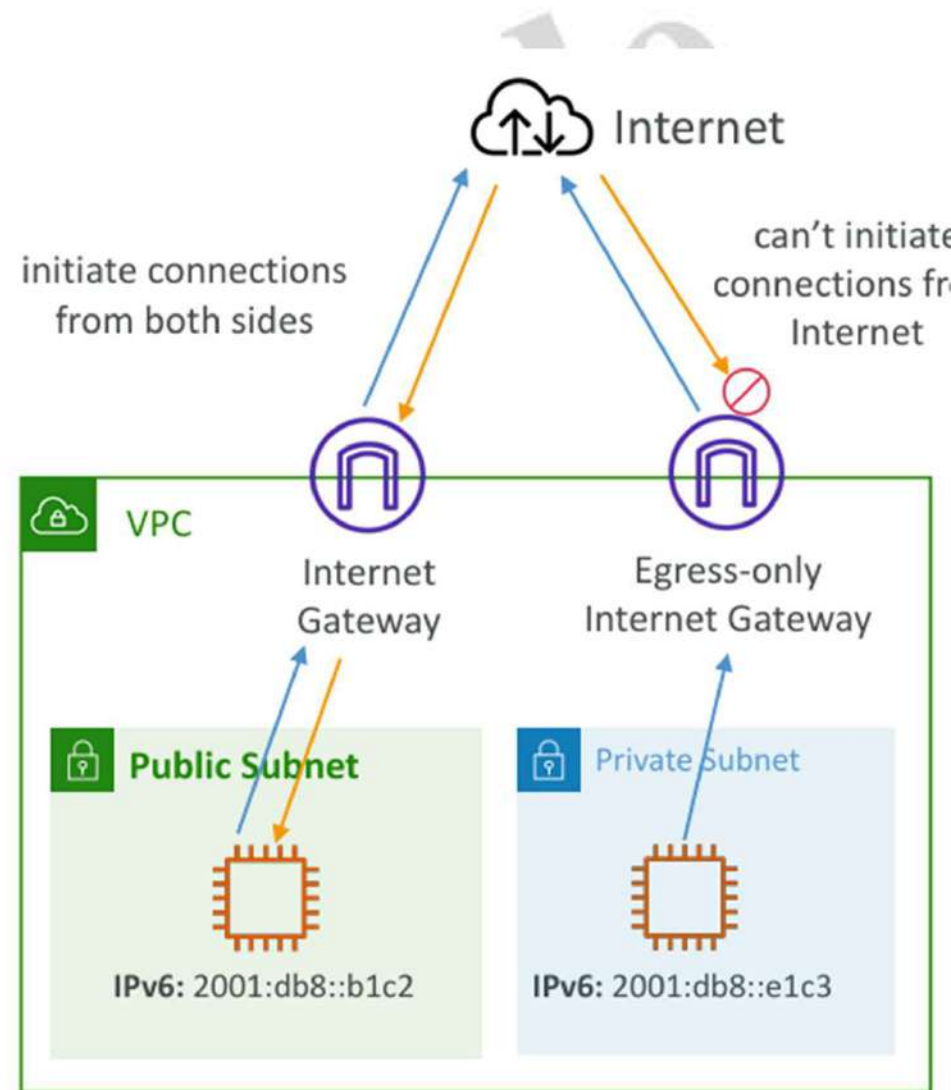
<https://www.youtube.com/@techwithtwinkle>

Egress-only Internet Gateway

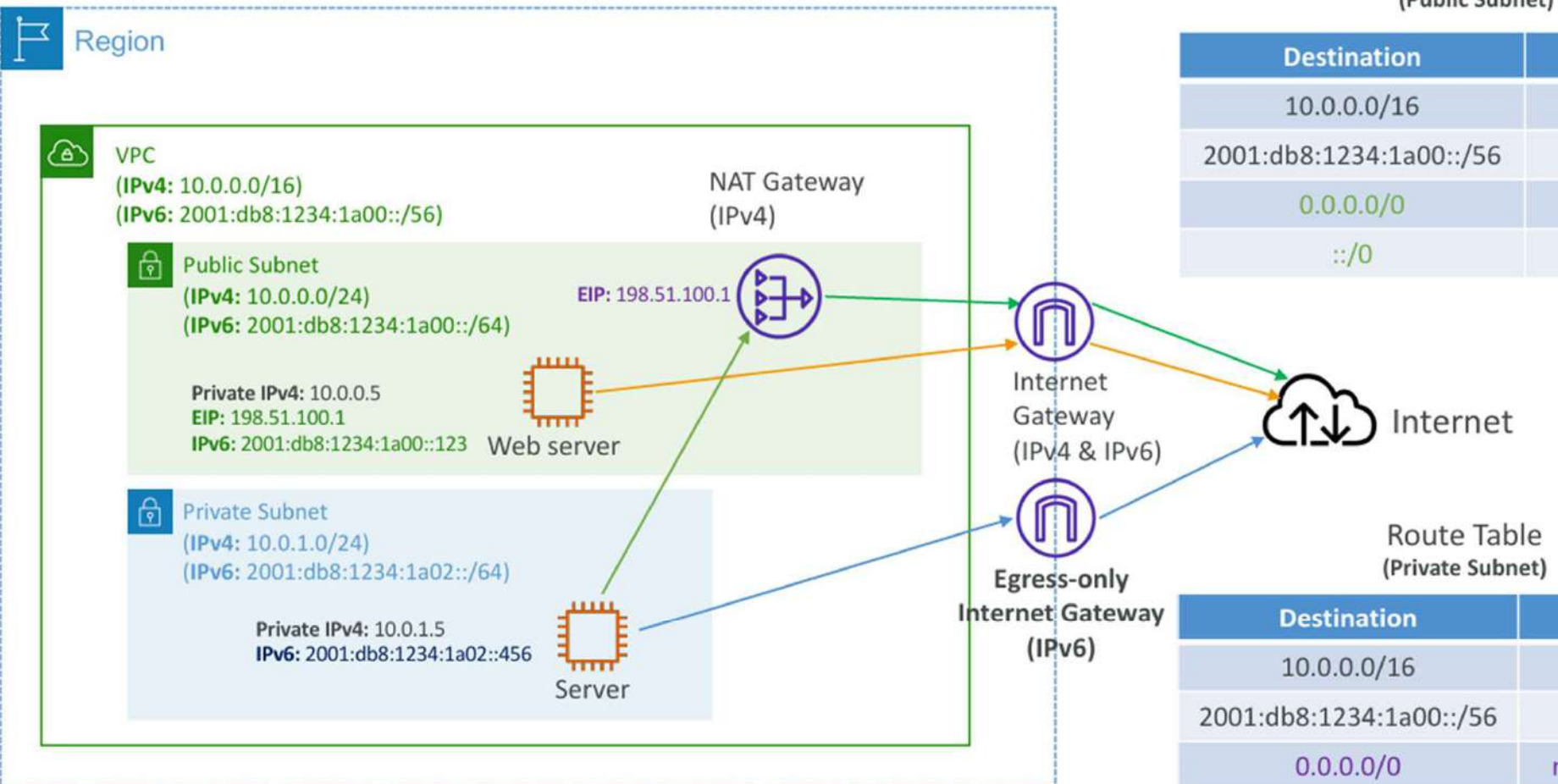


<https://www.youtube.com/@techwithtwinkle>

- ▶ Used for IPv6 only
- ▶ (similar to a NAT Gateway but for IPv6)
- ▶ Allows instances in your VPC outbound connections over IPv6 while preventing the internet to initiate an IPv6 connection to your instances
- ▶ **You must update the Route Tables**



IPv6 Routing



Route Table
(Public Subnet)

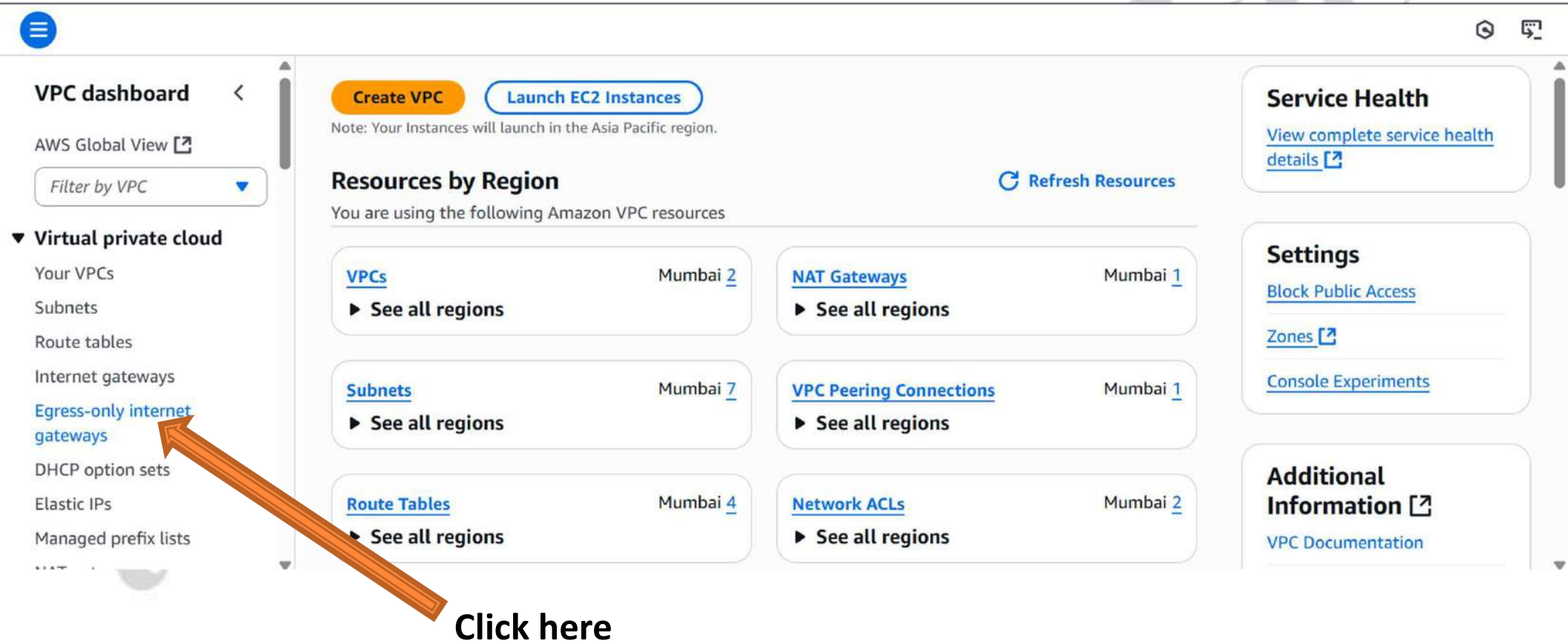
Destination	Target
10.0.0.0/16	local
2001:db8:1234:1a00::/56	local
0.0.0.0/0	igw-id
::/0	igw-id

Route Table
(Private Subnet)

Destination	Target
10.0.0.0/16	local
2001:db8:1234:1a00::/56	local
0.0.0.0/0	nat-gateway-id
::/0	eigw-id

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Egress-only Internet Gateway – Hands On



The screenshot shows the AWS VPC dashboard. On the left sidebar, under 'Virtual private cloud', the 'Egress-only internet gateways' link is highlighted with an orange arrow. The main content area shows 'Resources by Region' for Mumbai, listing VPCs (2), NAT Gateways (1), Subnets (7), VPC Peering Connections (1), Route Tables (4), and Network ACLs (2). The right sidebar contains 'Service Health', 'Settings', and 'Additional Information' sections.

VPC dashboard <

AWS Global View

Filter by VPC ▼

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways**
- DHCP option sets
- Elastic IPs
- Managed prefix lists

Create VPC **Launch EC2 Instances**

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region Refresh Resources

You are using the following Amazon VPC resources

Resource Type	Count
VPCs	Mumbai 2
▶ See all regions	
Subnets	Mumbai 7
▶ See all regions	
Route Tables	Mumbai 4
▶ See all regions	
NAT Gateways	Mumbai 1
▶ See all regions	
VPC Peering Connections	Mumbai 1
▶ See all regions	
Network ACLs	Mumbai 2
▶ See all regions	

Service Health
[View complete service health details](#)

Settings
[Block Public Access](#)
[Zones](#)
[Console Experiments](#)

Additional Information
[VPC Documentation](#)

Click here

<https://www.youtube.com/@techwithtwinkle>

 [VPC](#) > Egress only internet gateways    

VPC dashboard <

AWS Global View 

Filter by VPC ▾

▼ **Virtual private cloud**

- Your VPCs
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- Elastic IPs
- Managed prefix lists

Egress only internet gateways [Info](#)

 **Actions** ▾ **Create egress only internet gateway**

Name ▾	Egress only internet gateway ID ▾	Attached VPC ID ▾
No egress only internet gateway found		

Select an egress only internet gateway above  ▾

<https://www.youtube.com/@techwithtwinkle>

Click here

 [VPC](#) > [Egress only internet gateways](#) > Create egress only internet gateway   

Create egress only internet gateway [Info](#)

An Internet Gateway is a virtual router that connects a VPC to the internet.

Egress only internet gateway settings

Name - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

DemoEGW

Enter name

VPC

Attach the egress only internet gateway to this VPC.

vpc-06622afdfcf55aef7 (DemoVPC)

Select VPC

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - *optional*

Remove

Create a tag with a key or name and a value that you specify.

DemoEGW

VPC

Attach the egress only internet gateway to this VPC.

vpc-06622afdcf55aef7 (DemoVPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name



Value - optional

Q DemoEGW



Remove

Add new tag

You can add 49 more tags.

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Click here



Create egress only internet gateway

VPC > Route tables

Route tables (1/4) Info

Find route tables by attribute

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☒	PrivateRouteTable	rtb-01e7a7e0497c8595c	2 subnets	-	No
☐	-	rtb-0f04c3dbde0d20eb8	-	-	Yes
☐	PublicRouteTable	rtb-07fd3d246c91fa97b	2 subnets	-	No
☐	-	rtb-062183f5cc8205cab	-	-	Yes

rtb-01e7a7e0497c8595c / PrivateRouteTable

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (3)

Filter routes

Both Edit routes

Click on Edit routes

Edit routes

0.0.0.0/0
0.0.0.0/8
0.0.0.0/16
0.0.0.0/24
0.0.0.0/32
::/0
::/16
::/32
::/48

Q |

	Status	Propagated	Route Origin	
	Active	No	CreateRouteTable	
	Active	No	CreateRouteTable	
	Active	No	CreateRoute	Remove
	Active	No	CreateRoute	Remove

Add route

Select this

<https://www.youtube.com/@techwithtwinkle>

Edit routes

2406:da1a:c2e:7500::/56

10.0.0.0/16

Q 0.0.0.0/0 X

Q ::/0 X

Core Network
Egress Only Internet Gateway
Gateway Load Balancer Endp
Instance
Internet Gateway
local
NAT Gateway
Network Interface
Outpost Local Gateway

Egress Only Internet Gateway

	Status	Propagated	Route Origin	
	Active	No	CreateRouteTable	
	Active	No	CreateRouteTable	
	Active	No	CreateRoute	Remove
		No	CreateRoute	Remove

Add route

Select this

2406:da1a:c2e:7500::/56	local	Active	No	CreateRouteTable	
	Q local	X			
10.0.0.0/16	local	Active	No	CreateRouteTable	
	Q local	X			
Q 0.0.0.0/0	NAT Gateway	Active	No	CreateRoute	Remove
	Q nat-0e20925a019598c68	X			
Q ::/0	Egress Only Internet Gateway	-	No	CreateRoute	Remove
	Q eigw-	X			
	Use: "eigw-"				
	eigw-0874d21d2297970ba (DemoEGW)				

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Select this

Click here

Save changes

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VPC Section Summary (1/3)

- ▶ **CIDR** – IP Range
- ▶ **VPC** – Virtual Private Cloud => we define a list of IPv4 & IPv6 CIDR
- ▶ **Subnets** – tied to an AZ, we define a CIDR
- ▶ **Internet Gateway** – at the VPC level, provide IPv4 & IPv6 Internet Access
- ▶ **Route Tables** – must be edited to add routes from subnets to the IGW, VPC Peering Connections, VPC Endpoints, ...
- ▶ **Bastion Host** – public EC2 instance to SSH into, that has SSH connectivity to EC2 instances in private subnets
- ▶ **NAT Instances** – gives Internet access to EC2 instances in private subnets. Old, must be setup in a public subnet, disable Source / Destination check flag
- ▶ **NAT Gateway** – managed by AWS, provides scalable Internet access to private EC2 instances, when the target is an IPv4 address

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VPC Section Summary (2/3)

- ▶ **NACL** – stateless, subnet rules for inbound and outbound, don't forget Ephemeral Ports
- ▶ **Security Groups** – stateful, operate at the EC2 instance level
- ▶ **VPC Peering** – connect two VPCs with non overlapping CIDR, non-transitive
- ▶ **VPC Endpoints** – provide private access to AWS Services (S3, DynamoDB, CloudFormation, SSM) within a VPC
- ▶ **VPC Flow Logs** – can be setup at the VPC / Subnet / ENI Level, for ACCEPT and REJECT traffic, helps identifying attacks, analyze using Athena or CloudWatch Logs Insights
- ▶ **Site-to-Site VPN** – setup a Customer Gateway on DC, a Virtual Private Gateway on VPC, and site-to-site VPN over public Internet
- ▶ **AWS VPN CloudHub** – hub-and-spoke VPN model to connect your sites

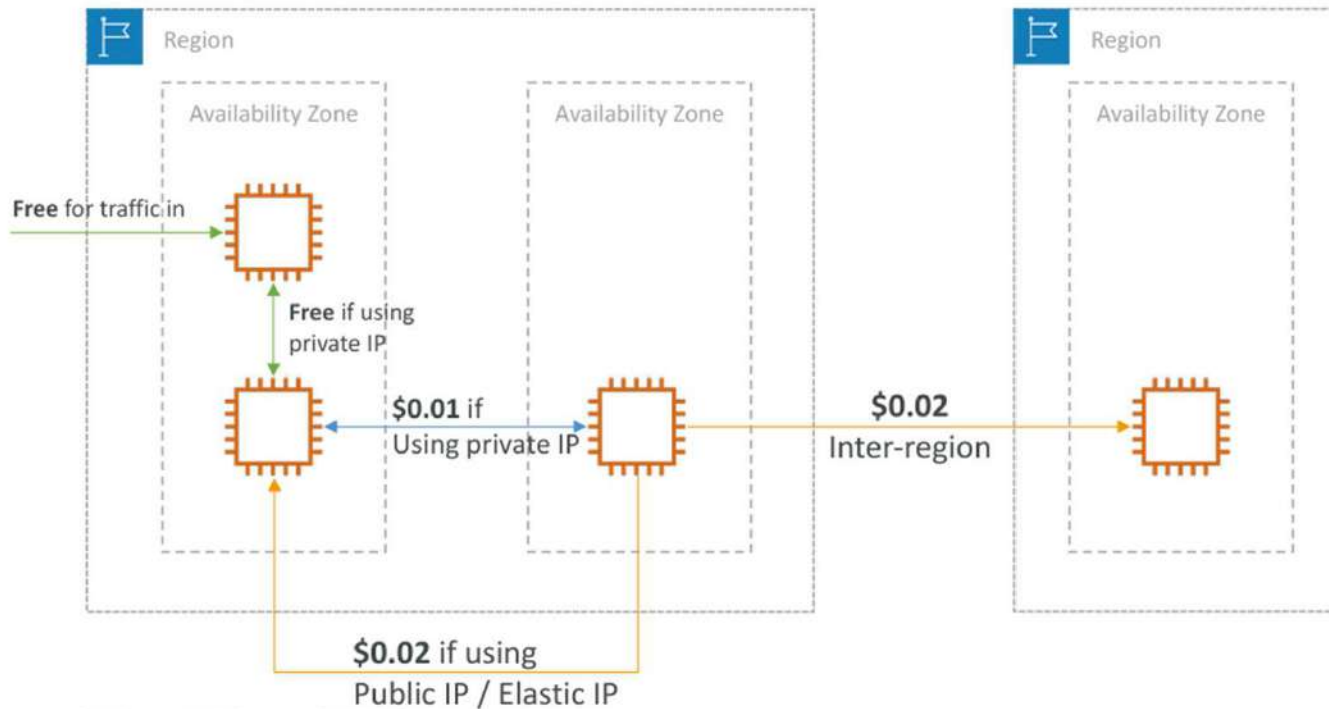
<https://www.youtube.com/@techwithtwinkle>

VPC Section Summary (3/3)

- ▶ **Direct Connect** – setup a Virtual Private Gateway on VPC, and establish a direct private connection to an AWS Direct Connect Location
- ▶ **Direct Connect Gateway** – setup a Direct Connect to many VPCs in different AWS regions
- ▶ **AWS PrivateLink / VPC Endpoint Services:**
 - Connect services privately from your service VPC to customers VPC
 - Doesn't need VPC Peering, public Internet, NAT Gateway, Route Tables
 - Must be used with Network Load Balancer & ENI
- ▶ **ClassicLink** – connect EC2-Classic EC2 instances privately to your VPC
- ▶ **Transit Gateway** – transitive peering connections for VPC, VPN & DX
- ▶ **Traffic Mirroring** – copy network traffic from ENIs for further analysis
- ▶ **Egress-only Internet Gateway** – like a NAT Gateway, but for IPv6 targets

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Networking Costs in AWS per GB - Simplified



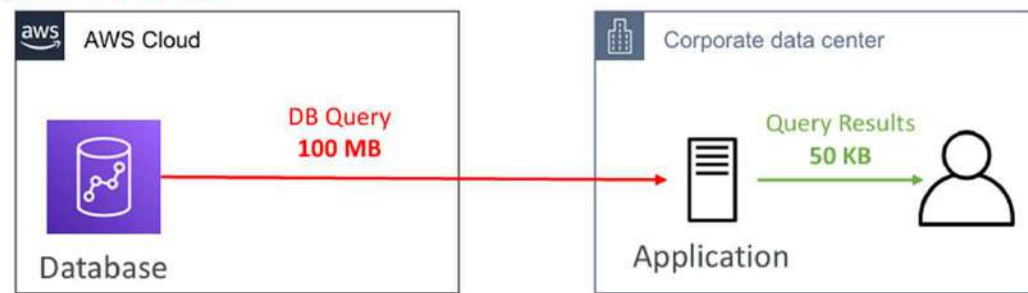
- ▶ Use Private IP instead of Public IP for good savings and better network performance
- ▶ Use same AZ for maximum savings (at the cost of high availability)

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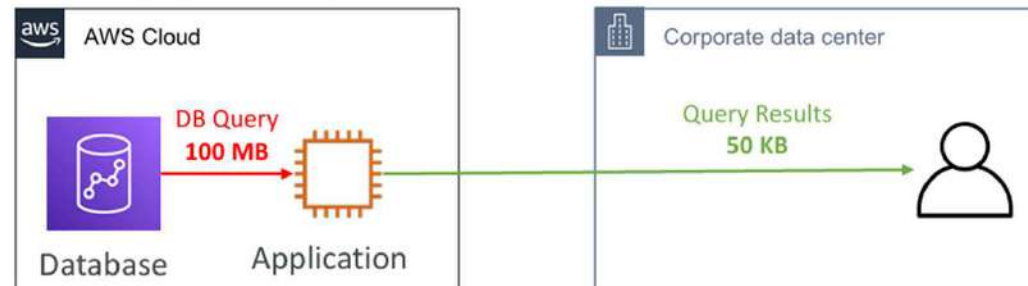
Minimizing egress traffic network cost

- ▶ **Egress traffic:** outbound traffic (from AWS to outside)
- ▶ **Ingress traffic:** inbound traffic - from outside to AWS (typically free)
- ▶ Try to keep as much internet traffic within AWS to minimize costs
- ▶ **Direct Connect location that are co-located in the same AWS Region result in lower cost for egress network**

Egress cost is high



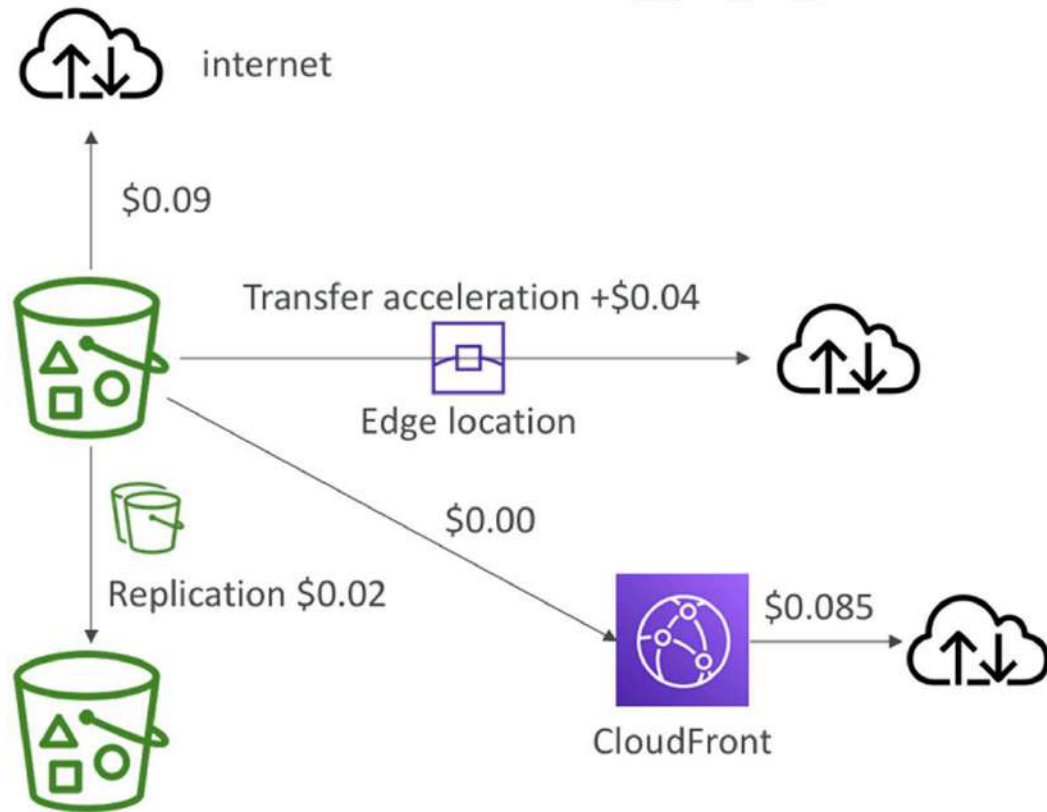
Egress cost is minimized



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S3 Data Transfer Pricing – Analysis for USA

- ▶ **S3 ingress:** free
- ▶ **S3 to Internet:** \$0.09 per GB
- ▶ **S3 Transfer Acceleration:**
 - Faster transfer times (50 to 500% better)
 - Additional cost on top of Data Transfer Pricing: +\$0.04 to \$0.08 per GB
- ▶ **S3 to CloudFront:** \$0.00 per GB
- ▶ **CloudFront to Internet:** \$0.085 per GB (slightly cheaper than S3)
- Caching capability (lower latency)
- Reduce costs associated with S3 Requests Pricing (7x cheaper with CloudFront)
- ▶ **S3 Cross Region Replication:** \$0.02 per GB



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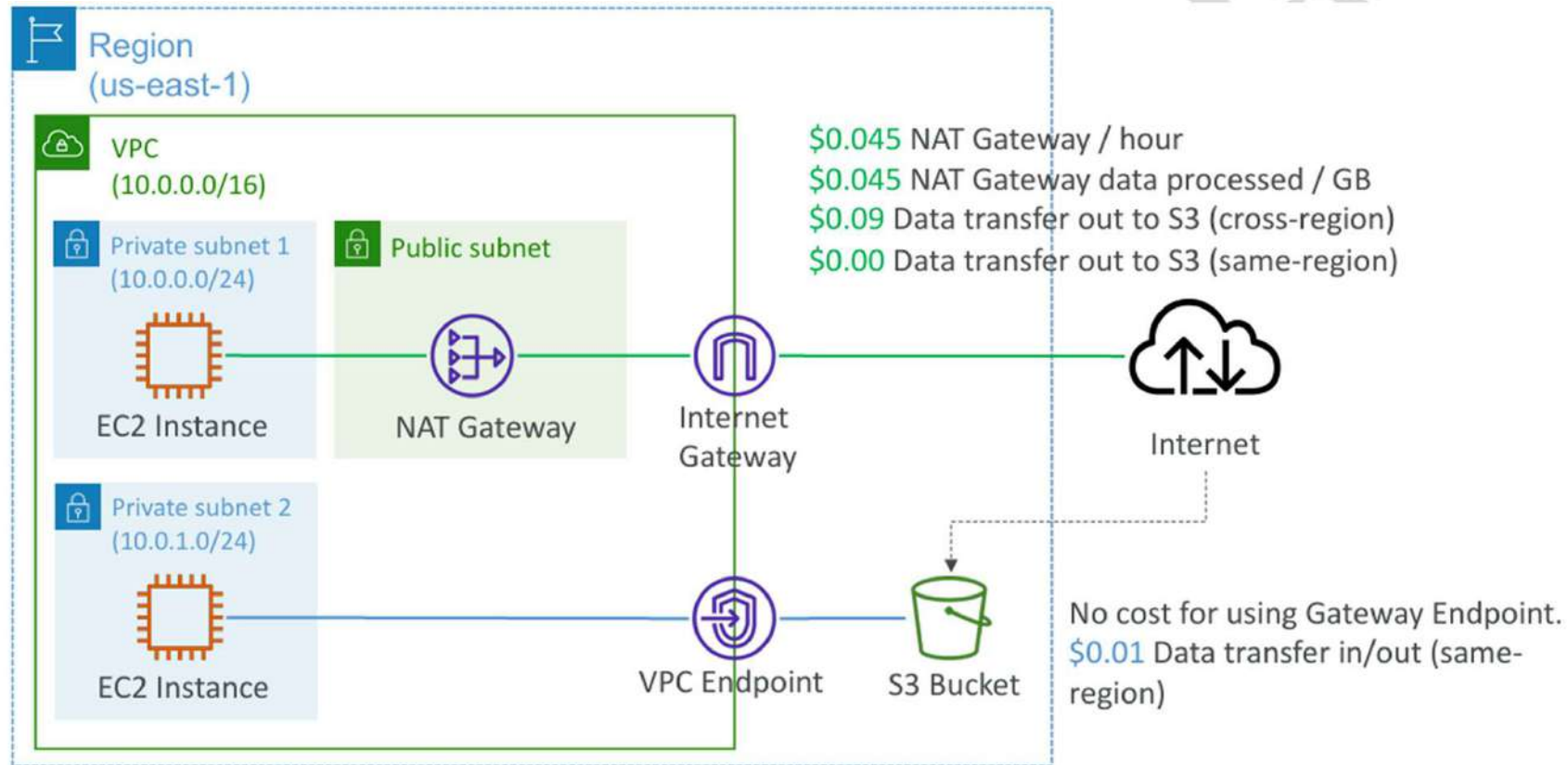
Pricing: NAT Gateway vs Gateway VPC Endpoint

Subnet 1 route table

Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	igw-id

Subnet 2 route table

Destination	Target
10.0.0.0/16	Local
pl-id for Amazon S3	vpce-id



<https://www.youtube.com/@techwithtwinkle>

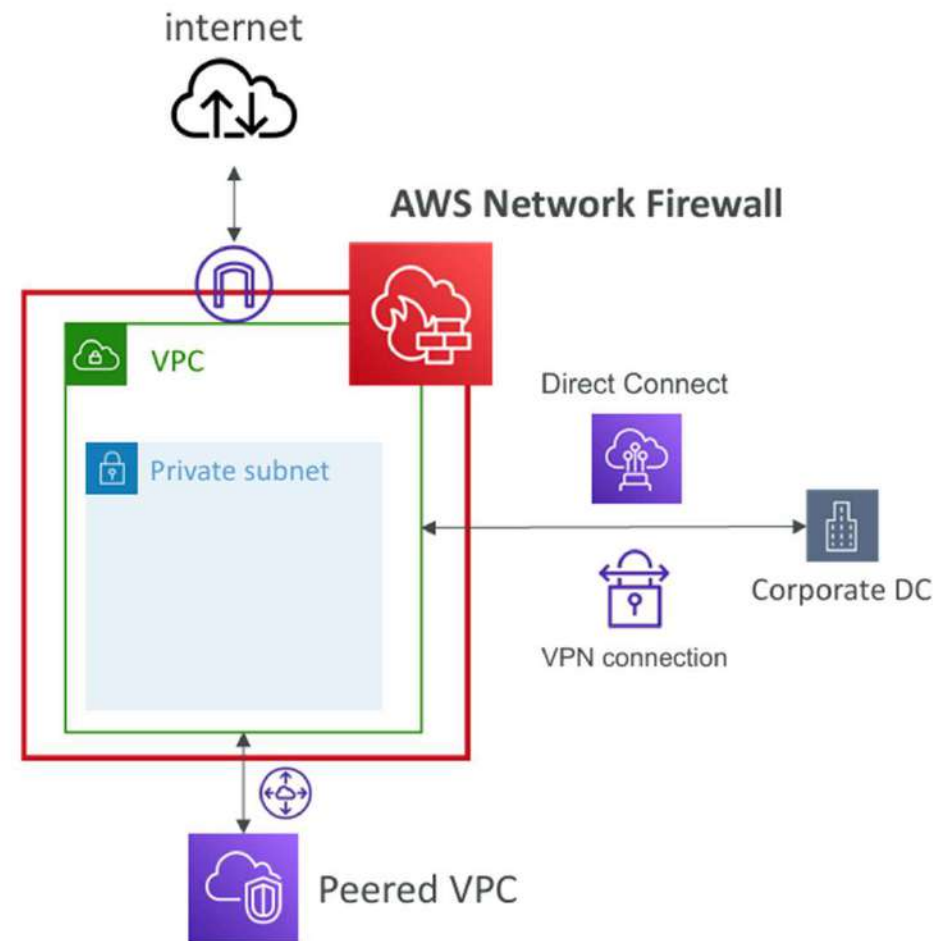
Network Protection on AWS

- ▶ To protect network on AWS, we've seen
- ▶ Network Access Control Lists (NACLs)
- ▶ Amazon VPC security groups
- ▶ AWS WAF (protect against malicious requests)
- ▶ AWS Shield & AWS Shield Advanced
- ▶ AWS Firewall Manager (to manage them across accounts)
- ▶ But what if we want to protect in a sophisticated way our entire VPC?

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AWS Network Firewall

- ▶ Protect your entire Amazon VPC
- ▶ From Layer 3 to Layer 7 protection
- ▶ Any direction, you can inspect
 - VPC to VPC traffic
 - Outbound to internet
 - Inbound from internet
 - To / from Direct Connect & Site-to-Site VPN
- ▶ Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- ▶ Rules can be centrally managed cross account by AWS Firewall Manager to apply to many VPCs



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Network Firewall – Fine Grained Controls



- ▶ Supports 1000s of rules
- IP & port - example: 10,000s of IPs filtering
- Protocol – example: block the SMB protocol for outbound communications
- Stateful domain list rule groups: only allow outbound traffic to *.mycorp.com or third-party software repo
- General pattern matching using regex
- ▶ **Traffic filtering:** Allow, drop, or alert for the traffic that matches the rules
- ▶ **Active flow inspection** to protect against network threats with intrusion prevention capabilities (like Gateway Load Balancer, but all managed by AWS)
- ▶ Send logs of rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose

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